

Vitamins, Minerals & Water – Page 423

Source: Lucent's General Knowledge – Biology

Biology: Vitamins, Minerals, Water & Balanced Diet

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the chemical name of Vitamin D?

- A. Tocopherol
- B. Calciferol
- C. Phylloquinone
- D. Retinol

Ans: B

Q2. Deficiency of Vitamin D in children causes:

- A. Scurvy
- B. Beriberi
- C. Rickets
- D. Night blindness

Ans: C

Q3. What is the chemical name of Vitamin E?

- A. Calciferol
- B. Ascorbic acid
- C. Tocopherol
- D. Thiamine

Ans: C

Q4. Deficiency of Vitamin E causes:

- A. Non-clotting of blood
- B. Less fertility and destruction of RBC
- C. Night blindness
- D. Rickets

Ans: B

Q5. Sources of Vitamin E include:

- A. Tomato and Soybean oil
- B. Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil
- C. Sea fish and Iodized salt
- D. Meat and Liver only

Ans: B

Q6. What is the chemical name of Vitamin K?

- A. Phylloquinone
- B. Riboflavin
- C. Niacin
- D. Pyridoxine

Ans: A

Q7. Deficiency of Vitamin K causes:

- A. Scurvy
- B. Pellagra
- C. Non-clotting of blood
- D. Anemia

Ans: C

Q8. Cobalt is found in which vitamin? A. Vitamin A B. Vitamin C C. Vitamin B12 D. Vitamin D	Ans: C
Q9. Synthesis of Vitamin D takes place through which substance in sunlight? A. Keratin B. Melanin C. Cholesterol (Ergosterol) D. Hemoglobin	Ans: C
Q10. Vitamin D2 is also known as: A. Calciferol B. Ergocalciferol C. Tocopherol D. Phylloquinone	Ans: B
Q11. Vitamin K is synthesized in our body by: A. Liver cells B. Kidney cells C. Bacteria in the colon D. Skin cells	Ans: C
Q12. Which vitamin is mainly stored in the liver? A. Vitamin C B. Vitamin B1 C. Vitamin A D. Vitamin E	Ans: C
Q13. Use of polished rice in human diet causes: A. Scurvy B. Rickets C. Beriberi D. Pellagra	Ans: C
Q14. Which is the richest source of Vitamin C? A. Milk B. Egg C. Amla D. Fish	Ans: C

Q15. Bleeding gum, falling of teeth, fragile bones, and delayed wound healing are symptoms of deficiency of: A. Vitamin A B. Vitamin D C. Vitamin C D. Vitamin K	Ans: C
Q16. Deficiency of calcium mainly occurs in the absence of which vitamin? A. Vitamin A B. Vitamin B C. Vitamin C D. Vitamin D	Ans: D
Q17. The daily requirement of Sodium (as sodium chloride) is: A. 20 mg B. 1 gram C. 2–5 grams D. 1.2 grams	Ans: C
Q18. Sodium is normally found in which part of the cell? A. Nucleus B. Protoplasm C. External fluid of cell D. Mitochondria	Ans: C
Q19. The daily requirement of Potassium is: A. 2–5 grams B. 1 gram C. 20 mg D. 1.2 grams	Ans: B
Q20. Potassium is normally found in: A. External fluid of cell B. Protoplasm C. Bones only D. Blood plasma only	Ans: B
Q21. The daily requirement of Calcium is approximately: A. 20 mg B. 1 gram C. 1.2 grams D. 2–5 grams	Ans: C

Q22. Which mineral provides strength to bones and teeth, and helps in blood clotting? A. Iron B. Calcium C. Zinc D. Iodine	Ans: B
Q23. The daily requirement of Iron for boys and girls respectively is: A. 20 mg for both B. 25 mg (boy) and 35 mg (girl) C. 35 mg (boy) and 25 mg (girl) D. 10 mg for both	Ans: B
Q24. Iron is an important component of: A. Thyroxin hormone B. Insulin C. Red Blood Corpuscles and Haemoglobin D. Bones and Teeth	Ans: C
Q25. Iodine is important for the synthesis of which hormone? A. Insulin B. Adrenaline C. Thyroxin hormone secreted by Thyroid gland D. Growth hormone	Ans: C
Q26. The daily requirement of Iodine is: A. 1 gram B. 1.2 grams C. 20 mg D. 2–5 grams	Ans: C
Q27. Which mineral is required for insulin functioning? A. Copper B. Magnesium C. Zinc D. Cobalt	Ans: C
Q28. Magnesium is important for functioning of: A. Thyroid gland B. Muscular system and nervous system C. Insulin production D. Blood clotting	Ans: B
Q29. Copper is needed for: A. Insulin functioning B. Thyroxin synthesis C. Formation of haemoglobin and bones, and as a conductor of electron D. Nerve impulse transmission only	Ans: C

Q30. Cobalt is essential for synthesis of: A. Thyroxin and Growth hormone B. RBC and Vitamin B12 C. Insulin and Glucagon D. Hemoglobin only	Ans: B
Q31. Deficiency of fluoride leads to: A. Anemia B. Goitre C. Mottling of teeth D. Night blindness	Ans: C
Q32. Presence of excess fluorine in water causes: A. Goitre B. Fluorosis C. Rickets D. Scurvy	Ans: B
Q33. To reduce tooth decay, most toothpastes contain: A. Calcium B. Iron C. Fluoride D. Zinc	Ans: C
Q34. The hard enamel layer of teeth is made of: A. Calcium carbonate B. Calcium hydroxy apatite C. Calcium oxalate D. Calcium phosphate only	Ans: B
Q35. Deficiency of calcium and iron is generally found in: A. Children B. Elderly men C. Pregnant women D. Athletes	Ans: C
Q36. A typical adult human body contains about how much magnesium? A. 5 grams B. 10 grams C. 25 grams D. 50 grams	Ans: C
Q37. Water is what percentage of our body weight? A. 40% B. 50% C. 60% D. 70%	Ans: C

Q38. Which of the following is NOT a main function of water in the body? A. Controls temperature by sweating and vapourization B. Important for excretion of excretory substances C. Synthesis of Vitamin D D. Maximum chemical reactions through hydrolysis	Ans: C
Q39. Minerals are: A. Organic compounds B. Homogenous inorganic material needed for metabolism C. Proteins D. Carbohydrates	Ans: B
Q40. Sources of Iodine include: A. Milk and cheese only B. Sea fish, sea food, green leafy vegetables, iodized salt C. Meat and liver only D. Eggs and bread only	Ans: B
Q41. Which mineral deficiency affects the activity of nitrogenase, nitrate reductase, and chlorate reductase in plants? A. Zinc B. Copper C. Molybdenum D. Magnesium	Ans: C
Q42. Sodium is related to which function in the body? A. Insulin functioning B. Contraction of muscles and transmission of nerve impulses C. Synthesis of thyroxin D. Formation of haemoglobin	Ans: B
Q43. Main sources of Calcium include: A. Sea fish and iodized salt B. Liver and fishes C. Milk, cheese, eggs, grains, fish D. Meat, fish, liver and grains	Ans: C
Q44. Phosphorus daily requirement is: A. 20 mg B. 1.2 grams C. 1 gram D. 25 mg	Ans: B

Q45. Phosphorus provides strength to bones and teeth in coordination with: A. Iron B. Zinc C. Calcium D. Iodine	Ans: C
Q46. Synthesis of vitamins can/cannot be done by cells of the body: A. Can be done easily B. Cannot be done; fulfilled by vitamin-containing food C. Only in liver D. Only in kidneys	Ans: B
Q47. Which vitamins can be synthesized in the human body? A. Vitamin A and B B. Vitamin B and C C. Vitamin D and K D. Vitamin E and K	Ans: C
Q48. Sources of Vitamin K include: A. Fish liver oil and milk B. Tomato, Soybean oil, Green vegetables, Alfalfa C. Eggs and bread D. Meat and liver	Ans: B
Q49. Sources of Iron include: A. Sea fish and iodized salt B. Vegetables only C. Albumen of egg, bread, Bajra, Banana, Spinach, Apple D. Milk and cheese only	Ans: C
Q50. Balance Diet is defined as: A. Eating only proteins B. Nutrition in which all important nutrients are available in sufficient quantity C. Eating only carbohydrates D. Fasting for health	Ans: B

Answer Key & Explanations

Q1. Answer: B — Calciferol

Page 423: Vitamin D chemical name is Calciferol.

Topic: Vitamins

Q2. Answer: C — Rickets

Page 423: Vitamin D deficiency causes Rickets in children and Osteomalacia/Osteoporosis in adults.

Topic: Vitamins

Q3. Answer: C — Tocopherol

Page 423: Vitamin E chemical name is Tocopherol.

Topic: Vitamins

Q4. Answer: B — Less fertility and destruction of RBC

Page 423: Vitamin E deficiency causes less fertility and destruction of RBC.

Topic: Vitamins

Q5. Answer: B — Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil

Page 423: Vitamin E sources include Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil, Wheat germ oil.

Topic: Vitamins

Q6. Answer: A — Phylloquinone

Page 423: Vitamin K chemical name is Phylloquinone.

Topic: Vitamins

Q7. Answer: C — Non-clotting of blood

Page 423: Vitamin K deficiency causes non-clotting of blood.

Topic: Vitamins

Q8. Answer: C — Vitamin B12

Page 423: Cobalt is found in Vitamin B12.

Topic: Vitamins

Q9. Answer: C — Cholesterol (Ergosterol)

Page 423: Synthesis of Vitamin D takes place through cholesterol (ergosterol) present in skin via ultraviolet rays of sunlight.

Topic: Vitamins

Q10. Answer: B — Ergocalciferol

Page 423: Vitamin D2 is known as ergocalciferol.

Topic: Vitamins

Q11. Answer: C — Bacteria in the colon

Page 423: Vitamin K is synthesized in our colon by bacteria and from there it is absorbed.

Topic: Vitamins

Q12. Answer: C — Vitamin A

Page 423: Vitamin A is mainly stored in liver.

Topic: Vitamins

Q13. Answer: C — Beriberi

Page 423: Use of polished rice in human diet causes Beriberi.

Topic: Vitamins

Q14. Answer: C — Amla

Page 423: Amla is the richest source of Vitamin C.

Topic: Vitamins

Q15. Answer: C — Vitamin C

Page 423: These symptoms occur due to deficiency of Vitamin C.

Topic: Vitamins

Q16. Answer: D — Vitamin D

Page 423: Deficiency of calcium mainly occurs in absence of Vitamin D.

Topic: Vitamins

Q17. Answer: C — 2–5 grams

Page 423: Daily requirement of Sodium is 2–5 grams (as sodium chloride).

Topic: Minerals

Q18. Answer: C — External fluid of cell

Page 423: Sodium is normally found in external fluid of cell.

Topic: Minerals

Q19. Answer: B — 1 gram

Page 423: Daily requirement of Potassium is approximately 1 gram.

Topic: Minerals

Q20. Answer: B — Protoplasm

Page 423: Potassium is normally found in protoplasm. It is important for different chemical reactions in cells.

Topic: Minerals

Q21. Answer: C — 1.2 grams

Page 423: Daily requirement of Calcium is approximately 1.2 grams.

Topic: Minerals

Q22. Answer: B — Calcium

Page 423: Calcium provides strength to bones and teeth with vitamin, important role in blood formation, muscular contraction, and help in clotting of blood.

Topic: Minerals

Q23. Answer: B — 25 mg (boy) and 35 mg (girl)

Page 423: Iron daily requirement is 25 mg for boys and 35 mg for girls.

Topic: Minerals

Q24. Answer: C — Red Blood Corpuscles and Haemoglobin

Page 423: Iron is important component of Red Blood Corpuscles and haemoglobin.

Topic: Minerals

Q25. Answer: C — Thyroxin hormone secreted by Thyroid gland

Page 423: Iodine is important for synthesis of thyroxin hormone secreted by Thyroid gland.

Topic: Minerals

Q26. Answer: C — 20 mg

Page 423: Daily requirement of Iodine is 20 mg.

Topic: Minerals

Q27. Answer: C — Zinc

Page 423: Zinc is required for insulin functioning. Sources include liver and fishes.

Topic: Minerals

Q28. Answer: B — Muscular system and nervous system

Page 423: Magnesium is for functioning of muscular system and nervous system.

Topic: Minerals

Q29. Answer: C — Formation of haemoglobin and bones, and as a conductor of electron

Page 423: Copper is needed for formation of haemoglobin and bones and as a conductor of electron.

Topic: Minerals

Q30. Answer: B — RBC and Vitamin B12

Page 423: Cobalt is essential for synthesis of RBC and Vitamin B12.

Topic: Minerals

Q31. Answer: C — Mottling of teeth

Page 423: Deficiency of fluoride leads to mottling of teeth.

Topic: Minerals

Q32. Answer: B — Fluorosis

Page 423: Presence of excess fluorine in water causes fluorosis.

Topic: Minerals

Q33. Answer: C — Fluoride

Page 423: To reduce tooth decay most tooth pastes contain fluoride.

Topic: Minerals

Q34. Answer: B — Calcium hydroxy apatite

Page 423: Hard enamel layer of teeth is calcium hydroxy apatite.

Topic: Minerals

Q35. Answer: C — Pregnant women

Page 423: Deficiency of calcium and iron is generally found in pregnant women.

Topic: Minerals

Q36. Answer: C — 25 grams

Page 423: A typical adult human body contains about 25 grams of magnesium.

Topic: Minerals

Q37. Answer: C — 60%

Page 423: Water is the important component of our body, 60% weight of the body is water.

Topic: Water

Q38. Answer: C — Synthesis of Vitamin D

Page 423: Water functions include temperature control, excretion, and hydrolysis. Vitamin D synthesis is by sunlight/cholesterol, not water.

Topic: Water

Q39. Answer: B — Homogenous inorganic material needed for metabolism

Page 423: Mineral is a homogenous inorganic material needed for body which control the metabolism of body.

Topic: Minerals

Q40. Answer: B — Sea fish, sea food, green leafy vegetables, iodized salt

Page 423: Iodine sources include sea fish, sea food, green leafy vegetables, iodized salt.

Topic: Minerals

Q41. Answer: C — Molybdenum

Page 423: Deficiency of molybdenum affects the activity of nitrogenase, nitrate reductase and chlorate reductase in plants.

Topic: Minerals

Q42. Answer: B — Contraction of muscles and transmission of nerve impulses

Page 423: Sodium is related to contraction of muscles and transmission of nerve impulses in nerve fiber.

Topic: Minerals

Q43. Answer: C — Milk, cheese, eggs, grains, fish

Page 423: Calcium sources include milk, cheese, eggs, grains, gram, fish etc.

Topic: Minerals

Q44. Answer: B — 1.2 grams

Page 423: Phosphorus daily requirement is 1.2 gram. Sources include milk, cheese, bajra, green leafy vegetables.

Topic: Minerals

Q45. Answer: C — Calcium

Page 423: Phosphorus provides strength to bones and teeth, in coordination with calcium.

Topic: Minerals

Q46. Answer: B — Cannot be done; fulfilled by vitamin-containing food

Page 423: Synthesis of vitamins cannot be done by the cells of body. It is fulfilled by vitamin containing food.

Topic: Vitamins

Q47. Answer: C — Vitamin D and K

Page 423: Synthesis of Vitamin D and K takes place in body (D via sunlight, K via colon bacteria).

Topic: Vitamins

Q48. Answer: B — Tomato, Soybean oil, Green vegetables, Alfalfa

Page 423: Vitamin K sources include Tomato, Soybean oil, Green vegetables, Alfalfa etc.

Topic: Vitamins

Q49. Answer: C — Albumen of egg, bread, Bajra, Banana, Spinach, Apple

Page 423: Iron sources include albumen of egg, bread, Bajra, Banana, Spinach apple.

Topic: Minerals

Q50. Answer: B — Nutrition in which all important nutrients are available in sufficient quantity

Page 423: Balance Diet is that nutrition in which all the important nutrients for organism are available in sufficient quantity.

Topic: Balanced Diet

Balanced Diet & Biotechnology – Page 424

Source: Lucent's General Knowledge – Biology

Biology: Diet, Biotechnology & Applied Biology

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Necessary calorie requirement for a male light worker is:

- A. 3000 calorie
- B. 2000 calorie
- C. 3600 calorie
- D. 2500 calorie

Ans: B

Q2. Necessary calorie requirement for a female eight-hour worker is:

- A. 2000 calorie
- B. 3000 calorie
- C. 2500 calorie
- D. 3600 calorie

Ans: C

Q3. A male hard worker requires how many calories?

- A. 2000
- B. 2500
- C. 3000
- D. 3600

Ans: D

Q4. Which substance is used as a preservative to preserve food items?

- A. Sodium chloride
- B. Sodium benzoate
- C. Calcium carbonate
- D. Potassium permanganate

Ans: B

Q5. Milk is not considered a complete food due to lack of:

- A. Calcium and Protein
- B. Vitamin C and Iron
- C. Vitamin A and D
- D. Fat and Carbohydrate

Ans: B

Q6. Which is a genetically modified vegetable recently made available in Indian market?

- A. Bt-Cotton
- B. Bt-Brinjal
- C. Bt-Corn
- D. Bt-Potato

Ans: B

Q7. Biotechnology deals with techniques of using:

- A. Only chemicals
- B. Live organisms or enzymes to produce useful products
- C. Only metals
- D. Nuclear energy

Ans: B

Q8. Genetic Engineering is the technique to alter: A. Physical structure of organism B. Chemistry of genetic material (DNA & RNA) to change phenotype C. Only protein structure D. Only cell membrane	Ans: B
Q9. Bioprocess Engineering involves maintenance of: A. Acidic environment B. Sterile ambience for desired microbes/eukaryotic cells C. High temperature only D. Vacuum conditions	Ans: B
Q10. Restriction enzyme is known as: A. Molecular glue B. Molecular scissors C. Molecular hammer D. Molecular tape	Ans: B
Q11. Plasmid is: A. A type of virus B. Extra small circular DNA in bacteria C. A type of protein D. Part of ribosome	Ans: B
Q12. Due to presence of plasmid, bacteria show resistance to: A. Heat B. Many antibiotics C. UV radiation D. Cold temperature	Ans: B
Q13. A plasmid can be used as a vector to deliver: A. Protein into host cell B. An alien piece of DNA into host organism C. Carbohydrate D. Lipids	Ans: B
Q14. DNA ligase is an enzyme that: A. Cuts DNA at specific sites B. Joins cut ends of DNA molecule together C. Unwinds DNA D. Copies DNA	Ans: B
Q15. Recombinant DNA is created by: A. Destroying DNA B. Combining two fragments of DNA from two different sources C. Heating DNA D. Freezing DNA	Ans: B

Q16. Cloning is the process of producing: A. Different versions of DNA B. Large number of identical copies of any template DNA C. Mutated DNA D. Shortened DNA	Ans: B
Q17. Restriction enzymes belong to a larger class of enzymes called: A. Ligases B. Nucleases C. Proteases D. Lipases	Ans: B
Q18. Exonuclease removes nucleotide from: A. The middle of DNA B. The end of DNA C. Both ends simultaneously D. Only RNA	Ans: B
Q19. Endonuclease cuts at: A. End of DNA only B. Specific position within the DNA C. Random positions D. Only RNA	Ans: B
Q20. Organisms whose genes have been altered by manipulation are called: A. Hybrid organisms B. Genetically Modified Organisms (GMO) C. Cloned organisms D. Mutant organisms	Ans: B
Q21. Gene therapy is: A. Treatment using antibiotics B. Collection of methods that allow correction of a gene defect in a child or embryo C. Surgical operation on genes D. Radiation therapy	Ans: B
Q22. The first clinical gene therapy was given in 1990 to a four-year-old girl with which deficiency? A. Insulin deficiency B. Adenosine deaminase (ADA) deficiency C. Iron deficiency D. Vitamin D deficiency	Ans: B

Q23. Genetically modified plants are useful in: A. Decreasing crop yields B. Increasing crop yields, reducing losses, and making crops stress tolerant C. Increasing pesticide use D. Reducing soil fertility	Ans: B
Q24. Papaver somniferum plant yields: A. Antibiotics B. Powerful analgesic C. Vitamins D. Minerals	Ans: B
Q25. The major source of sugar in the world is: A. Cocos nucifera B. Saccharum Officinarum C. Papaver somniferum D. Bt-Brinjal	Ans: B
Q26. Cocos nucifera is widely used in: A. Making medicines B. Preparation of culture medium C. Making clothes D. Building materials	Ans: B
Q27. To yield milk, cow is given which hormone? A. Insulin B. Prolactin C. Thyroxin D. Adrenaline	Ans: B
Q28. The best way to determine paternity is: A. Blood group test B. DNA finger printing C. X-ray D. CT scan	Ans: B
Q29. Chloromycetin is obtained from: A. Penicillium B. Streptomycetin venezuelae C. E. coli D. Yeast	Ans: B
Q30. In Recombinant Vector Vaccines, which are used as vectors? A. Chemicals B. Bacteria and virus C. Proteins D. Minerals	Ans: B

Q31. Bollgard I and Bollgard II technology are used in: A. Making vaccines B. Developing genetically modified crop plants C. Water purification D. Soil testing	Ans: B
Q32. The term 'ACE-2' is talked about in the context of: A. Cancer treatment B. Spread of viral diseases C. Diabetes management D. Heart surgery	Ans: B
Q33. A diagnostic test, Real Time PCR for COVID-19 detects: A. Protein of virus B. RNA of virus C. DNA of bacteria D. Antibodies only	Ans: B
Q34. Rapid Antibodies Test does not require sample from nose and throat and was approved by: A. WHO B. Indian Council of Medical Research C. FDA D. CDC	Ans: B
Q35. Vector vaccines work by: A. Killing all viruses directly B. Entering cells and enabling them to create spike protein C. Blocking all receptors D. Destroying antibodies	Ans: B
Q36. Sinovac given for COVID-19 is a: A. mRNA vaccine B. Whole virus vaccine C. Vector vaccine D. DNA vaccine	Ans: B
Q37. DNA Barcoding is a tool to: A. Create new DNA B. Distinguish among species that look alike C. Destroy harmful DNA D. Clone organisms	Ans: B

Q38. Covishield vaccine was produced by Serum Institute of India using which platform? A. Whole virus platform B. mRNA platform C. DNA platform D. Protein subunit platform	Ans: B
Q39. Sputnik V vaccine is manufactured by using: A. mRNA platform B. Vector base platform C. Whole virus D. Protein subunit	Ans: B
Q40. Applications of biotechnology include: A. Only medicine B. Agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment, energy production C. Only agriculture D. Only energy	Ans: B
Q41. Calorie requirement for a female light worker is: A. 2000 B. 2100 C. 2500 D. 3000	Ans: B
Q42. Calorie requirement for a male eight-hour worker is: A. 2000 B. 2500 C. 3000 D. 3600	Ans: C
Q43. Calorie requirement for a female hard worker is: A. 2100 B. 2500 C. 3000 D. 3600	Ans: C
Q44. In the balanced diet chart, the grain (wheat, rice) requirement for an adult male doing normal work is: A. 400 g B. 520 g C. 670 g D. 410 g	Ans: B

Q45. Making curd, bread or wine are examples of: A. Genetic engineering B. Biotechnology C. Gene therapy D. Cloning	Ans: B
Q46. Which technology creates unique character (antibiotic resistance) in bacteria? A. Gene therapy B. Plasmid presence C. Cloning D. PCR	Ans: B
Q47. Restriction enzyme cuts DNA at: A. Random locations B. Specific location C. Only at the ends D. Only the middle	Ans: B
Q48. The milk requirement for an adult male doing hard work is: A. 150 g B. 200 g C. 250 g D. 100 g	Ans: C
Q49. Pulses requirement for adult male normal worker is: A. 40 g B. 50 g C. 60 g D. 80 g	Ans: B
Q50. An example of GMO food crop approved in India is: A. Bt-Rice B. Bt-Brinjal C. Bt-Wheat D. Bt-Mango	Ans: B

Answer Key & Explanations

Q1. Answer: B — 2000 calorie

Page 424: Light worker male requires 2000 calorie, female 2100 calorie.

Topic: *Balanced Diet*

Q2. Answer: C — 2500 calorie

Page 424: Eight hours worker female requires 2500 calorie, male 3000 calorie.

Topic: *Balanced Diet*

Q3. Answer: D — 3600

Page 424: Hard worker male requires 3600 calorie, female 3000 calorie.

Topic: *Balanced Diet*

Q4. Answer: B — Sodium benzoate

Page 424: Sodium Benzoate is used as preservative to preserve food items.

Topic: *Food Science*

Q5. Answer: B — Vitamin C and Iron

Page 424: Milk is not considered as complete food due to lack of vitamin C and iron.

Topic: *Food Science*

Q6. Answer: B — Bt-Brinjal

Page 424: Bt-Brinjal is a genetically modified vegetable recently being made available in Indian market.

Topic: *Biotechnology*

Q7. Answer: B — Live organisms or enzymes to produce useful products

Page 424: Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce useful products (e.g., curd, bread, wine).

Topic: *Biotechnology*

Q8. Answer: B — Chemistry of genetic material (DNA & RNA) to change phenotype

Page 424: Genetic Engineering is the technique to alter the chemistry of genetic material (DNA & RNA) and introduce into host organism, changing the phenotype.

Topic: *Biotechnology*

Q9. Answer: B — Sterile ambience for desired microbes/eukaryotic cells

Page 424: Bioprocess Engineering is maintenance of sterile ambience for growth of desired microbes or eukaryotic cells for manufacture of biotechnological products.

Topic: *Biotechnology*

Q10. Answer: B — Molecular scissors

Page 424: Restriction enzyme is known as molecular scissors cutting the DNA at specific location.

Topic: *Biotechnology*

Q11. Answer: B — Extra small circular DNA in bacteria

Page 424: In addition to genomic DNA, many bacteria have extra small circular DNA known as plasmid.

Topic: Biotechnology

Q12. Answer: B — Many antibiotics

Page 424: Due to presence of plasmid, bacteria show resistance to many antibiotics. This creates a unique character in bacteria.

Topic: Biotechnology

Q13. Answer: B — An alien piece of DNA into host organism

Page 424: A plasmid can be used as vector to deliver an alien piece of DNA into host organism.

Topic: Biotechnology

Q14. Answer: B — Joins cut ends of DNA molecule together

Page 424: DNA ligase is an enzyme by which cut ends of DNA molecule join together.

Topic: Biotechnology

Q15. Answer: B — Combining two fragments of DNA from two different sources

Page 424: Recombinant DNA is a DNA which has been created by combining two fragments of DNA from two different sources.

Topic: Biotechnology

Q16. Answer: B — Large number of identical copies of any template DNA

Page 424: Cloning is producing large number of identical copies of any template DNA.

Topic: Biotechnology

Q17. Answer: B — Nucleases

Page 424: Restriction enzymes belong to larger class of enzymes called nucleases.

Topic: Biotechnology

Q18. Answer: B — The end of DNA

Page 424: Exonuclease removes nucleotide from the end of DNA.

Topic: Biotechnology

Q19. Answer: B — Specific position within the DNA

Page 424: Endonuclease cuts at specific position within the DNA.

Topic: Biotechnology

Q20. Answer: B — Genetically Modified Organisms (GMO)

Page 424: Plant, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO).

Topic: Biotechnology

Q21. Answer: B — Collection of methods that allow correction of a gene defect in a child or embryo

Page 424: Gene therapy is a collection of methods that allow correction of a gene defect that has been diagnosed in a child or embryo.

Topic: Biotechnology

Q22. Answer: B — Adenosine deaminase (ADA) deficiency

Page 424: The first clinical gene therapy was given in 1990 to a four-year-old girl with adenosine deaminase (ADA) deficiency.

Topic: Biotechnology

Q23. Answer: B — Increasing crop yields, reducing losses, and making crops stress tolerant

Page 424: GM plants are useful in increasing crop yields, reduce losses and make the crop more tolerant to stresses.

Topic: Biotechnology

Q24. Answer: B — Powerful analgesic

Page 424: Papaver somniferum plant yields powerful analgesic.

Topic: Biotechnology

Q25. Answer: B — Saccharum Officinarum

Page 424: Major source of sugar in the world is Saccharum Officinarum.

Topic: Biotechnology

Q26. Answer: B — Preparation of culture medium

Page 424: Cocos nucifera is a plant material widely used in preparation of culture medium.

Topic: Biotechnology

Q27. Answer: B — Prolactin

Page 424: To yield milk, cow is given prolactin.

Topic: Biotechnology

Q28. Answer: B — DNA finger printing

Page 424: The best way to determine paternity is DNA finger printing.

Topic: Biotechnology

Q29. Answer: B — Streptomycetin venezuelae

Page 424: Chloromycetin is obtained from streptomycetin venezuelae.

Topic: Biotechnology

Q30. Answer: B — Bacteria and virus

Page 424: Genetic engineering is applied in development of Recombinant Vector Vaccines in which bacteria and virus are used as vectors.

Topic: Biotechnology

Q31. Answer: B — Developing genetically modified crop plants

Page 424: Bollgard I and Bollgard II technology are used in developing genetically modified crop plants.

Topic: Biotechnology

Q32. Answer: B — Spread of viral diseases

Page 424: The term ACE-2 is talked about in the context of spread of viral diseases.

Topic: Biotechnology

Q33. Answer: B — RNA of virus

Page 424: Real Time PCR for COVID-19 detects RNA of virus.

Topic: Biotechnology

Q34. Answer: B — Indian Council of Medical Research

Page 424: Rapid Antibodies Test does not require sample from nose and throat in COVID-19 test, approved by Indian Council of Medical Research.

Topic: Biotechnology

Q35. Answer: B — Entering cells and enabling them to create spike protein

Page 424: Vector vaccines work to provide immunity by entering directly into the cells and enabling them to create spike protein.

Topic: Biotechnology

Q36. Answer: B — Whole virus vaccine

Page 424: Sinovac given for Covid-19 is a whole virus vaccine.

Topic: Biotechnology

Q37. Answer: B — Distinguish among species that look alike

Page 424: DNA Barcoding can be a tool to distinguish among species that look alike.

Topic: Biotechnology

Q38. Answer: B — mRNA platform

Page 424: The Serum Institute of India produced Covishield using mRNA platform.

Topic: Biotechnology

Q39. Answer: B — Vector base platform

Page 424: Sputnik V vaccine is manufactured by using vector base platform.

Topic: Biotechnology

Q40. Answer: B — Agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment, energy production

Page 424: Applications include agriculture, therapeutic, diagnostic, biopharmaceutical, processed food, waste treatment and energy production.

Topic: Biotechnology

Q41. Answer: B — 2100

Page 424: Female light worker requires 2100 calories.

Topic: Balanced Diet

Q42. Answer: C — 3000

Page 424: Male eight-hour worker requires 3000 calories.

Topic: Balanced Diet

Q43. Answer: C — 3000

Page 424: Female hard worker requires 3000 calories.

Topic: *Balanced Diet*

Q44. Answer: B — 520 g

Page 424: For adult male moderate (M) worker, grain requirement is 520 g.

Topic: *Balanced Diet*

Q45. Answer: B — Biotechnology

Page 424: Biotechnology examples include making curd, bread or wine in which microbes are used.

Topic: *Biotechnology*

Q46. Answer: B — Plasmid presence

Page 424: Due to presence of plasmid, bacteria show resistance to many antibiotics, creating unique character.

Topic: *Biotechnology*

Q47. Answer: B — Specific location

Page 424: Restriction enzyme cuts the DNA at specific location.

Topic: *Biotechnology*

Q48. Answer: C — 250 g

Page 424: For adult male hard worker, milk requirement is 250 g.

Topic: *Balanced Diet*

Q49. Answer: B — 50 g

Page 424: For adult male moderate worker, pulses requirement is 50 g.

Topic: *Balanced Diet*

Q50. Answer: B — Bt-Brinjal

Page 424: Bt-Brinjal is a genetically modified vegetable recently made available in Indian market.

Topic: *Biotechnology*

Proteins, Fats & Vitamins – Page 422

Source: Lucent's General Knowledge – Biology

Biology: Proteins, Fats & Vitamins

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. On the basis of chemical composition, proteins are divided into how many types?

- A. 2
- B. 3
- C. 4
- D. 5

Ans: B

Q2. Simple Protein consists of only:

- A. Fatty acids
- B. Amino acids
- C. Nucleic acids
- D. Carbohydrates

Ans: B

Q3. Which is an example of Conjugated Protein?

- A. Albumin
- B. Histones
- C. Haemoglobin (having some other chemical compound in addition to amino acids)
- D. Globulin

Ans: C

Q4. Derived Protein is obtained from:

- A. Fats
- B. Partial hydrolysis of natural protein
- C. Carbohydrates
- D. Minerals

Ans: B

Q5. Animal protein is called first class protein because:

- A. It is cheap
- B. It contains all the essential amino acids
- C. It is vegetarian
- D. It is easy to digest

Ans: B

Q6. Egg contains how many essential amino acids?

- A. 10
- B. 15
- C. 20 (all)
- D. 5

Ans: C

Q7. Proteins are important for:

- A. Only energy
- B. Physical growth, protoplasm and formation of cells, tissues
- C. Only bones
- D. Only blood

Ans: B

Q8. Lack of proteins hampers growth. This causes a disease called: A. Rickets B. Kwashiorkor C. Scurvy D. Beriberi	Ans: B
Q9. In Kwashiorkor disease, which parts of children are affected? A. Only legs B. Stomach comes out, hands and legs get slimmed C. Only hands D. Only head	Ans: B
Q10. Marasmus is caused by: A. Excess protein B. Loosened deficiency of protein and energy C. Excess vitamins D. Iron deficiency	Ans: B
Q11. Biological catalysts which speed up chemical reactions and remain unchanged at 37°C are called: A. Hormones B. Enzymes C. Vitamins D. Minerals	Ans: B
Q12. Human nails are made up of which protein? A. Albumin B. Keratin C. Casein D. Globulin	Ans: B
Q13. Soybean is the richest source of: A. Fat B. Protein C. Carbohydrate D. Vitamin C	Ans: B
Q14. When egg is heated, the transparent liquid portion turns solid. This happens due to: A. Evaporation B. Thermal denaturation of protein C. Chemical reaction with air D. Loss of water	Ans: B

Q15. Keratin is a protein which protects: A. Bones B. Epithelial cells from damage C. Blood cells D. Nerve cells	Ans: B
Q16. Fats are esters of: A. Amino acid and glucose B. Glycerol and fatty acid C. Nucleic acid and protein D. Starch and sugar	Ans: B
Q17. Fats are of how many types? A. 3 B. 2 – Saturated and Unsaturated fatty acids C. 4 D. 1	Ans: B
Q18. Saturated fatty acids are found in: A. Fish oil and vegetable oil B. Fat (solid at 20°C temperature) C. Only soybean oil D. Only coconut water	Ans: B
Q19. Unsaturated fatty acids are found in: A. Solid fats only B. Fish oil and vegetable oil (liquid form - called oil) C. Butter only D. Cheese only	Ans: B
Q20. An adult person should get approximately what percentage of energy from fat? A. 10-15% B. 20-30% C. 40-50% D. 5-10%	Ans: B
Q21. 1 gram of fat liberates how many kilo calories of energy? A. 4.1 B. 9.3 C. 7.0 D. 5.5	Ans: B
Q22. Fat stores maximum energy per: A. Milliliter B. Gram C. Kilogram D. Liter	Ans: B

Q23. Fat provides energy to the body and: A. Increases body weight only B. Prevents the loss of heat and remains under the skin C. Causes diseases D. Has no function	Ans: B
Q24. Rancidity is the oxidation of fat. It changes the: A. Color only B. Taste and smell of food C. Volume D. Weight	Ans: B
Q25. Due to lack of fat, skin gets: A. Darker B. Dry and body weight decreases C. Thicker D. Smoother	Ans: B
Q26. Vitamin was first coined/invented by: A. Louis Pasteur B. Sir F.G. Hopkins C. Robert Koch D. Alexander Fleming	Ans: B
Q27. The term 'Vitamine' was first used by: A. Hopkins B. Funk C. Pasteur D. Darwin	Ans: B
Q28. On the basis of solubility, vitamins are of how many types? A. 3 B. 2 – Water soluble and Fat soluble C. 4 D. 1	Ans: B
Q29. Which vitamins are water soluble? A. A and D B. B and C C. E and K D. A and K	Ans: B
Q30. Which vitamins are fat soluble? A. B and C B. A, D, E and K C. Only B12 D. Only C	Ans: B

Q31. Chemical name of Vitamin A is: A. Thiamine B. Retinol C. Riboflavin D. Niacin	Ans: B
Q32. Deficiency of Vitamin A causes: A. Beriberi B. Night blindness, Xerophthalmia C. Scurvy D. Pellagra	Ans: B
Q33. Chemical name of Vitamin B1 is: A. Riboflavin B. Thiamine C. Niacin D. Pyridoxine	Ans: B
Q34. Deficiency of Vitamin B2 (Riboflavin) causes: A. Night blindness B. Cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers C. Scurvy D. Rickets	Ans: B
Q35. Deficiency of Vitamin B3 (Nicotinamide/Niacin) causes: A. Rickets B. Whitening of hair, mental retardation, Pellagra C. Scurvy D. Night blindness	Ans: B
Q36. Deficiency of Vitamin B5 (Pantothenic acid) causes: A. Scurvy B. Burning sensation, Dermatitis C. Rickets D. Night blindness	Ans: B
Q37. Chemical name of Vitamin B6 is: A. Riboflavin B. Pyridoxine C. Thiamine D. Niacin	Ans: B
Q38. Deficiency of Vitamin B6 causes: A. Night blindness B. Anemia, skin disease, Depression C. Rickets D. Scurvy	Ans: B

Q39. Deficiency of Vitamin B7 (Biotin) causes: A. Scurvy B. Paralysis, body pain, hair falling, scaly and itchy skin C. Pellagra D. Rickets	Ans: B
Q40. Chemical name of Vitamin B9 is: A. Cyanocobalamin B. Folic acid C. Biotin D. Thiamine	Ans: B
Q41. Deficiency of Vitamin B9 (Folic acid) causes: A. Night blindness B. Pernicious Anemia, jaundice C. Rickets D. Pellagra	Ans: B
Q42. Chemical name of Vitamin B12 is: A. Folic acid B. Cyanocobalamin C. Pyridoxine D. Riboflavin	Ans: B
Q43. Deficiency of Vitamin B12 causes: A. Scurvy B. Anemia, dysentery C. Pellagra D. Rickets	Ans: B
Q44. Chemical name of Vitamin C is: A. Retinol B. Ascorbic acid C. Calciferol D. Tocopherol	Ans: B
Q45. Deficiency of Vitamin C causes: A. Rickets B. Scurvy, swelling of gums or bleeding gums C. Night blindness D. Beriberi	Ans: B
Q46. Sources of Vitamin A include: A. Sea fish only B. Milk, Egg, Cheese, Green vegetable, Carrot C. Only fruits D. Only grains	Ans: B

Q47. Kwashiorkor disease typically develops in children due to:

- A. Excess protein
- B. Protein deficiency while caloric intake may be adequate
- C. Vitamin D deficiency
- D. Iron deficiency

Ans: B

Q48. Proteins take part in formation of:

- A. Only bones
- B. Cells, protoplasm and tissues
- C. Only blood
- D. Only skin

Ans: B

Q49. Sources of Vitamin B1 (Thiamine) include:

- A. Fish liver oil
- B. Pulse, Liver, Egg, Milk, eggs etc.
- C. Only sea fish
- D. Only iodized salt

Ans: B

Q50. Vitamin B1 (Thiamine) deficiency causes which disease?

- A. Scurvy
- B. Beriberi
- C. Pellagra
- D. Rickets

Ans: B

Answer Key & Explanations

Q1. Answer: B — 3

Page 422: Proteins are divided into three types: Simple Protein, Conjugated Protein, and Derived Protein.

Topic: Proteins

Q2. Answer: B — Amino acids

Page 422: Simple Protein consists of only amino acids. Examples: Albumins, Globulins, Histones.

Topic: Proteins

Q3. Answer: C — Haemoglobin (having some other chemical compound in addition to amino acids)

Page 422: Conjugated Protein has some other chemical compound in addition to amino acids, e.g. Haemoglobin.

Topic: Proteins

Q4. Answer: B — Partial hydrolysis of natural protein

Page 422: Derived Protein is derived from the partial hydrolysis of natural protein. Examples: Peptone, Peptide, Proteases.

Topic: Proteins

Q5. Answer: B — It contains all the essential amino acids

Page 422: Animal protein is called first class protein because it contains all the essential amino acids.

Topic: Proteins

Q6. Answer: C — 20 (all)

Page 422: Egg contains all the essential amino acids.

Topic: Proteins

Q7. Answer: B — Physical growth, protoplasm and formation of cells, tissues

Page 422: Proteins are important for physical growth, protoplasm and formation of cells, tissues.

Topic: Proteins

Q8. Answer: B — Kwashiorkor

Page 422: Lack of proteins causes Kwashiorkor and Marasmus in children.

Topic: Proteins

Q9. Answer: B — Stomach comes out, hands and legs get slimmed

Page 422: In Kwashiorkor, the stomach comes out, and hands and legs of children get slimmed. Caused by protein deficiency.

Topic: Proteins

Q10. Answer: B — Loosened deficiency of protein and energy

Page 422: Marasmus is caused by loosened deficiency of protein and energy together.

Topic: Proteins

Q11. Answer: B — Enzymes

Page 422: Biological catalysts (enzymes) speed up chemical reactions and remain unchanged after reaction at 37°C.

Topic: Proteins

Q12. Answer: B — Keratin

Page 422: Human nails are made up of protein Keratin.

Topic: Proteins

Q13. Answer: B — Protein

Page 422: Soybean is the richest source of protein.

Topic: Proteins

Q14. Answer: B — Thermal denaturation of protein

Page 422: When egg is heated, the protein turns solid due to thermal denaturation of protein (denaturation around at normal solution).

Topic: Proteins

Q15. Answer: B — Epithelial cells from damage

Page 422: Keratin is the protein which protects the epithelial cells from damage.

Topic: Proteins

Q16. Answer: B — Glycerol and fatty acid

Page 422: Fats are esters of glycerol and fatty acid. They contain carbon, hydrogen and oxygen but proportionally less oxygen.

Topic: Fats

Q17. Answer: B — 2 – Saturated and Unsaturated fatty acids

Page 422: Only two types – Saturated fatty acids (solid at 20°C, e.g. this is called fat) and Unsaturated fatty acids (liquid form, called oil).

Topic: Fats

Q18. Answer: B — Fat (solid at 20°C temperature)

Page 422: Saturated fatty acids are solid at 20°C temperature, this is called fat.

Topic: Fats

Q19. Answer: B — Fish oil and vegetable oil (liquid form - called oil)

Page 422: Unsaturated fatty acids are found in fish oil and vegetable oil. Examples: coconut oil, palm oil.

Topic: Fats

Q20. Answer: B — 20-30%

Page 422: Normally an adult person should get approximately 20-30% of energy from fat.

Topic: Fats

Q21. Answer: B — 9.3

Page 422: 9.3 kilo calorie energy is liberated from 1 gram of fat.

Topic: Fats

Q22. Answer: B — Gram

Page 422: Fat stores maximum energy per gram.

Topic: Fats

Q23. Answer: B — Prevents the loss of heat and remains under the skin

Page 422: Fat provides energy to the body and prevents the loss of heat, remaining under the skin.

Topic: Fats

Q24. Answer: B — Taste and smell of food

Page 422: Rancidity takes place due to oxidation of fat and oils. It changes the taste and smell of food.

Topic: Fats

Q25. Answer: B — Dry and body weight decreases

Page 422: Due to the lack of fat, skin gets fatty/dry, the body weight of the body gets checked/decreases and development of the body is affected.

Topic: Fats

Q26. Answer: B — Sir F.G. Hopkins

Page 422: Vitamin was coined/invented by Sir F.G. Hopkins.

Topic: Vitamins

Q27. Answer: B — Funk

Page 422: The term vitamin was coined by Funk.

Topic: Vitamins

Q28. Answer: B — 2 – Water soluble and Fat soluble

Page 422: On the basis of solubility, vitamins are of two types: (a) Vitamin soluble in water: Vitamin-B and Vitamin-C, (b) Vitamin soluble in fat: Vitamin-A, Vitamin-D, Vitamin-E and Vitamin-K.

Topic: Vitamins

Q29. Answer: B — B and C

Page 422: Water soluble vitamins are Vitamin B and Vitamin C.

Topic: Vitamins

Q30. Answer: B — A, D, E and K

Page 422: Fat soluble vitamins are Vitamin A, D, E and K.

Topic: Vitamins

Q31. Answer: B — Retinol

Page 422: Vitamin A chemical name is Retinol.

Topic: Vitamins

Q32. Answer: B — Night blindness, Xerophthalmia

Page 422: Vitamin A deficiency causes night blindness and xerophthalmia.

Topic: Vitamins

Q33. Answer: B — Thiamine

Page 422: Vitamin B1 chemical name is Thiamine.

Topic: Vitamins

Q34. Answer: B — Cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers

Page 422: Vitamin B2 deficiency causes cracking of skin, reddish eye, cracking at corners of lips, mouth ulcers.

Topic: Vitamins

Q35. Answer: B — Whitening of hair, mental retardation, Pellagra

Page 422: Vitamin B3 deficiency causes whitening of hair, mental retardation, Pellagra.

Topic: Vitamins

Q36. Answer: B — Burning sensation, Dermatitis

Page 422: Vitamin B5 (Pantothenic acid) deficiency causes burning sensation, dermatitis.

Topic: Vitamins

Q37. Answer: B — Pyridoxine

Page 422: Vitamin B6 chemical name is Pyridoxine.

Topic: Vitamins

Q38. Answer: B — Anemia, skin disease, Depression

Page 422: Vitamin B6 deficiency causes Anemia, skin disease, Depression.

Topic: Vitamins

Q39. Answer: B — Paralysis, body pain, hair falling, scaly and itchy skin

Page 422: Vitamin B7 (Biotin) deficiency causes paralysis, body pain, hair falling, scaly and itchy skin.

Topic: Vitamins

Q40. Answer: B — Folic acid

Page 422: Vitamin B9 chemical name is Folic acid.

Topic: Vitamins

Q41. Answer: B — Pernicious Anemia, jaundice

Page 422: Vitamin B9 deficiency causes Pernicious Anemia, jaundice.

Topic: Vitamins

Q42. Answer: B — Cyanocobalamin

Page 422: Vitamin B12 chemical name is Cyanocobalamin (or Cobalamin).

Topic: Vitamins

Q43. Answer: B — Anemia, dysentery

Page 422: Vitamin B12 deficiency causes Anemia, dysentery.

Topic: Vitamins

Q44. Answer: B — Ascorbic acid

Page 422: Vitamin C chemical name is Ascorbic acid.

Topic: Vitamins

Q45. Answer: B — Scurvy, swelling of gums or bleeding gums

Page 422: Vitamin C deficiency causes Scurvy, swelling of gums or bleeding gums.

Topic: Vitamins

Q46. Answer: B — Milk, Egg, Cheese, Green vegetable, Carrot

Page 422: Vitamin A sources include Milk, Egg, Cheese, Green vegetables, Carrot.

Topic: Vitamins

Q47. Answer: B — Protein deficiency while caloric intake may be adequate

Page 422: Kwashiorkor is caused by protein deficiency, especially when caloric intake is not severely affected.

Topic: Proteins

Q48. Answer: B — Cells, protoplasm and tissues

Page 422: Proteins take part in formation of cells, protoplasm and tissues.

Topic: Proteins

Q49. Answer: B — Pulse, Liver, Egg, Milk, eggs etc.

Page 422: Sources of Vitamin B1 include Pulse, Liver, Egg, Milk, eggs.

Topic: Vitamins

Q50. Answer: B — Beriberi

Page 422: Vitamin B1 (Thiamine) deficiency causes Beriberi.

Topic: Vitamins

Respiratory System – Page 420

Source: Lucent's General Knowledge – Biology

Biology: Respiratory System

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The opening of larynx is called:

- A. Pharynx
- B. Glottis
- C. Bronchi
- D. Alveoli

Ans: B

Q2. The cartilaginous flap structure that prevents the entry of food during swallowing is:

- A. Larynx
- B. Epiglottis
- C. Trachea
- D. Bronchi

Ans: B

Q3. A pair of vocal cords that help in producing sound are present inside the:

- A. Pharynx
- B. Larynx
- C. Trachea
- D. Bronchi

Ans: B

Q4. Trachea passes through the neck and from the base of larynx reaches the middle of:

- A. Abdomen
- B. Thoracic cavity (chest)
- C. Cranium
- D. Pelvis

Ans: B

Q5. Trachea is supported by:

- A. Bones
- B. C-shaped cartilaginous rings
- C. Muscles only
- D. Ligaments only

Ans: B

Q6. The inner lining of trachea is secreted by:

- A. Sweat glands
- B. Ciliated epithelium (mucus secreting)
- C. Bone cells
- D. Nerve cells

Ans: B

Q7. Trachea divides into two branches called:

- A. Alveoli
- B. Bronchi
- C. Pharynx
- D. Larynx

Ans: B

Q8. Bronchi further divide into smaller branches called: A. Alveoli B. Bronchioles C. Trachea D. Larynx	Ans: B
Q9. Each lung is surrounded by a membrane called: A. Pericardium B. Pleural membrane C. Peritoneum D. Meninges	Ans: B
Q10. There is a network of blood capillaries on: A. Trachea B. Alveoli C. Pharynx D. Larynx	Ans: B
Q11. Each lung has colour: A. Red B. Pink, spongy C. Yellow D. Blue	Ans: B
Q12. Right lung is divided into: A. 2 lobes B. 3 lobes C. 4 lobes D. 5 lobes	Ans: B
Q13. Nitrogen percentage in the constitution of air is: A. 21% B. 79% C. 0.03% D. 4%	Ans: B
Q14. Oxygen percentage in the constitution of air is: A. 79% B. 21% C. 4% D. 0.03%	Ans: B
Q15. Carbon dioxide percentage in exhaled air is approximately: A. 0.03% B. 4% C. 21% D. 79%	Ans: B

Q16. The maximum volume of air which can be inhaled through human breathing is: A. 200 mL B. 400 mL C. 600 mL D. 800 mL	Ans: B
Q17. An adult breathes approximately 12 to 16 times in a: A. Hour B. Minute C. Second D. Day	Ans: B
Q18. During inspiration, the diaphragm: A. Moves up B. Flattens (moves down) C. Stays same D. Disappears	Ans: B
Q19. During expiration, air pressure inside and outside the body becomes: A. Very different B. Equal C. Inside is much higher D. Outside is zero	Ans: B
Q20. The process of respiration can be divided into how many parts? A. 2 B. 4 parts C. 6 D. 3	Ans: B
Q21. External respiration is the exchange of gases in: A. Cells B. Lungs C. Blood D. Stomach	Ans: B
Q22. Exchange of gases between blood and tissue fluid is called: A. External respiration B. Internal respiration C. Cellular respiration D. Anaerobic respiration	Ans: B

Q23. Cellular respiration is where glucose is oxidised by oxygen. This process is called: A. Fermentation B. Cellular respiration C. Photosynthesis D. Osmosis	Ans: B
Q24. There are how many types of cellular respiration? A. 3 B. 2 – Anaerobic and Aerobic C. 4 D. 1	Ans: B
Q25. Anaerobic respiration occurs in the absence of: A. Water B. Oxygen C. Carbon dioxide D. Nitrogen	Ans: B
Q26. During anaerobic respiration, only how many ATP molecules are produced from one molecule of glucose? A. 38 B. 2 C. 4 D. 36	Ans: B
Q27. In anaerobic respiration in yeast, the final product is: A. Lactic acid B. Ethyl alcohol and CO₂ C. Water D. Oxygen	Ans: B
Q28. In anaerobic respiration in animals, the final product is: A. Ethyl alcohol B. Lactic acid C. Acetic acid D. Ethanol	Ans: B
Q29. A sprint athlete feels cramps due to accumulation of: A. Glucose B. Lactic acid C. Ethanol D. CO₂	Ans: B

Q30. Aerobic respiration takes place in presence of oxygen. The complete process produces how many ATP? A. 2 B. 38 C. 4 D. 36	Ans: B
Q31. Glycolysis (cytoplasm) was first studied by: A. Kreb B. Embden, Meyerhof, Parnas (EMP path) C. Watson D. Mendel	Ans: B
Q32. Glycolysis is present in both types of respiration and takes place in: A. Mitochondria B. Cytoplasm C. Nucleus D. Ribosome	Ans: B
Q33. Kreb's Cycle was described by: A. Embden B. Hans Krebs in 1937 C. Watson in 1953 D. Mendel in 1865	Ans: B
Q34. Kreb's Cycle is completed inside the: A. Cytoplasm B. Mitochondria C. Nucleus D. Ribosome	Ans: B
Q35. Kreb's cycle is also called: A. EMP pathway B. Citric Acid Cycle or TCA cycle C. Calvin cycle D. Nitrogen cycle	Ans: B
Q36. Breakdown of pyruvate into CO₂, H₂O and energy takes place in: A. Cytoplasm B. Mitochondria C. Nucleus D. Cell membrane	Ans: B

Q37. In glycolysis, 4 atoms of hydrogen formed are used in converting NAD to: A. ATP B. 2NADH₂ C. FAD D. GTP	Ans: B
Q38. The enzymes which take part in glycolysis and respiration are found in: A. Nucleus B. Cell cytoplasm C. Ribosome D. Golgi body	Ans: B
Q39. To start glycolysis, how many molecules of ATP are utilized? A. 4 B. 2 C. 1 D. 6	Ans: B
Q40. As a result of glycolysis, how many ATP are obtained net? A. 4 B. 2 (net, since 4 produced minus 2 used) C. 38 D. 36	Ans: B
Q41. There is no need of oxygen in: A. Aerobic respiration B. Glycolysis C. Kreb's cycle D. Electron transport chain	Ans: B
Q42. Breaking down of glucose in cytoplasm results in formation of: A. Ethanol B. Pyruvate (pyruvic acid) C. Lactic acid D. Acetic acid	Ans: B
Q43. ATP stands for: A. Adenine Tri Phosphate B. Adenosine Triphosphate C. Amino Tri Phosphate D. Adenosine Tri Protein	Ans: B
Q44. Transportation of oxygen from lungs to the cells of body takes place by: A. White blood cells B. Haemoglobin (in RBC) C. Plasma only D. Platelets	Ans: B

Q45. Carbon dioxide from cells to lungs is mainly transported as: A. Attached to hemoglobin B. Bicarbonates (70% as HCO_3) C. Dissolved in plasma only D. In WBCs	Ans: B
Q46. Carbon dioxide after mixing with plasma forms: A. Oxygen B. Carbonic acid C. Lactic acid D. Acetic acid	Ans: B
Q47. Respiration is controlled by: A. Cerebrum B. Medulla in second (hindbrain) C. Cerebellum D. Hypothalamus	Ans: B
Q48. Respiration is a: A. Anabolic process B. Catabolic process (it also reduces the weight of the body) C. Neither D. Synthetic process	Ans: B
Q49. Cyanide poisoning causes death by: A. Stopping digestion B. Breaking down the electron transport chain system C. Stopping heartbeat D. Blocking kidneys	Ans: B
Q50. The gaseous exchange in lungs is called: A. Internal respiration B. External respiration C. Cellular respiration D. Anaerobic respiration	Ans: B

Answer Key & Explanations

Q1. Answer: B — Glottis

Page 420: The opening of larynx is called glottis.

Topic: Respiratory Anatomy

Q2. Answer: B — Epiglottis

Page 420: Epiglottis is the cartilaginous flap that prevents food entry during swallowing.

Topic: Respiratory Anatomy

Q3. Answer: B — Larynx

Page 420: A pair of vocal cords are present inside the larynx.

Topic: Respiratory Anatomy

Q4. Answer: B — Thoracic cavity (chest)

Page 420: Trachea from the base of larynx reaches the middle of thoracic cavity.

Topic: Respiratory Anatomy

Q5. Answer: B — C-shaped cartilaginous rings

Page 420: Trachea is called wind pipe, supported by C-shaped cartilaginous rings.

Topic: Respiratory Anatomy

Q6. Answer: B — Ciliated epithelium (mucus secreting)

Page 420: The inner lining of trachea is ciliated epithelium (mucus secreting).

Topic: Respiratory Anatomy

Q7. Answer: B — Bronchi

Page 420: Trachea divides into two branches called bronchi, each entering one lung.

Topic: Respiratory Anatomy

Q8. Answer: B — Bronchioles

Page 420: Bronchi further divide into smaller branches called bronchioles, finally giving alveoli.

Topic: Respiratory Anatomy

Q9. Answer: B — Pleural membrane

Page 420: Each lung is surrounded by a membrane called pleural membrane.

Topic: Respiratory Anatomy

Q10. Answer: B — Alveoli

Page 420: There is a network of blood capillaries on alveoli.

Topic: Respiratory Anatomy

Q11. Answer: B — Pink, spongy

Page 420: Each lung is pink, spongy in comparison to left lung.

Topic: Respiratory Anatomy

Q12. Answer: B — 3 lobes

Page 420: Right lung has three lobes and left lung has two lobes.

Topic: Respiratory Anatomy

Q13. Answer: B — 79%

Page 420: Constitution of air: Nitrogen 79%, Oxygen 21%, Carbon dioxide 0.03%.

Topic: Respiration

Q14. Answer: B — 21%

Page 420: Constitution of air: Oxygen 21%.

Topic: Respiration

Q15. Answer: B — 4%

Page 420: In exhaled air, CO₂ is about 4% (vs 0.03% in inhaled air).

Topic: Respiration

Q16. Answer: B — 400 mL

Page 420: Maximum volume of air inhaled is approximately 400 mL.

Topic: Respiration

Q17. Answer: B — Minute

Page 420: An adult breathes approximately 12 to 16 times in a minute.

Topic: Respiration

Q18. Answer: B — Flattens (moves down)

Page 420: During inspiration, the diaphragm flattens/contracts, creating low pressure, allowing air to enter lungs.

Topic: Respiration

Q19. Answer: B — Equal

Page 420: In expiration, diaphragm relaxes, air pressure inside and outside body becomes equal.

Topic: Respiration

Q20. Answer: B — 4 parts

Page 420: Process of respiration is divided into 4 parts: External, Internal, Transportation of gases, Cellular respiration.

Topic: Respiration

Q21. Answer: B — Lungs

Page 420: External respiration is exchange of gases in lungs.

Topic: Respiration

Q22. Answer: B — Internal respiration

Page 420: Internal respiration is exchange of gases between blood and tissue fluid.

Topic: Respiration

Q23. Answer: B — Cellular respiration

Page 420: Cellular respiration: glucose is oxidised by oxygen, reached into the cell.

Topic: Respiration

Q24. Answer: B — 2 – Anaerobic and Aerobic

Page 420: Two types of cellular respiration: Anaerobic and Aerobic.

Topic: Respiration

Q25. Answer: B — Oxygen

Page 420: Anaerobic respiration is if the oxidation of food takes place in absence of oxygen.

Topic: Respiration

Q26. Answer: B — 2

Page 420: During anaerobic respiration, only 2 ATP molecules are produced.

Topic: Respiration

Q27. Answer: B — Ethyl alcohol and CO₂

Page 420: In yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy (ethyl alcohol or ethanol)}$.

Topic: Respiration

Q28. Answer: B — Lactic acid

Page 420: In animals: anaerobic respiration produces lactic acid. Accumulation causes muscle pain (cramps).

Topic: Respiration

Q29. Answer: B — Lactic acid

Page 420: After a sprint, lactic acid accumulates causing pain in the thigh muscles.

Topic: Respiration

Q30. Answer: B — 38

Page 420: Complete aerobic respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 2870 \text{ kJ energy (38 ATP)}$.

Topic: Respiration

Q31. Answer: B — Embden, Meyerhof, Parnas (EMP path)

Page 420: Glycolysis was first studied by Embden, Meyerhof, Parnas. Also called EMP pathway.

Topic: Respiration

Q32. Answer: B — Cytoplasm

Page 420: Glycolysis is present in both aerobic and anaerobic respiration, takes place in cytoplasm.

Topic: Respiration

Q33. Answer: B — Hans Krebs in 1937

Page 420: Krebs's Cycle was described by Hans Krebs in 1937. Also called Citric Acid Cycle or TCA cycle.

Topic: Respiration

Q34. Answer: B — Mitochondria

Page 420: Krebs's cycle is completed inside the Mitochondria in presence of specific enzymes.

Topic: Respiration

Q35. Answer: B — Citric Acid Cycle or TCA cycle

Page 420: Krebs's cycle is also called Citric Acid Cycle or Tricarboxylic Acid Cycle (TCA cycle).

Topic: Respiration

Q36. Answer: B — Mitochondria

Page 420: Breakdown of pyruvate takes place in mitochondria.

Topic: Respiration

Q37. Answer: B — 2NADH₂

Page 420: 4 atoms of hydrogen formed in glycolysis are used in converting NAD to 2NADH₂.

Topic: Respiration

Q38. Answer: B — Cell cytoplasm

Page 420: Enzymes for glycolysis are found in cell cytoplasm.

Topic: Respiration

Q39. Answer: B — 2

Page 420: To start the process of glycolysis, 2 molecules of ATP are utilised.

Topic: Respiration

Q40. Answer: B — 2 (net, since 4 produced minus 2 used)

Page 420: As a result of glycolysis, 4 molecules of ATP are obtained. But 2 were used, so net gain is 2 ATP.

Topic: Respiration

Q41. Answer: B — Glycolysis

Page 420: There is no need of oxygen in glycolysis. This process is similar in both anaerobic and aerobic respiration.

Topic: Respiration

Q42. Answer: B — Pyruvate (pyruvic acid)

Page 420: Breaking down of glucose in cytoplasm results in formation of two molecules of pyruvate and energy.

Topic: Respiration

Q43. Answer: B — Adenosine Triphosphate

Page 420: ATP = Adenosine Triphosphate.

Topic: Respiration

Q44. Answer: B — Haemoglobin (in RBC)

Page 420: Transportation of oxygen takes place by haemoglobin in blood.

Topic: Respiration

Q45. Answer: B — Bicarbonates (70% as HCO_3)

Page 420: 70% of CO_2 is transported as bicarbonates, rest by mixing with plasma and combining with haemoglobin.

Topic: Respiration

Q46. Answer: B — Carbonic acid

Page 420: Carbon dioxide after mixing with plasma forms carbonic acid.

Topic: Respiration

Q47. Answer: B — Medulla in second (hindbrain)

Page 420: Respiration is controlled by medulla in the hindbrain.

Topic: Respiration

Q48. Answer: B — Catabolic process (it also reduces the weight of the body)

Page 420: Respiration is a catabolic process. It also reduces the weight of the body.

Topic: Respiration

Q49. Answer: B — Breaking down the electron transport chain system

Page 420: Cyanide poisoning causes death by breaking down the electron transport chain system.

Topic: Respiration

Q50. Answer: B — External respiration

Page 420: The gaseous exchange in lungs is called external respiration.

Topic: Respiration

Nutrients & Carbohydrates – Page 421

Source: Lucent's General Knowledge – Biology

Biology: Nutrients, Carbohydrates & Energy Production

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The function of food in providing basic nutrition is called:

- A. Digestion
- B. Nutrition
- C. Absorption
- D. Excretion

Ans: B

Q2. Nutrients include:

- A. Only proteins
- B. Carbohydrates, proteins, fats, minerals, vitamins and water
- C. Only vitamins
- D. Only minerals

Ans: B

Q3. Carbohydrates are organic compounds made of:

- A. Nitrogen and Hydrogen
- B. Carbon, Hydrogen and Oxygen
- C. Sulphur and Phosphorus
- D. Only Carbon

Ans: B

Q4. Carbohydrate is the major intake in animals and humans, providing what percentage of energy?

- A. 20-30%
- B. 50 to 75%
- C. 90%
- D. 10%

Ans: B

Q5. Carbohydrate containing aldehyde group is called:

- A. Ketose
- B. Aldose
- C. Lactose
- D. Sucrose

Ans: B

Q6. Carbohydrates are classified into how many main groups?

- A. 2
- B. 3 – Monosaccharides, Oligosaccharides, Polysaccharides
- C. 4
- D. 5

Ans: B

Q7. Monosaccharides are the simplest sugars composed of:

- A. Multiple units
- B. Single polyhydroxy or ketone units
- C. Proteins
- D. Fats

Ans: B

Q8. Most abundant monosaccharide found in nature is: A. Fructose B. Glucose C. Galactose D. Ribose	Ans: B
Q9. Glucose is a type of: A. Disaccharide B. Hexose sugar C. Polysaccharide D. Amino acid	Ans: B
Q10. The sweetest sugar is: A. Glucose B. Fructose C. Sucrose D. Lactose	Ans: B
Q11. When 2 to 10 monosaccharides join together they form: A. Polysaccharides B. Oligosaccharides C. Amino acids D. Fatty acids	Ans: B
Q12. Maltose, Sucrose and Lactose are usually: A. Monosaccharides B. Disaccharides (crystalline, sweet in taste) C. Polysaccharides D. Proteins	Ans: B
Q13. Polysaccharides are formed when more than how many monosaccharides join? A. 2 B. 10 C. 5 D. 3	Ans: B
Q14. Polysaccharides are large, insoluble, tasteless. Examples include: A. Glucose and Fructose B. Starch, glycogen, cellulose, chitin C. Maltose and Sucrose D. Ribose and Deoxyribose	Ans: B
Q15. Glucose is the source of immediate energy production in: A. Plants only B. The cell C. Bones D. Hair	Ans: B

Q16. Carbohydrate is used as fuel. During the process of respiration, glucose breaks into CO₂ and H₂O with the help of: A. Vitamins B. Oxygen C. Minerals D. Hormones	Ans: B
Q17. One gram of glucose gives how many kilo calories of energy? A. 9.3 B. 4.2 C. 7.0 D. 2.1	Ans: B
Q18. Nucleic acids are polymers of: A. Amino acids B. Nucleosides and nucleotides C. Fatty acids D. Glucose	Ans: B
Q19. Lactose of milk is formed from: A. Two glucose molecules B. Glucose and galactose C. Fructose and glucose D. Two fructose molecules	Ans: B
Q20. Glucose is used for the formation of fat and: A. Vitamins B. Amino acid C. Minerals D. Hormones	Ans: B
Q21. Carbon skeleton of fats, chitin, cellulose is used in: A. Respiration only B. Formation of fat and amino acid in the body C. Blood production D. Bone formation	Ans: B
Q22. Sources of carbohydrate include: A. Meat and fish only B. Potato, wheat, rice, maize, sweet potato, and other plant and animal materials C. Eggs only D. Milk only	Ans: B

Q23. Protein was first used by: A. Hopkins B. J. Berzelius C. Funk D. Pasteur	Ans: B
Q24. Proteins are complex organic compounds made of: A. Fatty acids B. Amino acids (approximately 16% of the human body is protein) C. Nucleic acids D. Glucose units	Ans: B
Q25. How many types of amino acids are necessary for protein synthesis in the human body? A. 10 B. 20 C. 30 D. 15	Ans: B
Q26. Out of 20 amino acids, how many are synthesized by the body itself? A. 8 B. 12 C. 15 D. 20	Ans: B
Q27. Nitrogen is present in protein at approximately: A. 10% B. 16% C. 25% D. 5%	Ans: B
Q28. Production of energy: By oxidation of pyruvic acid, from one molecule NADH, how many molecules of ATP are formed? A. 2 B. 3 C. 1 D. 5	Ans: B
Q29. From one molecule of NADH₂, how many molecules of ATP are formed? A. 2 B. 3 C. 1 D. 5	Ans: B

Q30. Total ATP molecules produced from one molecule of glucose in aerobic respiration is: A. 36 B. 38 C. 2 D. 4	Ans: B
Q31. From glycolysis, how many molecules of NADH are produced from one glucose molecule? A. 4 B. 2 C. 1 D. 6	Ans: B
Q32. Kreb's cycle produces how many ATP molecules per glucose? A. 38 B. 24 C. 2 D. 4	Ans: B
Q33. The main respiratory substances are: A. Only fats B. Carbohydrate, fat, and protein C. Only minerals D. Only vitamins	Ans: B
Q34. The part of eye which is responsible for the conversion of image into neural signal is: A. Cornea B. Retina C. Iris D. Lens	Ans: B
Q35. Retina of eye contains cells called: A. Neurons B. Photoreceptor cells C. Muscle cells D. Blood cells	Ans: B
Q36. The optic nerve carries image information to the: A. Spinal cord B. Brain C. Heart D. Lungs	Ans: B

Q37. Cyanide poisoning causes death by affecting: A. Digestive system B. Electron transport chain system C. Skeletal system D. Endocrine system	Ans: B
Q38. Organisms get energy from: A. Only water B. Nutrients (mainly carbohydrate and fat used for biosynthesis and energy) C. Only air D. Only minerals	Ans: B
Q39. Assimilation means: A. Eating food B. Absorption of food and utilization for body functions C. Excretion of waste D. Breathing	Ans: B
Q40. Undigested foods are included in: A. Nutrition B. Excretion/egestion C. Assimilation D. Absorption	Ans: B
Q41. Monosaccharides containing six carbon atoms are called: A. Pentoses B. Hexoses C. Trioses D. Tetroses	Ans: B
Q42. Starch is a: A. Monosaccharide B. Polysaccharide C. Disaccharide D. Amino acid	Ans: B
Q43. Glycogen is the storage form of carbohydrate in: A. Plants B. Animals C. Bacteria D. Fungi only	Ans: B
Q44. Cellulose is found in: A. Animal muscles B. Plant cell walls C. Blood D. Bones	Ans: B

Q45. Pentose sugars include: A. Glucose and Fructose B. Ribose and Deoxyribose C. Sucrose and Maltose D. Starch and Glycogen	Ans: B
Q46. Sucrose is formed by joining: A. Two glucose molecules B. Glucose and Fructose C. Glucose and Galactose D. Two fructose molecules	Ans: B
Q47. Maltose is formed by joining: A. Glucose and fructose B. Two glucose molecules C. Glucose and galactose D. Two fructose molecules	Ans: B
Q48. Which polysaccharide forms the exoskeleton of arthropods? A. Starch B. Chitin C. Glycogen D. Cellulose	Ans: B
Q49. Essential amino acids are those which: A. Body can synthesize B. Must be obtained from food (8 in number) C. Are not needed D. Are only in plants	Ans: B
Q50. During complete aerobic respiration of one glucose molecule, how many calories of energy is produced? A. 1000 kJ B. 2870 kJ (686 kcal) C. 500 kJ D. 100 kJ	Ans: B

Answer Key & Explanations

Q1. Answer: B — Nutrition

Page 421: To maintain life, organisms need nutrition. Nutrition is the basic function of food.

Topic: Nutrients

Q2. Answer: B — Carbohydrates, proteins, fats, minerals, vitamins and water

Page 421: Nutrients include carbohydrates, proteins, fats, minerals, vitamins and water.

Topic: Nutrients

Q3. Answer: B — Carbon, Hydrogen and Oxygen

Page 421: Carbohydrates are organic compounds in which the ratio of Carbon, Hydrogen and Oxygen is 1:2:1.

Topic: Carbohydrates

Q4. Answer: B — 50 to 75%

Page 421: Carbohydrate is the major intake providing 50 to 75% energy.

Topic: Carbohydrates

Q5. Answer: B — Aldose

Page 421: Carbohydrate containing aldehyde group is called aldose and with ketone group is called ketose.

Topic: Carbohydrates

Q6. Answer: B — 3 – Monosaccharides, Oligosaccharides, Polysaccharides

Page 421: Carbohydrates are classified into three main groups: Monosaccharides, Oligosaccharides, Polysaccharides.

Topic: Carbohydrates

Q7. Answer: B — Single polyhydroxy or ketone units

Page 421: Monosaccharides are the simplest carbohydrates, single polyhydroxy or ketone units.

Topic: Carbohydrates

Q8. Answer: B — Glucose

Page 421: Most abundant monosaccharide found in nature is glucose. Containing 6 carbon atoms.

Topic: Carbohydrates

Q9. Answer: B — Hexose sugar

Page 421: Glucose is a type of hexose sugar (6 carbon atoms).

Topic: Carbohydrates

Q10. Answer: B — Fructose

Page 421: The sweetest sugar is fructose.

Topic: Carbohydrates

Q11. Answer: B — Oligosaccharides

Page 421: Oligosaccharides are formed when 2 to 10 monosaccharides join together.

Topic: Carbohydrates

Q12. Answer: B — Disaccharides (crystalline, sweet in taste)

Page 421: They are usually crystalline in nature and sweet in test. Maltose, sucrose, lactose are disaccharides.

Topic: Carbohydrates

Q13. Answer: B — 10

Page 421: Polysaccharides are formed when more than 10 monosaccharides join together.

Topic: Carbohydrates

Q14. Answer: B — Starch, glycogen, cellulose, chitin

Page 421: Polysaccharides examples: starch, glycogen, cellulose, chitin etc. They are large, insoluble, tasteless.

Topic: Carbohydrates

Q15. Answer: B — The cell

Page 421: Glucose is the source of immediate energy production in the cell.

Topic: Carbohydrates

Q16. Answer: B — Oxygen

Page 421: Carbohydrate is used as fuel. During respiration, glucose breaks into CO₂ and H₂O with help of oxygen releasing 4.2 kilo calories energy.

Topic: Carbohydrates

Q17. Answer: B — 4.2

Page 421: One gram of glucose gives 4.2 kilo calories energy.

Topic: Carbohydrates

Q18. Answer: B — Nucleosides and nucleotides

Page 421: Nucleic acids are polymers of nucleosides and nucleotides.

Topic: Carbohydrates

Q19. Answer: B — Glucose and galactose

Page 421: Lactose of milk is formed from glucose and galactose.

Topic: Carbohydrates

Q20. Answer: B — Amino acid

Page 421: Glucose is used for the formation of fat and amino acid.

Topic: Carbohydrates

Q21. Answer: B — Formation of fat and amino acid in the body

Page 421: Carbon skeleton of fats, chitin, cellulose etc. is used in the body for formation of structures.

Topic: Carbohydrates

Q22. Answer: B — Potato, wheat, rice, maize, sweet potato, and other plant and animal materials

Page 421: Sources include potato, wheat, rice, maize, sweet potato, and other plant and animal sources.

Topic: Carbohydrates

Q23. Answer: B — J. Berzelius

Page 421: Protein word was first used by J. Berzelius.

Topic: Proteins

Q24. Answer: B — Amino acids (approximately 16% of the human body is protein)

Page 421: Protein is a complex organic compound made up of amino acids. Approximately 16% of human body is protein.

Topic: Proteins

Q25. Answer: B — 20

Page 421: Twenty types of amino acid are necessary for protein synthesis.

Topic: Proteins

Q26. Answer: B — 12

Page 421: Out of 20, 12 are synthesized by body itself and remaining 8 are obtained from food (essential amino acids).

Topic: Proteins

Q27. Answer: B — 16%

Page 421: Nitrogen is 16% present in protein.

Topic: Proteins

Q28. Answer: B — 3

Page 421: By the oxidation of NADH, 3 molecules of ATP are formed.

Topic: Energy Production

Q29. Answer: B — 3

Page 421: One molecule of NADH₂ gives 3 ATP molecules.

Topic: Energy Production

Q30. Answer: B — 38

Page 421: ATP glucose total = 2 + 36 = 38 ATP molecules from complete oxidation of glucose.

Topic: Energy Production

Q31. Answer: B — 2

Page 421: From glycolysis, 2 NADH molecules are produced.

Topic: Energy Production

Q32. Answer: B — 24

Page 421: Kreb's cycle produces 24 ATP (from complete cycle for both pyruvate molecules).

Topic: Energy Production

Q33. Answer: B — Carbohydrate, fat, and protein

Page 421: Main respiratory substances are carbohydrate, fat and protein. At first, oxidation of glucose takes place, then fat, then protein.

Topic: Respiration

Q34. Answer: B — Retina

Page 421: The part of eye responsible for conversion of image into neural signal is Retina. It contains photoreceptor cells.

Topic: Eye/Vision

Q35. Answer: B — Photoreceptor cells

Page 421: Retina contains photoreceptor cells and sends information to brain to decide what the picture is.

Topic: Eye/Vision

Q36. Answer: B — Brain

Page 421: Retina side of eye processes and sends information to the brain through optic nerve.

Topic: Eye/Vision

Q37. Answer: B — Electron transport chain system

Page 421: Cyanide poisoning causes death by breaking down the electron transport chain/equalising the pressure on the system.

Topic: Respiration

Q38. Answer: B — Nutrients (mainly carbohydrate and fat used for biosynthesis and energy)

Page 421: Organisms get energy from nutrients. Carbohydrate and fat are used for biosynthesis and energy.

Topic: Nutrients

Q39. Answer: B — Absorption of food and utilization for body functions

Page 421: Assimilation is the absorption of nutrients and their utilization for body functions.

Topic: Nutrients

Q40. Answer: B — Excretion/egestion

Page 421: Function of food includes nutrition, digestion, absorption, assimilation and egestion of undigested foods.

Topic: Nutrients

Q41. Answer: B — Hexoses

Page 421: Monosaccharides with 6 carbon atoms are called hexoses (e.g., glucose, fructose).

Topic: Carbohydrates

Q42. Answer: B — Polysaccharide

Page 421: Starch is a polysaccharide. It is the storage form of carbohydrate in plants.

Topic: Carbohydrates

Q43. Answer: B — Animals

Page 421: Glycogen is the storage form of carbohydrate in animals.

Topic: Carbohydrates

Q44. Answer: B — Plant cell walls

Page 421: Cellulose is found in plant cell walls. It is a structural polysaccharide.

Topic: Carbohydrates

Q45. Answer: B — Ribose and Deoxyribose

Page 421: Monosaccharides with 5 carbons include ribose (in RNA) and deoxyribose (in DNA).

Topic: Carbohydrates

Q46. Answer: B — Glucose and Fructose

Page 421: Sucrose (table sugar) is formed by joining glucose and fructose.

Topic: Carbohydrates

Q47. Answer: B — Two glucose molecules

Page 421: Maltose is formed by joining two glucose molecules.

Topic: Carbohydrates

Q48. Answer: B — Chitin

Page 421: Chitin is the polysaccharide that forms exoskeleton of arthropods.

Topic: Carbohydrates

Q49. Answer: B — Must be obtained from food (8 in number)

Page 421: Essential amino acids are 8 in number and must be obtained from food.

Topic: Proteins

Q50. Answer: B — 2870 kJ (686 kcal)

Page 421: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 2870 \text{ kJ energy}$.

Topic: Energy Production

Digestive System – Page 414

Source: Lucent's General Knowledge – Biology

Biology: Digestive System

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The enzyme Amylase is present in:

- A. Gastric juice
- B. Saliva
- C. Pancreatic juice only
- D. Intestinal juice only

Ans: B

Q2. Saliva converts starch into:

- A. Glucose
- B. Maltose
- C. Protein
- D. Fat

Ans: B

Q3. Gastric juice contains which enzymes?

- A. Amylase and Lipase
- B. Pepsin and Rennin
- C. Trypsin and Lipase
- D. Lactase and Sucrase

Ans: B

Q4. Pepsin in gastric juice converts:

- A. Starch into Maltose
- B. Protein into Polypeptides
- C. Fat into Fatty acid
- D. Sucrose into Glucose

Ans: B

Q5. Rennin in gastric juice converts:

- A. Starch to sugar
- B. Casein (milk protein) into Calcium paracasein
- C. Fat to fatty acid
- D. Maltose to glucose

Ans: B

Q6. Pancreatic juice contains:

- A. Pepsin and Rennin
- B. Trypsin, Amylase, and Lipase
- C. Sucrase only
- D. Lactase only

Ans: B

Q7. Intestinal juice contains:

- A. Pepsin
- B. Erepsin, Maltase, Lactase, Sucrase, Lipase
- C. Rennin only
- D. Amylase only

Ans: B

Q8. Maltase converts Maltose into: A. Protein B. Glucose C. Fat D. Amino acid	Ans: B
Q9. Sucrase converts Sucrose into: A. Maltose B. Glucose and Fructose C. Protein D. Fatty acid	Ans: B
Q10. Lactase converts Lactose into: A. Maltose and Sucrose B. Glucose and Galactose C. Protein and Fat D. Amino acid and Glycerol	Ans: B
Q11. Lipase converts fat into: A. Glucose B. Fatty acid and Glycerol C. Amino acid D. Starch	Ans: B
Q12. Erepsin converts: A. Starch to maltose B. Remaining protein into amino acid C. Fat to glycerol D. Sucrose to glucose	Ans: B
Q13. Enzymes are: A. Inorganic catalysts B. Biocatalysts (organic biological catalysts) C. Minerals D. Vitamins	Ans: B
Q14. The root of tooth is covered by: A. Enamel B. Cementum C. Dentin D. Pulp	Ans: B
Q15. The pouch of caecum is connected to: A. Stomach B. Junction of small and large intestine C. Esophagus D. Pharynx	Ans: B

Q16. Complete digestion of food takes place in: A. Stomach B. Small intestine C. Large intestine D. Mouth	Ans: B
Q17. Digestion of fat takes place in: A. Mouth B. Small and large intestine C. Stomach only D. Esophagus	Ans: B
Q18. Small intestine is longer than: A. Esophagus B. Large intestine C. Trachea D. All of above	Ans: B
Q19. Undigested food reaches into large intestine where bacteria turns it into: A. Vitamins B. Faeces C. Blood D. Glucose	Ans: B
Q20. Absorption of digested food takes place through: A. Stomach wall B. Wall of small intestine into blood C. Large intestine only D. Esophagus	Ans: B
Q21. Intestinal juice is alkaline in nature. In healthy people approximately how many litres are secreted daily? A. 1 litre B. 2 litres C. 5 litres D. 500 mL	Ans: B
Q22. The liver converts the emulsified fats into: A. Protein B. Glycerol and fatty acid C. Starch D. Amino acid	Ans: B

Q23. Liver converts the maltose into: A. Starch B. Glucose C. Protein D. Fat	Ans: B
Q24. Liver converts the lactose into: A. Maltose B. Glucose and galactose C. Sucrose D. Fructose	Ans: B
Q25. Liver converts the sucrose into: A. Maltose B. Glucose and fructose C. Starch D. Protein	Ans: B
Q26. The liver is the largest gland in the human body. Its weight is approximately: A. 500 grams B. 1500 grams C. 2500 grams D. 3000 grams	Ans: B
Q27. Bile is secreted from: A. Pancreas B. Liver C. Stomach D. Small intestine	Ans: B
Q28. Liver converts excess of amino acid into ammonia by: A. Hydrolysis B. Deamination C. Oxidation D. Reduction	Ans: B
Q29. Ammonia is further converted into urea by: A. Glycolysis B. Ornithine cycle C. Krebs's cycle D. Calvin cycle	Ans: B
Q30. Bile of liver accelerates the action of: A. Amylase B. Enzymes present in intestine C. Pepsin D. Rennin	Ans: B

Q31. The main functions of bile include making the pH of food as: A. Acidic B. Alkaline and facilitates action of bile C. Neutral D. Very acidic	Ans: B
Q32. Bile neutralizes the acidic pH of food coming from: A. Mouth B. Stomach C. Esophagus D. Large intestine	Ans: B
Q33. The pH value of bile is: A. 5.7 B. 7.7 C. 3.7 D. 9.7	Ans: B
Q34. Bile contains approximately what percentage of water? A. 75% B. 85% C. 95% D. 65%	Ans: B
Q35. Bile is a yellowish-green coloured and: A. Acidic liquid B. Alkaline liquid C. Neutral liquid D. Oily liquid	Ans: B
Q36. Bile comes from the bile duct into the: A. Stomach B. Duodenum C. Large intestine D. Esophagus	Ans: B
Q37. Gall bladder of liver is a: A. Kidney shaped structure B. Pear shaped sac C. Round ball D. Tube shaped	Ans: B
Q38. Liver is an important organ in regulating body: A. Movement B. Temperature C. Height D. Hair growth	Ans: B

Q39. Liver helps in investigation of a person's death due to: A. Heart attack B. Poisoning (it has been done due to investigating a person's death) C. Natural death D. Old age	Ans: B
Q40. Dead RBCs are destroyed by: A. Kidneys B. Liver C. Heart D. Lungs	Ans: B
Q41. Liver reserves some quantity of: A. Calcium only B. Iron, copper, and vitamins C. Sodium only D. Potassium only	Ans: B
Q42. Liver produces which protein that helps in clotting of blood? A. Albumin B. Fibrinogen and Heparin C. Insulin D. Glucagon	Ans: B
Q43. In case of obstruction in bile duct, liver cells stop working. Bilirubin from blood spreads throughout the body causing: A. Anemia B. Jaundice C. Diabetes D. Scurvy	Ans: B
Q44. Pancreatic juice secretes out of pancreas. The liquid's pH value is: A. 5.5-6.5 B. 7.5-8.3 C. 3.5-4.5 D. 9.5-10.5	Ans: B
Q45. Pancreas is the second largest gland of the human body. It functions as both: A. Only endocrine B. Exocrine and endocrine gland C. Only exocrine D. Neither	Ans: B

Q46. Islets of Langerhans are present in: A. Liver B. Pancreas C. Kidney D. Stomach	Ans: B
Q47. Insulin is secreted by which cells of Islets of Langerhans? A. α-cells B. β-cells C. δ-cells D. γ-cells	Ans: B
Q48. Glucagon is secreted by which cells of Islets of Langerhans? A. β-cells B. α-cells C. δ-cells D. γ-cells	Ans: B
Q49. Insulin was discovered by: A. Pasteur B. Medical scientist Langerhans C. Fleming D. Darwin	Ans: B
Q50. In case of excess glucose in blood, liver converts glucose into glycogen. If not, it converts to: A. Protein B. Fat C. Amino acid D. Vitamin	Ans: B

Answer Key & Explanations

Q1. Answer: B — Saliva

Page 414: Saliva contains enzyme Amylase which converts starch into maltose.

Topic: Digestive Enzymes

Q2. Answer: B — Maltose

Page 414: Saliva (Amylase) converts Starch into Maltose.

Topic: Digestive Enzymes

Q3. Answer: B — Pepsin and Rennin

Page 414: Gastric juice contains Pepsin and Rennin.

Topic: Digestive Enzymes

Q4. Answer: B — Protein into Polypeptides

Page 414: Pepsin converts Protein into Polypeptides.

Topic: Digestive Enzymes

Q5. Answer: B — Casein (milk protein) into Calcium paracasein

Page 414: Rennin converts Casein (milk protein) into Calcium paracasein.

Topic: Digestive Enzymes

Q6. Answer: B — Trypsin, Amylase, and Lipase

Page 414: Pancreatic juice contains Trypsin (Protein→Sugar), Amylase (Starch→Amino acid and glycerol), and Lipase (Fat→Fatty acid and glycerol).

Topic: Digestive Enzymes

Q7. Answer: B — Erepsin, Maltase, Lactase, Sucrase, Lipase

Page 414: Intestinal juice contains Erepsin, Maltase, Lactase, Sucrase, Lipase etc.

Topic: Digestive Enzymes

Q8. Answer: B — Glucose

Page 414: Maltase converts Maltose into Glucose.

Topic: Digestive Enzymes

Q9. Answer: B — Glucose and Fructose

Page 414: Sucrase converts Sucrose into Glucose and Fructose.

Topic: Digestive Enzymes

Q10. Answer: B — Glucose and Galactose

Page 414: Lactase converts Lactose into Glucose and Galactose.

Topic: Digestive Enzymes

Q11. Answer: B — Fatty acid and Glycerol

Page 414: Lipase converts Fat into Fatty acid and Glycerol.

Topic: Digestive Enzymes

Q12. Answer: B — Remaining protein into amino acid

Page 414: Erepsin in intestinal juice converts remaining protein (peptones) into amino acids.

Topic: Digestive Enzymes

Q13. Answer: B — Biocatalysts (organic biological catalysts)

Page 414: Enzymes are biocatalysts.

Topic: Digestive Enzymes

Q14. Answer: B — Cementum

Page 414: Root of tooth is covered by cementum.

Topic: Digestive System

Q15. Answer: B — Junction of small and large intestine

Page 414: The pouch (caecum) is connected to the junction of small and large intestine.

Topic: Digestive System

Q16. Answer: B — Small intestine

Page 414: Complete digestion of food takes place in small intestine.

Topic: Digestive System

Q17. Answer: B — Small and large intestine

Page 414: Digestion of fat takes place in small and large intestine.

Topic: Digestive System

Q18. Answer: B — Large intestine

Page 414: Small intestine is longer than large intestine.

Topic: Digestive System

Q19. Answer: B — Faeces

Page 414: Undigested food reaches large intestine where bacteria turns it into faeces, which is excreted.

Topic: Digestive System

Q20. Answer: B — Wall of small intestine into blood

Page 414: Absorption of digested food gets mixed into blood through the wall of small intestine.

Topic: Digestive System

Q21. Answer: B — 2 litres

Page 414: Intestinal juice is alkaline. In healthy people approximately 2 litres of intestinal juice is secreted everyday.

Topic: Digestive System

Q22. Answer: B — Glycerol and fatty acid

Page 414: Liver converts the emulsified fats into glycerol and fatty acid.

Topic: Liver

Q23. Answer: B — Glucose

Page 414: Liver converts maltose into glucose.

Topic: Liver

Q24. Answer: B — Glucose and galactose

Page 414: Liver converts lactose into glucose and galactose.

Topic: Liver

Q25. Answer: B — Glucose and fructose

Page 414: Liver converts sucrose into glucose and fructose.

Topic: Liver

Q26. Answer: B — 1500 grams

Page 414: Liver is the largest gland of the human body. Weight is approximately 1500 grams.

Topic: Liver

Q27. Answer: B — Liver

Page 414: Bile is secreted from liver.

Topic: Liver

Q28. Answer: B — Deamination

Page 414: Liver converts excess of amino acid into ammonia by deamination.

Topic: Liver

Q29. Answer: B — Ornithine cycle

Page 414: Ammonia is further converted into urea by ornithine cycle. Urea comes out of body through kidney.

Topic: Liver

Q30. Answer: B — Enzymes present in intestine

Page 414: Bile accelerates the action of enzymes present in intestine.

Topic: Liver

Q31. Answer: B — Alkaline and facilitates action of bile

Page 414: Bile makes pH of food alkaline and facilitates action of enzymes.

Topic: Liver

Q32. Answer: B — Stomach

Page 414: Bile neutralizes the acidic pH of food coming from stomach.

Topic: Liver

Q33. Answer: B — 7.7

Page 414: Bile pH value is 7.7.

Topic: Liver

Q34. Answer: B — 85%

Page 414: Bile contains 85% water and pigments 12%.

Topic: Liver

Q35. Answer: B — Alkaline liquid

Page 414: Bile is a yellowish-green coloured and alkaline liquid. Its pH is 7.7.

Topic: Liver

Q36. Answer: B — Duodenum

Page 414: Bile comes from the bile duct into the duodenum.

Topic: Liver

Q37. Answer: B — Pear shaped sac

Page 414: Gall bladder of liver is a pear shaped sac. Bile is stored here.

Topic: Liver

Q38. Answer: B — Temperature

Page 414: Liver is an important organ in regulating the body temperature.

Topic: Liver

Q39. Answer: B — Poisoning (it has been done due to investigating a person's death)

Page 414: Liver helps in investigating a person's death (especially in poisoning cases).

Topic: Liver

Q40. Answer: B — Liver

Page 414: Dead RBCs are destroyed by the liver.

Topic: Liver

Q41. Answer: B — Iron, copper, and vitamins

Page 414: Liver reserves some quantity of iron, copper and vitamins.

Topic: Liver

Q42. Answer: B — Fibrinogen and Heparin

Page 414: Liver produces Fibrinogen protein (clotting) and Heparin (anti-clotting).

Topic: Liver

Q43. Answer: B — Jaundice

Page 414: Obstruction in bile duct → bilirubin spreads → Jaundice.

Topic: Liver

Q44. Answer: B — 7.5-8.3

Page 414: Pancreatic juice pH is 7.5-8.3. It contains 98% water and enzymes which can digest all three types of food.

Topic: Pancreas

Q45. Answer: B — Exocrine and endocrine gland

Page 414: Pancreas is the second largest gland, functions as both exocrine (digestive juice) and endocrine (hormones).

Topic: Pancreas

Q46. Answer: B — Pancreas

Page 414: Islets of Langerhans are part of the Pancreas.

Topic: Pancreas

Q47. Answer: B — β -cells

Page 414: Insulin is secreted by β -cells of Islets of Langerhans.

Topic: Pancreas

Q48. Answer: B — α -cells

Page 414: Glucagon is secreted by α -cells. Somatostatin hormones are secreted from δ -cells.

Topic: Pancreas

Q49. Answer: B — Medical scientist Langerhans

Page 414: It was discovered by the medical scientist Langerhans.

Topic: Pancreas

Q50. Answer: B — Fat

Page 414: Liver regulates blood glucose: excess $\rightarrow$ glycogen $\rightarrow$ if more excess $\rightarrow$ fat.

Topic: Liver

Excretory & Nervous System – Page 416

Source: Lucent's General Knowledge – Biology

Biology: Excretory System, Nervous System & Brain

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Lymphocytes present in lymph help to prevent the body from diseases by:

- A. Producing hormones
- B. Killing harmful bacteria and other toxic foreign bodies
- C. Producing glucose
- D. Absorbing nutrients

Ans: B

Q2. The node found in lymph vessels is called:

- A. Alveoli
- B. Lymph node
- C. Nephron
- D. Neuron

Ans: B

Q3. Lymph helps in:

- A. Digestion only
- B. Healing the wounds
- C. Producing hormones
- D. Blood clotting only

Ans: B

Q4. Excretion is the removal of:

- A. Digested food
- B. Waste products formed during metabolism
- C. Blood
- D. Oxygen

Ans: B

Q5. The excretory organs of human are:

- A. Only kidneys
- B. Kidneys, Skin, Liver and Lungs
- C. Only liver
- D. Only skin

Ans: B

Q6. The main excretory organ in human is:

- A. Skin
- B. Kidneys
- C. Liver
- D. Lungs

Ans: B

Q7. Each kidney is made up of approximately how many kidney ducts (nephrons)?

- A. 50,000
- B. 1,30,00,000 (13 lakh)
- C. 1000
- D. 100

Ans: B

Q8. Nephron is the structural and functional unit of: A. Liver B. Kidney C. Heart D. Lung	Ans: B
Q9. The cup-like structure in every nephron is called: A. Glomerulus B. Bowman's capsule C. Loop of Henle D. Collecting duct	Ans: B
Q10. Glomerulus is made up of: A. Muscle fibers B. Thin blood vessels found in Bowman's capsule C. Nerve fibers D. Bone cells	Ans: B
Q11. The process of filtration of liquid into cavity of Bowman's capsule is called: A. Absorption B. Ultra filtration C. Osmosis D. Diffusion	Ans: B
Q12. The main function of the kidneys is: A. Producing hormones B. Purification of blood plasma by excreting nitrogenous waste through urination C. Producing enzymes D. Digesting food	Ans: B
Q13. In kidneys, average blood filtrated per minute is: A. 50 mL B. 125 mL C. 250 mL D. 500 mL	Ans: B
Q14. How many litres of urine is formed daily? A. 0.5 litres B. 1.45 litres C. 5 litres D. 10 litres	Ans: B
Q15. Urine is an aqueous solution in which water constitutes: A. 80% B. 95% C. 70% D. 99%	Ans: B

Q16. The colour of urine is light yellow due to presence of: A. Bilirubin B. Urochromes (formed by dissociation of haemoglobin) C. Melanin D. Carotene	Ans: B
Q17. The correct sequence of urine flow in human body is: A. Ureter → Kidney → Bladder → Urethra B. Kidney → Ureter → Urinary bladder → Urethra C. Urethra → Bladder → Ureter → Kidney D. Bladder → Kidney → Ureter → Urethra	Ans: B
Q18. Urine pH is: A. 7 (neutral) B. 6 (acidic) C. 8 (alkaline) D. 5 (very acidic)	Ans: B
Q19. Kidney stones are made up of: A. Calcium carbonate B. Calcium oxalate C. Calcium chloride D. Sodium chloride	Ans: B
Q20. Excretory parts of Arthropods are: A. Nephridia B. Malpighian tubules C. Flame cells D. Kidneys	Ans: B
Q21. Excretory parts of Flat worm are: A. Kidneys B. Flame cells C. Malpighian tubules D. Nephridia	Ans: B
Q22. Excretory parts of Animals of Annelida Phylum are: A. Flame cells B. Nephridia C. Malpighian tubules D. Kidneys	Ans: B
Q23. Hemodialysis is the process of: A. Blood transfusion B. Removal of excess urea from blood using artificial kidney C. Blood typing D. Blood clotting	Ans: B

Q24. Excretory parts of Curstaceans (crabs, lobsters) are: A. Flame cells B. Antennal gland C. Nephridia D. Kidneys	Ans: B
Q25. Liver cells play the main role in excretion by converting amino acids and ammonia into: A. Glucose B. Urea C. Fat D. Protein	Ans: B
Q26. Lungs excrete which gaseous substances? A. Oxygen only B. Carbon dioxide and water vapour C. Nitrogen only D. Hydrogen only	Ans: B
Q27. Nervous System of human is divided into how many parts? A. 2 B. 3 – Central, Peripheral, Autonomic C. 4 D. 1	Ans: B
Q28. Central Nervous System is made up of: A. Only Brain B. Brain and Spinal Cord C. Only Spinal Cord D. Only Nerves	Ans: B
Q29. Brain is covered by a membrane called: A. Peritoneum B. Meninges C. Pericardium D. Pleura	Ans: B
Q30. Brain is situated in a bony box called: A. Vertebra B. Cranium C. Sternum D. Pelvis	Ans: B

Q31. Brain is divided into three parts: A. Left, Right, Middle B. Fore brain, Mid brain, Hind brain C. Upper, Lower, Middle D. Inner, Outer, Core	Ans: B
Q32. The weight of the human brain is: A. 500 grams B. 1336 grams C. 2000 grams D. 750 grams	Ans: B
Q33. Fore brain includes: A. Cerebellum and Pons B. Cerebrum, Thalamus and Hypothalamus C. Medulla only D. Corpora quadrigemina	Ans: B
Q34. Cerebrum is the centre of: A. Only movement B. Wisdom, will power, memory, knowledge and thinking C. Only hearing D. Only vision	Ans: B
Q35. Cerebrum is the largest part of the brain and is responsible for: A. Only balance B. Analysis, coordination of muscular movements from sense organs C. Only breathing D. Only heartbeat	Ans: B
Q36. The function of Thalamus is: A. Memory B. It is the centre of feeling pain, cold and heat C. Balance D. Breathing	Ans: B
Q37. Hind brain consists of: A. Cerebrum and Thalamus B. Cerebellum, Pons and Medulla C. Only Cerebrum D. Only Thalamus	Ans: B
Q38. Outer part of kidney is called: A. Medulla B. Cortex C. Pelvis D. Capsule	Ans: B

Q39. Which arteriole carries blood TO the glomerulus? A. Efferent arteriole B. Afferent arteriole C. Renal vein D. Collecting duct	Ans: B
Q40. Blood supply to the kidneys compared to other organs is: A. Very less B. In large quantity C. Same D. Zero	Ans: B
Q41. Unicellular animals excrete by: A. Kidneys B. Diffusion through general body surface and contractile vacuole C. Flame cells D. Malpighian tubules	Ans: B
Q42. Animals of Porifera Phylum excrete by: A. Kidneys B. General body surface C. Nephridia D. Flame cells	Ans: B
Q43. Coelenterates excrete: A. By kidneys B. Directly by cells C. By nephridia D. By Malpighian tubules	Ans: B
Q44. Mollusca excrete by: A. Kidneys B. Urinary organ (kidney-like organ) C. Flame cells D. Antennal gland	Ans: B
Q45. Vertebrates mainly excrete by: A. Flame cells B. Kidneys C. Antennal gland D. Malpighian tubules	Ans: B
Q46. Oil gland and sweat glands found in skin are part of: A. Digestive system B. Excretory system (Skin) C. Respiratory system D. Nervous system	Ans: B

Q47. The excretion of some spices like garlic, onion happens through:

- A. Kidneys
- B. Lungs (as vapour)
- C. Skin
- D. Liver

Ans: B

Q48. Input and output nerves meet at:

- A. Heart
- B. Central nervous system
- C. Liver
- D. Kidneys

Ans: B

Q49. Mid brain contains:

- A. Cerebrum
- B. Corpora quadrigemina
- C. Cerebellum
- D. Medulla

Ans: B

Q50. Lymph helps in transportation of:

- A. Only oxygen
- B. Different materials like oxygen, protein, and nutrients to tissues
- C. Only carbon dioxide
- D. Only hormones

Ans: B

Answer Key & Explanations

Q1. Answer: B — Killing harmful bacteria and other toxic foreign bodies

Page 416: Lymphocytes in lymph prevent diseases by killing harmful bacteria and toxic foreign bodies.

Topic: Lymph

Q2. Answer: B — Lymph node

Page 416: The node found in lymph vessels is called lymph node. It works as a filter of microbes.

Topic: Lymph

Q3. Answer: B — Healing the wounds

Page 416: Lymph helps in healing the wounds.

Topic: Lymph

Q4. Answer: B — Waste products formed during metabolism

Page 416: Excretion is removal of waste products formed during metabolism from the body.

Topic: Excretory System

Q5. Answer: B — Kidneys, Skin, Liver and Lungs

Page 416: Excretory organs: 1. Kidneys, 2. Skin, 3. Liver, 4. Lungs.

Topic: Excretory System

Q6. Answer: B — Kidneys

Page 416: Kidneys are the main excretory organ in human and other mammals. Weight is 140 grams.

Topic: Excretory System

Q7. Answer: B — 1,30,00,000 (13 lakh)

Page 416: Each kidney is made up of approximately 1,30,00,000 kidney ducts called nephrons.

Topic: Excretory System

Q8. Answer: B — Kidney

Page 416: Nephron is the structural and functional unit of the kidney.

Topic: Excretory System

Q9. Answer: B — Bowman's capsule

Page 416: There is a cup-like structure in every nephron called Bowman's capsule.

Topic: Excretory System

Q10. Answer: B — Thin blood vessels found in Bowman's capsule

Page 416: Glomerulus is made up of thin blood vessels found in Bowman's capsule.

Topic: Excretory System

Q11. Answer: B — Ultra filtration

Page 416: Process of filtration of liquid into cavity of Bowman's capsule is called ultra filtration.

Topic: Excretory System

Q12. Answer: B — Purification of blood plasma by excreting nitrogenous waste through urination

Page 416: Main function of kidneys is purification of blood plasma, excreting unwanted nitrogenous waste substances through urination.

Topic: Excretory System

Q13. Answer: B — 125 mL

Page 416: In kidneys, average 125 mL per minute blood is filtrated, i.e., 180 litres per day.

Topic: Excretory System

Q14. Answer: B — 1.45 litres

Page 416: Out of 180 litres filtered, 1.45 litres urine is formed daily and remaining is absorbed back.

Topic: Excretory System

Q15. Answer: B — 95%

Page 416: Urine is an aqueous solution in which 95% is water. Other constituents include urea, uric acid, chloride, sodium, potassium, creatinine.

Topic: Excretory System

Q16. Answer: B — Urochromes (formed by dissociation of haemoglobin)

Page 416: Colour of urine is light yellow due to presence of urochromes, formed by dissociation of haemoglobin.

Topic: Excretory System

Q17. Answer: B — Kidney → Ureter → Urinary bladder → Urethra

Page 416: Correct sequence: Kidney → Ureter → Urinary bladder → Urethra.

Topic: Excretory System

Q18. Answer: B — 6 (acidic)

Page 416: Urine is acidic. Its pH value is 6.

Topic: Excretory System

Q19. Answer: B — Calcium oxalate

Page 416: The stone formed in kidneys is made up of calcium oxalate.

Topic: Excretory System

Q20. Answer: B — Malpighian tubules

Page 416: Arthropods excrete by Malpighian tubules.

Topic: Excretory System

Q21. Answer: B — Flame cells

Page 416: Flat worm excretory parts are flame cells.

Topic: Excretory System

Q22. Answer: B — Nephridia

Page 416: Annelida Phylum excretory parts are Nephridia.

Topic: Excretory System

Q23. Answer: B — Removal of excess urea from blood using artificial kidney

Page 416: Hemodialysis is removal of excess urea from the blood of patient using artificial kidney.

Topic: Excretory System

Q24. Answer: B — Antennal gland

Page 416: Crustaceans excretory parts are Antennal gland.

Topic: Excretory System

Q25. Answer: B — Urea

Page 416: Liver cells play main role in excretion by converting amino acids and ammonia of blood into urea.

Topic: Excretory System

Q26. Answer: B — Carbon dioxide and water vapour

Page 416: Lungs excrete carbon dioxide and water vapour.

Topic: Excretory System

Q27. Answer: B — 3 – Central, Peripheral, Autonomic

Page 416: Nervous System is divided into 3 parts: 1. CNS, 2. PNS, 3. Autonomic Nervous System.

Topic: Nervous System

Q28. Answer: B — Brain and Spinal Cord

Page 416: CNS of human is made up of two parts - Brain and Spinal Cord.

Topic: Nervous System

Q29. Answer: B — Meninges

Page 416: Brain is covered by membrane called meninges. It is situated in a bony box called cranium.

Topic: Nervous System

Q30. Answer: B — Cranium

Page 416: Brain is situated in a bony box called cranium which protects it from external injury.

Topic: Nervous System

Q31. Answer: B — Fore brain, Mid brain, Hind brain

Page 416: Brain is divided into Fore brain, Mid brain, and Hind brain.

Topic: Nervous System

Q32. Answer: B — 1336 grams

Page 416: Weight of the brain of human is 1336 grams.

Topic: Nervous System

Q33. Answer: B — Cerebrum, Thalamus and Hypothalamus

Page 416: Fore brain includes Cerebrum, Thalamus and Hypothalamus.

Topic: Nervous System

Q34. Answer: B — Wisdom, will power, memory, knowledge and thinking

Page 416: Cerebrum is the most developed part of brain. Centre of wisdom, will power, movements, memory, knowledge and thinking.

Topic: Nervous System

Q35. Answer: B — Analysis, coordination of muscular movements from sense organs

Page 416: Cerebrum is the largest part. It analyses and coordinates muscular movements received from sense organs.

Topic: Nervous System

Q36. Answer: B — It is the centre of feeling pain, cold and heat

Page 416: Thalamus is the centre of feeling pain, cold and heat.

Topic: Nervous System

Q37. Answer: B — Cerebellum, Pons and Medulla

Page 416: Hind brain consists of Cerebellum, Pons and Medulla.

Topic: Nervous System

Q38. Answer: B — Cortex

Page 416: Outer part is called cortex and inner part is called medulla.

Topic: Excretory System

Q39. Answer: B — Afferent arteriole

Page 416: Afferent arteriole carries blood to the glomerulus. Efferent arteriole takes blood out.

Topic: Excretory System

Q40. Answer: B — In large quantity

Page 416: Supply of blood to kidneys takes place in large quantity in comparison to other organs.

Topic: Excretory System

Q41. Answer: B — Diffusion through general body surface and contractile vacuole

Page 416: Unicellular animals excrete by diffusion through general body surface and contractile vacuole.

Topic: Excretory System

Q42. Answer: B — General body surface

Page 416: Porifera Phylum animals excrete by general body surface.

Topic: Excretory System

Q43. Answer: B — Directly by cells

Page 416: Coelenterates excrete directly by cells.

Topic: Excretory System

Q44. Answer: B — Urinary organ (kidney-like organ)

Page 416: Mollusca excrete by urinary organ (by urinary organ/kidney-like structure).

Topic: Excretory System

Q45. Answer: B — Kidneys

Page 416: Vertebrates mainly excrete by kidneys.

Topic: Excretory System

Q46. Answer: B — Excretory system (Skin)

Page 416: Skin has oil glands and sweat glands. Skin is the second excretory organ.

Topic: Excretory System

Q47. Answer: B — Lungs (as vapour)

Page 416: Lungs excrete vapour component of some substances like garlic, onion and some spices.

Topic: Excretory System

Q48. Answer: B — Central nervous system

Page 416: Input and output nerves meet at central nervous system.

Topic: Nervous System

Q49. Answer: B — Corpora quadrigemina

Page 416: Mid brain contains Corpora quadrigemina.

Topic: Nervous System

Q50. Answer: B — Different materials like oxygen, protein, and nutrients to tissues

Page 416: Lymph helps in transportation of different materials like oxygen, protein and other nutrients to the tissues.

Topic: Lymph

Computer Introduction & Generations – Page 488

Source: Lucent's General Knowledge – Computer

Computer: Introduction, Generations & Types

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The word 'Computer' is derived from the English word:

- A. Complete
- B. Compute
- C. Compare
- D. Compose

Ans: B

Q2. Who is called the 'Father of Modern Computer'?

- A. Blaise Pascal
- B. Charles Babbage
- C. John Von Newman
- D. Howard Ekin

Ans: B

Q3. Blaise Pascal developed the first mechanical calculator in:

- A. 1833 AD
- B. 1642 AD
- C. 1937 AD
- D. 1946 AD

Ans: B

Q4. The credit of developing first computer program goes to:

- A. Charles Babbage
- B. Lady Ada Augusta
- C. Herman Holorith
- D. Howard Ekin

Ans: B

Q5. Herman Holorith prepared an electronic tabulating machine in 1880 using:

- A. Transistors
- B. Punch Card
- C. IC chips
- D. Vacuum tubes

Ans: B

Q6. Howard Ekin developed the first Mechanical Computer called:

- A. ENIAC-I
- B. Mark-I
- C. EDVAC
- D. UNIVAC

Ans: B

Q7. J.P. Ekart and John Moschley invented world's first electronic computer in 1946 called:

- A. EDVAC
- B. ENIAC-I
- C. UNIVAC
- D. Mark-I

Ans: B

Q8. Electronic Valve or Vacuum Tube was used as a switch in: A. Second generation B. First generation computer C. Third generation D. Fourth generation	Ans: B
Q9. John Von Newman invented EDVAC (Electronic Discrete Variable Computer) in: A. 1946 B. 1951 C. 1937 D. 1642	Ans: B
Q10. The credit of using Binary System in computers goes to: A. Babbage B. John Von Newman C. Pascal D. Ada Augusta	Ans: B
Q11. The main electronic component of First Generation (1940-52) computers was: A. Transistor B. Electronic Valve / Vacuum Tube C. Integrated Circuit D. Microprocessor	Ans: B
Q12. The main electronic component of Second Generation (1952-64) computers was: A. Vacuum Tube B. Transistor C. Integrated Circuit D. Optical Fibre	Ans: B
Q13. The main electronic component of Third Generation (1964-71) computers was: A. Transistor B. Integrated Circuit (IC) C. Vacuum Tube D. Optical Fibre	Ans: B
Q14. The main electronic component of Fourth Generation (1971-) computers was: A. Transistor B. Largely Integrated Circuit (LSI/VLSI) C. Vacuum Tube D. IC only	Ans: B
Q15. Fifth Generation computers will use: A. Transistors B. Optical Fibre and Artificial Intelligence C. Vacuum Tubes D. Simple ICs	Ans: B

Q16. Computers used by individual persons for domestic and official purposes are called: A. Super Computer B. Micro Computer (PC) C. Main Frame Computer D. Mini Computer	Ans: B
Q17. Mini Computer is how many times more powerful than Micro Computer? A. 2 to 5 times B. 5 to 50 times C. 100 times D. 1000 times	Ans: B
Q18. In Main Frame Computer, more than how many people can work at a time using Time Sharing and Multi Tasking? A. 10 B. 100 C. 50 D. 500	Ans: B
Q19. Super Computers are the most expensive and the fastest computers, able to process: A. Simple calculations only B. Most complex jobs with very high speed C. Only text documents D. Only graphics	Ans: B
Q20. In Quantum Computers, Q-Bit will be used in lieu of: A. Bytes B. Binary Bits C. Kilobytes D. Megabytes	Ans: B
Q21. The first programming language (1st Generation, 1940-52) was: A. COBOL B. FORTRAN-i C. BASIC D. Pascal	Ans: B
Q22. FORTRAN-ii, ALGOL-60, COBOL, LISP belong to which generation of programming languages? A. 1st Generation B. 2nd Generation (1952-64) C. 3rd Generation D. 4th Generation	Ans: B

Q23. PL/I, ALGOL-W, ALGOL-68, Pascal, SIMULA-67, APL, SNOBOL, 4 BASIC, C belong to: A. 2nd Generation B. 3rd Generation (1964-71) C. 4th Generation D. 5th Generation	Ans: B
Q24. 4th Generation (1971-90) programming languages include: A. Only FORTRAN B. CLUE, ALFARD, UCLID, Reformed Pascal, MODULA, EDA, ORACLE C. Only C D. Only Python	Ans: B
Q25. 5th Generation (Since 1990) uses: A. Only FORTRAN B. Artificial Intelligence Languages C. Only Assembly D. Only COBOL	Ans: B
Q26. A computer is an electronic machine that helps to: A. Only play games B. Process data and solve problems in almost all fields C. Only print documents D. Only send emails	Ans: B
Q27. A computer is an information-processing and: A. Food preparation tool B. Information-accessing tool C. Musical instrument D. Cooking device	Ans: B
Q28. Information processing is the essence of: A. Cooking B. Computing C. Painting D. Singing	Ans: B
Q29. Computer is an electronic device which processes the input informations according to: A. Random instructions B. Given set of instructions called program C. No instructions D. Only voice commands	Ans: B

Q30. John Von Newman used which concept in EDVAC? A. Punch cards B. Stored Program C. Vacuum tubes only D. Paper tape	Ans: B
Q31. IBM-700, IBM-1401, IBM-1620, CDC-3600, ATLAS, ICL-1901 are examples of which generation? A. First B. Second (1952-64) C. Third D. Fourth	Ans: B
Q32. IBM-360, IBM-370, NCR-395, CDC-1700, ICL-2903 are examples of which generation? A. Second B. Third (1964-71) C. Fourth D. Fifth	Ans: B
Q33. APPLE and DCM are examples of which generation computers? A. Third B. Fourth (1971-) C. Second D. First	Ans: B
Q34. Desktop, Laptop, Tablets are examples of: A. Super computers B. Micro computers (PCs) C. Main frame computers D. Mini computers	Ans: B
Q35. The era of computers started in which centuries? A. 18th B. 20th and 21st C. 15th D. 10th	Ans: B
Q36. Computer and Information Technology (IT) in recent years has become: A. Obsolete B. An integral part of our life C. Only for scientists D. Only for military	Ans: B

Q37. One of the greatest things that man has ever created is: A. Television B. The Computer C. Radio D. Telephone	Ans: B
Q38. EDSAC, EDVAC, UNIVAC are computers of which generation? A. Second B. First (1940-52) C. Third D. Fourth	Ans: B
Q39. The first mechanical calculator 'Pascalene' was developed by: A. Babbage B. Blaise Pascal C. Newman D. Ekin	Ans: B
Q40. The Second Revolution of computers was given a right direction by: A. Mark-I B. EDVAC and Stored Program concept by Newman C. Punch Cards D. ENIAC only	Ans: B
Q41. Computer is used in areas of: A. Only education B. Administration, medicine, education, sports, defence, shops, home, markets C. Only sports D. Only defence	Ans: B
Q42. Quantum Computers are probably more advanced than: A. Calculators B. Human brain C. Abacus D. Typewriters	Ans: B
Q43. The development of Quantum Computer is in: A. Completed stage B. Final stage C. Not started D. Abandoned	Ans: B
Q44. A computer accepts information or data from the outside world and: A. Deletes it B. Processes it to produce a new information C. Ignores it D. Stores it permanently without processing	Ans: B

Q45. The period of Third Generation computers was: A. 1940-52 B. 1964-71 C. 1952-64 D. 1971-90	Ans: B
Q46. The period of First Generation computers was: A. 1952-64 B. 1940-52 C. 1964-71 D. 1971-90	Ans: B
Q47. The period of Second Generation computers was: A. 1940-52 B. 1952-64 C. 1964-71 D. 1971-90	Ans: B
Q48. Herman Holorith's tabulating machine was automatically functional with the help of: A. Electricity only B. Punch Card C. Transistor D. IC chip	Ans: B
Q49. The first electronic computer ENIAC-I paved the way for first revolution in the field of: A. Music B. Calculating machine or computer C. Cooking D. Agriculture	Ans: B
Q50. Main Frame Computers use which techniques to allow multiple users? A. Only batch processing B. Time Sharing and Multi Tasking C. Only serial processing D. Only parallel processing	Ans: B

Answer Key & Explanations

Q1. Answer: B — Compute

Page 488: The word computer has derived from an English word 'Compute', which means 'to calculate'.

Topic: Introduction

Q2. Answer: B — Charles Babbage

Page 488: British scientist Charles Babbage was the first person to conceive an automatic calculator or a computer in 1833. He is called the 'Father of Modern Computer'.

Topic: History

Q3. Answer: B — 1642 AD

Page 488: Blaise Pascal had developed the first mechanical calculator in 1642 AD, which is called 'Pascalene'.

Topic: History

Q4. Answer: B — Lady Ada Augusta

Page 488: The credit of developing first computer program goes to Lady Ada Augusta, a student of Babbage.

Topic: History

Q5. Answer: B — Punch Card

Page 488: Herman Holorith prepared an electronic tabulating machine in 1880, automatically functional with the help of Punch Card.

Topic: History

Q6. Answer: B — Mark-I

Page 488: Howard Ekin developed the first Mechanical Computer 'Mark-I' in 1937.

Topic: History

Q7. Answer: B — ENIAC-I

Page 488: J.P. Ekart and John Moschley invented world's first electronic computer 'ENIAC-I' in 1946.

Topic: History

Q8. Answer: B — First generation computer

Page 488: Electronic Valve or Vacuum Tube was used as a switch in the first generation computer (ENIAC-I).

Topic: History

Q9. Answer: B — 1951

Page 488: John Von Newman invented EDVAC in 1951, in which he used Stored Program. The credit of using Binary System in computers also goes to him.

Topic: History

Q10. Answer: B — John Von Newman

Page 488: The credit of using Binary System in computers goes to John Von Newman.

Topic: History

Q11. Answer: B — Electronic Valve / Vacuum Tube

Page 488: First Generation (1940-52) used Electronic Valve/Vacuum Tube. Examples: EDSAC, EDVAC, UNIVAC.

Topic: Generations

Q12. Answer: B — Transistor

Page 488: Second Generation (1952-64) used Transistor. Examples: IBM-700, IBM-1401, IBM-1620.

Topic: Generations

Q13. Answer: B — Integrated Circuit (IC)

Page 488: Third Generation (1964-71) used Integrated Circuit. Examples: IBM-360, IBM-370, NCR-395.

Topic: Generations

Q14. Answer: B — Largely Integrated Circuit (LSI/VLSI)

Page 488: Fourth Generation (1971-) uses Largely Integrated Circuit. Examples: APPLE, DCM.

Topic: Generations

Q15. Answer: B — Optical Fibre and Artificial Intelligence

Page 488: Fifth Generation uses Optical Fibre and Artificial Intelligence.

Topic: Generations

Q16. Answer: B — Micro Computer (PC)

Page 488: Micro Computers are used by individual, also called PC or Personal Computer, for domestic and official purposes.

Topic: Types

Q17. Answer: B — 5 to 50 times

Page 488: Mini Computer is comparatively larger and 5 to 50 times more powerful than Micro Computer.

Topic: Types

Q18. Answer: B — 100

Page 488: Main Frame Computer: By Time Sharing and Multi Tasking techniques many people rather more than 100 people can work at a time.

Topic: Types

Q19. Answer: B — Most complex jobs with very high speed

Page 488: Super Computers are very powerful, most expensive and fastest, able to process most complex jobs with very high speed.

Topic: Types

Q20. Answer: B — Binary Bits

Page 488: In Quantum Computers Q-Bit will be used in lieu of Binary Bits.

Topic: Types

Q21. Answer: B — FORTRAN-i

Page 488: 1st Generation (1940-52) programming language was FORTRAN-i.

Topic: Programming

Q22. Answer: B — 2nd Generation (1952-64)

Page 488: 2nd Generation (1952-64) languages: FORTRAN-ii, ALGOL-60, COBOL, LISP.

Topic: Programming

Q23. Answer: B — 3rd Generation (1964-71)

Page 488: 3rd Generation (1964-71) languages include PL/I, ALGOL-W, ALGOL-68, Pascal, SIMULA-67, APL, SNOBOL, 4 BASIC, C.

Topic: Programming

Q24. Answer: B — CLUE, ALFARD, UCLID, Reformed Pascal, MODULA, EDA, ORACLE

Page 488: 4th Generation (1971-90): CLUE, ALFARD, UCLID, Reformed Pascal, MODULA, EDA, ORACLE.

Topic: Programming

Q25. Answer: B — Artificial Intelligence Languages

Page 488: 5th Generation (Since 1990) uses Artificial Intelligence Languages.

Topic: Programming

Q26. Answer: B — Process data and solve problems in almost all fields

Page 488: A computer is an electronic machine that helps to process data and solve problems in education, home, medicine, science, technology, research, designing, publishing, communication etc.

Topic: Introduction

Q27. Answer: B — Information-accessing tool

Page 488: A computer is an information-processing and information-accessing tool.

Topic: Introduction

Q28. Answer: B — Computing

Page 488: Information processing is the essence of computing.

Topic: Introduction

Q29. Answer: B — Given set of instructions called program

Page 488: Computer processes input informations according to the given set of instructions, called program.

Topic: Introduction

Q30. Answer: B — Stored Program

Page 488: John Von Newman invented EDVAC in 1951, in which he used Stored Program concept.

Topic: History

Q31. Answer: B — Second (1952-64)

Page 488: These are Second Generation computers using Transistors.

Topic: Generations

Q32. Answer: B — Third (1964-71)

Page 488: These are Third Generation computers using Integrated Circuits.

Topic: Generations

Q33. Answer: B — Fourth (1971-)

Page 488: APPLE, DCM are Fourth Generation computers using Largely Integrated Circuits.

Topic: Generations

Q34. Answer: B — Micro computers (PCs)

Page 488: PCs are used for domestic and official purposes like Desktop, Laptop, Tablets etc.

Topic: Types

Q35. Answer: B — 20th and 21st

Page 488: The era of 20th and 21st century has witnessed rapid developments in science and technology.

Topic: Introduction

Q36. Answer: B — An integral part of our life

Page 488: Computer and Information Technology (IT) in recent years has become an integral part of our life.

Topic: Introduction

Q37. Answer: B — The Computer

Page 488: One of the greatest things that man has ever created is, perhaps, 'the Computer'.

Topic: Introduction

Q38. Answer: B — First (1940-52)

Page 488: EDSAC, EDVAC, UNIVAC are First Generation (1940-52) computers using Electronic Valve/Vacuum Tube.

Topic: Generations

Q39. Answer: B — Blaise Pascal

Page 488: Blaise Pascal developed the first mechanical calculator 'Pascalene' in 1642 AD.

Topic: History

Q40. Answer: B — EDVAC and Stored Program concept by Newman

Page 488: Newman's EDVAC and Stored Program concept gave a right direction to the Computer Revolution (Second Revolution).

Topic: History

Q41. Answer: B — Administration, medicine, education, sports, defence, shops, home, markets

Page 488: Computer is being used in areas of administration, medicine, education, sports, defence, shops, home, markets and many more.

Topic: Introduction

Q42. Answer: B — Human brain

Page 488: Probably Quantum Computers will be more advanced than that of human brain.

Topic: Types

Q43. Answer: B — Final stage

Page 488: The development of Quantum Computer is in final stage.

Topic: Types

Q44. Answer: B — Processes it to produce a new information

Page 488: A computer accepts information/data from outside world, processes it to produce new information.

Topic: Introduction

Q45. Answer: B — 1964-71

Page 488: Third Generation period was 1964-71, using Integrated Circuits.

Topic: Generations

Q46. Answer: B — 1940-52

Page 488: First Generation period was 1940-52.

Topic: Generations

Q47. Answer: B — 1952-64

Page 488: Second Generation period was 1952-64.

Topic: Generations

Q48. Answer: B — Punch Card

Page 488: Herman Holorith's machine in 1880 was automatically functional with the help of Punch Card.

Topic: History

Q49. Answer: B — Calculating machine or computer

Page 488: ENIAC-I paved the way for first revolution in the field of calculating machine or computer.

Topic: History

Q50. Answer: B — Time Sharing and Multi Tasking

Page 488: Main Frame Computer: By Time Sharing and Multi Tasking techniques, many people can work at different terminals.

Topic: Types

Important Computer Facts & Super Computers – Page 489

Source: Lucent's General Knowledge – Computer

Computer: Important Facts, Super Computers & Firsts

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. December 2 is observed as:

- A. World Internet Day
- B. Computer Literacy Day
- C. Science Day
- D. Technology Day

Ans: B

Q2. India announced New Computer Policy in:

- A. 1990
- B. 1984
- C. 1978
- D. 2000

Ans: B

Q3. First computer made in India is:

- A. PARAM
- B. Siddharth
- C. PACE
- D. MACH

Ans: B

Q4. First computer in India was installed in the Main Post Office of Bangalore on:

- A. January 26, 1980
- B. August 16, 1986
- C. March 15, 1990
- D. December 2, 1984

Ans: B

Q5. First Pollution Free Computerized Petrol Pump of India is in:

- A. Delhi
- B. Mumbai
- C. Bangalore
- D. Chennai

Ans: B

Q6. First Computer University (in Private Sector) in India is:

- A. IIT Delhi
- B. Rajeev Gandhi Computer University
- C. IIIT Hyderabad
- D. NIT Trichy

Ans: B

Q7. Bangalore (now Bengaluru) is also known as the:

- A. Garden City of India
- B. Silicon Valley of India
- C. IT Capital of India
- D. Computer City

Ans: B

Q8. First Indian News Paper to be available on Internet is: A. Times of India B. The Hindu C. Hindustan Times D. Indian Express	Ans: B
Q9. First Indian magazine to be available on Internet is: A. Outlook B. India Today C. Frontline D. The Week	Ans: B
Q10. First Indian political party which created its website on internet is: A. Congress B. Bhartiya Janata Party (BJP) C. CPI D. AAP	Ans: B
Q11. There are mainly three types of Computers: A. Small, Medium, Large B. Digital, Analog and Hybrid C. Fast, Slow, Medium D. Old, New, Latest	Ans: B
Q12. First Super Computer of the world is: A. Deep Blue B. CRAY K-1-S C. PARAM D. Blue Gene	Ans: B
Q13. 'Deep Blue' is a Super Computer which had defeated World Chess Champion: A. Viswanathan Anand B. Garry Kasparov C. Magnus Carlsen D. Bobby Fischer	Ans: B
Q14. Deep Blue is able to do work equivalent to how many computers? A. 16 B. 32 C. 64 D. 128	Ans: B
Q15. Seymour Cray is called the Father of: A. Personal Computer B. Super Computer C. Laptop D. Internet	Ans: B

Q16. CRAY KIS super computer was manufactured by: A. IBM Co., USA B. CRAY K Research Co., USA C. Cambridge University, UK D. C-DAC, Pune	Ans: B
Q17. Deep Blue and Blue Gene were manufactured by: A. CRAY K Research B. IBM Co., USA C. Cambridge University D. NAL Bangalore	Ans: B
Q18. COSMOS super computer was developed by: A. IBM Co., USA B. Cambridge University, UK C. CRAY K Research D. C-DAC Pune	Ans: B
Q19. First Electronic Digital Computer of the world is: A. UNIVAC B. ENIAC C. EDVAC D. Mark-I	Ans: B
Q20. Most popular Operating System in smartphones is: A. Windows B. Android C. iOS D. Linux	Ans: B
Q21. Most Super Computers use which operating system? A. Windows B. Linux C. Android D. Mac OS	Ans: B
Q22. USENET is a link to connect all the: A. Schools B. Universities C. Hospitals D. Banks	Ans: B
Q23. First book on Personal Computer was written by: A. Bill Gates B. Ted Nelson C. Tim Berners-Lee D. Steve Jobs	Ans: B

Q24. First magazine on Computer is: A. PC World B. Computer and Automotion C. Byte D. Wired	Ans: B
Q25. First home Computer is: A. Apple II B. Comodor VIC / 20 C. IBM PC D. Altair 8800	Ans: B
Q26. First Practical Digital computer is: A. ENIAC B. UNIVAC C. EDVAC D. Mark-I	Ans: B
Q27. FORTRAN is the first: A. Operating System B. Programming Language C. Computer virus D. Web browser	Ans: B
Q28. PROLOG is the language of the: A. First generation B. Fifth generation C. Second generation D. Third generation	Ans: B
Q29. J.S. Kilbi developed the: A. Transistor B. IC chips C. Vacuum tube D. Optical fibre	Ans: B
Q30. A computer error is known as: A. Virus B. Bug C. Worm D. Spam	Ans: B

Q31. C-DAC (Centre for Development and Advanced Computing) was established in Pune in: A. 1984 B. 1988 C. 1990 D. 1995	Ans: B
Q32. FLO SOLVER super computer was developed by: A. C-DAC, Pune B. NAL, Bangalore C. IIT Bombay D. DRDO	Ans: B
Q33. PACE super computer was developed by: A. NAL Bangalore B. DRDO C. C-DAC Pune D. IIS Bangalore	Ans: B
Q34. PARAM-10000 super computer was developed by: A. NAL Bangalore B. C-DAC, Pune C. DRDO D. IIT Bombay	Ans: B
Q35. Super computer PARAM-10000 has the capacity of 1 billion calculations per second. The main credit goes to: A. APJ Abdul Kalam B. Dr. Vijay P. Bhaskar, Executive Director of C-DAC, Pune C. Narayan Murthy D. Nandan Nilekani	Ans: B
Q36. PARAM KAMRUPA Super Computer facility was inaugurated at IIT Guahati on: A. March 28, 1998 B. Oct. 13, 2022 C. August 16, 1986 D. December 2, 1984	Ans: B
Q37. National Aeronautics Laboratories, Bangalore was the first in India to develop a Super Computer named: A. PARAM B. FLO SOLVER C. PACE D. MACH	Ans: B

Q38. Laser Printers are the: A. Slowest printers B. Fastest printers C. Cheapest printers D. Oldest printers	Ans: B
Q39. IBM stands for: A. Indian Business Machine B. International Business Machine C. Integrated Business Module D. Internal Binary Machine	Ans: B
Q40. Computer virus is a man made digital parasite which corrupts the file and is known as: A. Worm B. File corrupter C. Bug D. Spam	Ans: B
Q41. Modem is a device which connects computers and works based on: A. Radio waves B. Telephone lines C. Satellite D. Bluetooth	Ans: B
Q42. Y-2K was a technical problem associated with the calendar known as: A. Date Bug B. Millenium Bug C. Year Bug D. Time Bug	Ans: B
Q43. The development of computer started in India since: A. 1970 B. 1955 C. 1946 D. 1980	Ans: B
Q44. Indian Institute of Science, Bangalore has developed a small palm sized touch screen computer called: A. PARAM B. Simputer C. Tablet D. PACE	Ans: B

Q45. First computerarium in India is in: A. Mumbai B. Bangaluru (Karnataka) C. Delhi D. Chennai	Ans: B
Q46. Vellanad of Thiruvananthapuram district in Kerala has been declared the first fully computerised village of: A. World B. India C. Asia D. Kerala only	Ans: B
Q47. Linus Benedict Torvalds, creator of Unix like Operating System (Linux), was born in Helsinki, Finland on: A. January 1, 1970 B. December 28, 1969 C. March 15, 1975 D. August 16, 1971	Ans: B
Q48. On Sept. 17, 1991, Linus completed the first version of: A. Windows B. Linux C. Mac OS D. DOS	Ans: B
Q49. CHIPP-16 super computer was developed by: A. NAL, Bangalore B. C-Dot, Bangalore C. DRDO D. IBM	Ans: B
Q50. MULTIMICRO and MACH super computers were developed in: A. NAL, Bangalore and DRDO B. IIS, Bangalore and IIT, Bombay C. C-DAC and C-Dot D. IBM and CRAY	Ans: B

Answer Key & Explanations

Q1. Answer: B — Computer Literacy Day

Page 489: December 2 is observed as Computer Literacy Day.

Topic: Computer Facts

Q2. Answer: B — 1984

Page 489: India has announced New Computer Policy in 1984.

Topic: Computer Facts

Q3. Answer: B — Siddharth

Page 489: First computer (made in India) is 'Siddharth', manufactured by Electronics Corporation of India.

Topic: Computer Facts

Q4. Answer: B — August 16, 1986

Page 489: First computer in India was installed in the Main Post Office of Bangalore on August 16, 1986.

Topic: Computer Facts

Q5. Answer: B — Mumbai

Page 489: First Pollution Free Computerized Petrol Pump of India is in Mumbai.

Topic: Computer Facts

Q6. Answer: B — Rajeev Gandhi Computer University

Page 489: First Computer University (in Private Sector) in India is Rajeev Gandhi Computer University.

Topic: Computer Facts

Q7. Answer: B — Silicon Valley of India

Page 489: Bangalore (now Bengaluru) is also known as the Silicon Valley of India.

Topic: Computer Facts

Q8. Answer: B — The Hindu

Page 489: First Indian News Paper to be available on Internet is 'The Hindu'.

Topic: Computer Facts

Q9. Answer: B — India Today

Page 489: First Indian magazine to be available on Internet is 'India Today'.

Topic: Computer Facts

Q10. Answer: B — Bhartiya Janata Party (BJP)

Page 489: First Indian political party which has created its website on internet is 'Bhartiya Janata Party (BJP)'.

Topic: Computer Facts

Q11. Answer: B — Digital, Analog and Hybrid

Page 489: Mainly there are three types of Computers — Digital, Analog and Hybrid.

Topic: Computer Facts

Q12. Answer: B — CRAY K-1-S

Page 489: First Super Computer of the world is CRAY K-1-S, developed by Cray K Company of U.S.A.

Topic: Super Computers

Q13. Answer: B — Garry Kasparov

Page 489: 'Deep Blue' is a Super Computer which had defeated World Chess Champion Garry Kasparov.

Topic: Super Computers

Q14. Answer: B — 32

Page 489: Deep Blue is able to do the work equivalent to 32 computers and can think 20 crore steps of chess in 1 second.

Topic: Super Computers

Q15. Answer: B — Super Computer

Page 489: Seymour Cray is called the Father of Super Computer.

Topic: Super Computers

Q16. Answer: B — CRAY K Research Co., USA

Page 489: CRAY KIS manufactured by CRAY K Research Co., USA.

Topic: Super Computers

Q17. Answer: B — IBM Co., USA

Page 489: Deep Blue and Blue Gene manufactured by IBM Co., USA.

Topic: Super Computers

Q18. Answer: B — Cambridge University, UK

Page 489: COSMOS developed by Cambridge University, UK.

Topic: Super Computers

Q19. Answer: B — ENIAC

Page 489: First Electronic Digital Computer of the world is 'ENIAC'.

Topic: Computer Facts

Q20. Answer: B — Android

Page 489: Most popular OS: in PC—Windows, in smartphones—Android, in Tablets—iOS. Android is used in highest number of devices across the world.

Topic: Computer Facts

Q21. Answer: B — Linux

Page 489: Most Super Computers use Linux.

Topic: Computer Facts

Q22. Answer: B — Universities

Page 489: USENET is a link to connect all the universities.

Topic: Computer Facts

Q23. Answer: B — Ted Nelson

Page 489: First book on Personal Computer was written by Ted Nelson. The book 'Soul of New Machine' won Pulitzer Prize.

Topic: Computer Facts

Q24. Answer: B — Computer and Automotion

Page 489: First magazine on Computer is 'Computer and Automotion'.

Topic: Computer Facts

Q25. Answer: B — Comodor VIC / 20

Page 489: First home Computer is Comodor VIC / 20.

Topic: Computer Facts

Q26. Answer: B — UNIVAC

Page 489: First Practical Digital computer is UNIVAC.

Topic: Computer Facts

Q27. Answer: B — Programming Language

Page 489: FORTRAN is the first Programming Language.

Topic: Computer Facts

Q28. Answer: B — Fifth generation

Page 489: PROLOG is the language of the fifth generation of computer.

Topic: Computer Facts

Q29. Answer: B — IC chips

Page 489: J.S. Kilbi developed the IC chips.

Topic: Computer Facts

Q30. Answer: B — Bug

Page 489: A computer error is known as Bug.

Topic: Computer Facts

Q31. Answer: B — 1988

Page 489: C-DAC was established in Pune in 1988.

Topic: Super Computers

Q32. Answer: B — NAL, Bangalore

Page 489: FLO SOLVER developed by NAL, Bangalore.

Topic: Super Computers

Q33. Answer: B — DRDO

Page 489: PACE developed by DRDO.

Topic: Super Computers

Q34. Answer: B — C-DAC, Pune

Page 489: PARAM-10000 developed by C-DAC, Pune.

Topic: Super Computers

Q35. Answer: B — Dr. Vijay P. Bhaskar, Executive Director of C-DAC, Pune

Page 489: PARAM-10000 was made on March 28, 1998. Main credit goes to Dr. Vijay P. Bhaskar, Executive Director of C-DAC, Pune.

Topic: Super Computers

Q36. Answer: B — Oct. 13, 2022

Page 489: PARAM KAMRUPA was inaugurated by President Droupadi Murmu at IIT Guahati on Oct. 13, 2022.

Topic: Super Computers

Q37. Answer: B — FLO SOLVER

Page 489: National Aeronautics Laboratories, Bangalore was the first in India to develop a Super Computer named FLO SOLVER.

Topic: Super Computers

Q38. Answer: B — Fastest printers

Page 489: Laser Printers are the fastest printers.

Topic: Computer Facts

Q39. Answer: B — International Business Machine

Page 489: IBM (International Business Machine) is an American computer company.

Topic: Computer Facts

Q40. Answer: B — File corrupter

Page 489: Computer virus is a man made digital parasite which corrupts (infects) the file and known as 'File corrupter'.

Topic: Computer Facts

Q41. Answer: B — Telephone lines

Page 489: Modem is a device which connects the computers and works based on telephone lines.

Topic: Computer Facts

Q42. Answer: B — Millenium Bug

Page 489: Y-2K was a technical problem associated with the calendar (Date, Month and Year) known as 'Millenium Bug'.

Topic: Computer Facts

Q43. Answer: B — 1955

Page 489: The development of computer started in India since 1955.

Topic: Computer Facts

Q44. Answer: B — Simputer

Page 489: Indian Institute of Science, Bangalore has developed 'Simputer', a small palm sized touch screen computer.

Topic: Computer Facts

Q45. Answer: B — Bangaluru (Karnataka)

Page 489: First computerarium in India is in Bangaluru (Karnataka).

Topic: Computer Facts

Q46. Answer: B — India

Page 489: Vellanad of Thiruvananthapuram district in Kerala has been declared the first fully computerised village of India.

Topic: Computer Facts

Q47. Answer: B — December 28, 1969

Page 489: Linus Benedict Torvalds was born in Helsinki, Finland on December 28, 1969.

Topic: Computer Facts

Q48. Answer: B — Linux

Page 489: On Sept. 17, 1991, Linus completed the first version of Linux. He released it under General Public Licence.

Topic: Computer Facts

Q49. Answer: B — C-Dot, Bangalore

Page 489: CHIPP-16 developed by C-Dot, Bangalore.

Topic: Super Computers

Q50. Answer: B — IIS, Bangalore and IIT, Bombay

Page 489: MULTIMICRO by IIS, Bangalore. MACH by IIT, Bombay.

Topic: Super Computers

Hardware, Binary System & Data Units – Page 490

Source: Lucent's General Knowledge – Computer

Computer: Hardware, Binary System, Data Measurement & Logic Gates

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. A computer has two main parts:

- A. Monitor and Keyboard
- B. Hardware and Software
- C. CPU and Memory
- D. Input and Output

Ans: B

Q2. Hardware comprises of the physical units of a computer system. These units constitute the:

- A. Software
- B. Hardware of a computer
- C. Programs
- D. Data

Ans: B

Q3. A computer has three main units:

- A. ALU, CU, Memory
- B. Input unit, Processing unit, Output unit
- C. Keyboard, Mouse, Monitor
- D. CPU, RAM, HDD

Ans: B

Q4. The computer has its own internal:

- A. Voice
- B. Language
- C. Music
- D. Video

Ans: B

Q5. The computer is essentially made of electronic components that can generate either a low (0 volt) or a high (say 5 volts). These states are represented by:

- A. Alphabets
- B. Binary digits (bits) - 1s and 0s
- C. Decimal numbers
- D. Roman numerals

Ans: B

Q6. Computers always work with:

- A. Decimal numbers
- B. Bits
- C. Words
- D. Pictures

Ans: B

Q7. Computer scientists use combination of how many bits to represent various symbols?

- A. 4
- B. 8
- C. 16
- D. 32

Ans: B

Q8. Using 8 bits, the computer can represent how many different symbols? A. 128 B. 256 C. 64 D. 512	Ans: B
Q9. A combination of 8 bits is called a: A. Nibble B. Byte C. Word D. Kilobyte	Ans: B
Q10. In computers, information is represented using multiples of 8 (2^3) bits. Higher units are expressible in multiples of: A. 2^2 B. 2^3 C. 10 D. 100	Ans: B
Q11. The power of two closest to a kilo (1000) is $2^{10} =$: A. 512 B. 1024 C. 2048 D. 256	Ans: B
Q12. 1 Mega Byte is equal to: A. 1024 Bytes B. 1 million bytes or 1000 Kilo Bytes C. 1 billion bytes D. 512 KB	Ans: B
Q13. A Giga Byte refers to approximately: A. 1000 bytes B. 1000 megabytes or 1 million kilobytes C. 100 megabytes D. 10 megabytes	Ans: B
Q14. 4 Bit = : A. 1 Byte B. 1 Nibble C. 1 KB D. 1 Word	Ans: B

Q15. 2 Nibble (8 Bit) = : A. 1 Nibble B. 1 Byte C. 1 KB D. 1 MB	Ans: B
Q16. 1024 Byte = : A. 1 MB B. 1 Kilo Byte (KB) C. 1 GB D. 1 Nibble	Ans: B
Q17. 1024 KB = : A. 1 GB B. 1 Mega Byte (MB) C. 1 TB D. 1 Byte	Ans: B
Q18. 1024 MB = : A. 1 TB B. 1 Giga Byte (GB) C. 1 PB D. 1 KB	Ans: B
Q19. 1024 GB = : A. 1 PB B. 1 Tera Byte (TB) C. 1 MB D. 1 KB	Ans: B
Q20. 1024 TB = : A. 1 GB B. 1 Peta Byte (PB) C. 1 MB D. 1 EB	Ans: B
Q21. The basic elements of computers that can signal a 1 or 0 are called: A. Transistors B. Flip-flops C. Resistors D. Capacitors	Ans: B
Q22. Modern computers use a very tiny set of flip-flops known as: A. RAM B. Register C. Cache D. ROM	Ans: B

Q23. The entire independent circuits can be designed on a small piece of semi conductor material like: A. Copper B. Silicon C. Gold D. Iron	Ans: B
Q24. Silicon is obtained from sand and is a poor conductor of electricity. Miniature circuits made on silicon are called: A. Transistors B. Integrated Circuits (IC) C. Resistors D. Capacitors	Ans: B
Q25. By 1971, engineers were able to build a complete computer on a single chip of silicon called: A. RAM chip B. Microprocessor C. ROM chip D. GPU	Ans: B
Q26. The computer converts all decimal numbers into: A. Octal numbers B. Binary numbers or combinations of bits C. Hexadecimal numbers D. Roman numerals	Ans: B
Q27. The internal circuits that perform mathematical operations on bits are usually made of: A. Resistors B. Logic gates C. Capacitors D. Inductors	Ans: B
Q28. The three basic logic gates are: A. ADD, SUB, MUL B. AND, OR and NOT C. IF, THEN, ELSE D. NAND, NOR, XOR	Ans: B
Q29. A computer is organized into three basic units: A. Hardware, Software, Data B. CPU, Memory Unit, Input/Output Unit C. Monitor, Keyboard, Mouse D. RAM, ROM, Cache	Ans: B

Q30. The CPU is the part of a computer that performs the main function of: A. Data storage B. Information processing C. Printing D. Networking	Ans: B
Q31. While earliest machines were 8-bit, contemporary computers can work with: A. 4 bits B. 16, 32 and even 64 bits C. Only 8 bits D. 128 bits	Ans: B
Q32. The earliest computers were designed so that their components could work with how many bits at a time? A. 4 B. 8 C. 16 D. 32	Ans: B
Q33. Hertz is a unit of: A. Power B. Frequency C. Length D. Weight	Ans: B
Q34. 1 MHz equals: A. 1000 ticks per second B. 1 million ticks per second C. 1 billion ticks per second D. 100 ticks per second	Ans: B
Q35. 1 GHz = : A. 100 MHz B. 1000 MHz C. 10 MHz D. 10000 MHz	Ans: B
Q36. The ticks per second of the internal clock are measured in: A. Watts B. Megahertz and Gigahertz C. Meters D. Kilograms	Ans: B

Q37. Higher the clock-speed of CPU: A. Slower the computer B. Faster the computer C. No effect D. Less memory	Ans: B
Q38. A flip-flop is a simple: A. Mechanical device B. Electrical device C. Optical device D. Chemical device	Ans: B
Q39. The most important characteristic of registers is that binary digits can be stored in them using: A. Magnets B. Certain voltages C. Light signals D. Sound waves	Ans: B
Q40. The binary number system is similar to decimal system. The only difference is in: A. Operations B. The number of digits used C. Speed D. Size	Ans: B
Q41. In binary system, only which digits are used? A. 0 to 9 B. 0 and 1 C. 0 to 7 D. 0 to F	Ans: B
Q42. 8 bits are enough to cover our entire range of alphabets, numbers and special characters like \$, @, +, ■. This is because $2^8 =$: A. 128 B. 256 C. 512 D. 64	Ans: B
Q43. To build complex information like paragraphs, we need larger numbers of bytes called: A. Bits and Nibbles B. Kilobyte, Megabyte and Gigabyte C. Only Nibbles D. Only Bits	Ans: B

Q44. Logic gates are components that generate a 1 or a 0 depending on: A. Time B. The input C. Temperature D. Pressure	Ans: B
Q45. Computer scientists use a special convention: A high is represented by '1' and a low is represented by: A. 2 B. 0 C. 9 D. 5	Ans: B
Q46. It is difficult to talk always in terms of currents and voltages. Therefore computer scientists represent information using: A. Analog signals B. Binary convention (1s and 0s) C. Sound waves D. Light pulses	Ans: B
Q47. The term 'binary' refers to something that has: A. Three parts B. Two parts C. Four parts D. One part	Ans: B
Q48. Linus released Linux under: A. MIT License B. General Public Licence (open and free to all) C. Proprietary license D. Commercial license	Ans: B
Q49. A computer may be used to control purely mechanical action. It has: A. Only software B. Two main parts: Hardware and Software C. Only hardware D. Only data	Ans: B
Q50. The computer supplies processed information back to the users using special: A. Input devices B. Output devices C. Storage devices D. Network devices	Ans: B

Answer Key & Explanations

Q1. Answer: B — Hardware and Software

Page 490: It has two main parts: (i) Hardware and (ii) Software.

Topic: Hardware

Q2. Answer: B — Hardware of a computer

Page 490: A computer has three main units: Input unit, Processing unit, Output unit. These constitute the hardware.

Topic: Hardware

Q3. Answer: B — Input unit, Processing unit, Output unit

Page 490: A computer has three main units: 1. Input unit, 2. Processing unit and 3. Output unit.

Topic: Hardware

Q4. Answer: B — Language

Page 490: The computer has its own internal 'language'.

Topic: Hardware

Q5. Answer: B — Binary digits (bits) - 1s and 0s

Page 490: A high is represented by '1' and a low by '0'. The 1s and 0s are known as binary digits or 'bits'.

Topic: Binary System

Q6. Answer: B — Bits

Page 490: Computers always work with bits. They do not understand any other form.

Topic: Binary System

Q7. Answer: B — 8

Page 490: Computer scientists use combination of 8 bits taken together to represent various symbols.

Topic: Binary System

Q8. Answer: B — 256

Page 490: Because every bit can take one of 2 possible values, using 8 bits, the computer can represent 256 different symbols.

Topic: Binary System

Q9. Answer: B — Byte

Page 490: Such a combination of 8 bits is called a byte.

Topic: Binary System

Q10. Answer: B — 2^3

Page 490: Since 8 bits are the smallest unit of information, higher units are expressible in multiples of 2^3 .

Topic: Data Units

Q11. Answer: B — 1024

Page 490: The power of two closest to a kilo (1000) is $2^{10} = 1024$.

Topic: Data Units

Q12. Answer: B — 1 million bytes or 1000 Kilo Bytes

Page 490: Mega Bytes is used in conventional sense equal to 1 million bytes or 1000 Kilo Bytes.

Topic: Data Units

Q13. Answer: B — 1000 megabytes or 1 million kilobytes

Page 490: A Giga Byte refers to 1000 megabytes approximately or 1 million kilobytes.

Topic: Data Units

Q14. Answer: B — 1 Nibble

Page 490: 4 Bit = 1 Nibble.

Topic: Data Units

Q15. Answer: B — 1 Byte

Page 490: 2 Nibble (8 Bit) = 1 Byte.

Topic: Data Units

Q16. Answer: B — 1 Kilo Byte (KB)

Page 490: 1024 Byte = 1 Kilo Byte (KB).

Topic: Data Units

Q17. Answer: B — 1 Mega Byte (MB)

Page 490: 1024 KB = 1 Mega Byte (MB).

Topic: Data Units

Q18. Answer: B — 1 Giga Byte (GB)

Page 490: 1024 MB = 1 Giga Byte (GB).

Topic: Data Units

Q19. Answer: B — 1 Tera Byte (TB)

Page 490: 1024 GB = 1 Tera Byte (TB).

Topic: Data Units

Q20. Answer: B — 1 Peta Byte (PB)

Page 490: 1024 TB = 1 Peta Byte (PB).

Topic: Data Units

Q21. Answer: B — Flip-flops

Page 490: The basic elements that can signal a 1 or 0 are called flip-flops. It is a simple electrical device.

Topic: Hardware

Q22. Answer: B — Register

Page 490: Modern computers use a very tiny set of flip-flops known as a register.

Topic: Hardware

Q23. Answer: B — Silicon

Page 490: The entire independent circuits can be designed on a small piece of semi conductor material like silicon.

Topic: Hardware

Q24. Answer: B — Integrated Circuits (IC)

Page 490: Silicon from sand is a poor conductor. Such miniature circuits are called Integrated Circuits (IC).

Topic: Hardware

Q25. Answer: B — Microprocessor

Page 490: By 1971, engineers built a complete computer on a single silicon chip called the microprocessor.

Topic: Hardware

Q26. Answer: B — Binary numbers or combinations of bits

Page 490: The computer converts all decimal numbers into binary numbers or combinations of bits.

Topic: Binary System

Q27. Answer: B — Logic gates

Page 490: The internal circuits are usually made of two or more logic gates. Logic gates generate a 1 or 0 depending on input.

Topic: Logic Gates

Q28. Answer: B — AND, OR and NOT

Page 490: The three basic logic gates are AND, OR and NOT.

Topic: Logic Gates

Q29. Answer: B — CPU, Memory Unit, Input/Output Unit

Page 490: A computer is organized into three basic units: 1. CPU, 2. Memory Unit (MU), 3. Input/Output Unit.

Topic: Hardware

Q30. Answer: B — Information processing

Page 490: The CPU is the part that performs the main function of information processing.

Topic: CPU

Q31. Answer: B — 16, 32 and even 64 bits

Page 490: While earliest machines were 8-bit, contemporary computers can work with 16, 32 and even 64 bits. This is called data width.

Topic: Hardware

Q32. Answer: B — 8

Page 490: The earliest computers were designed so that their components could work with 8 bits at a time.

Topic: Hardware

Q33. Answer: B — Frequency

Page 490: Hertz is a unit of frequency.

Topic: Hardware

Q34. Answer: B — 1 million ticks per second

Page 490: 1 MHz = 1 million 'ticks' per second.

Topic: Hardware

Q35. Answer: B — 1000 MHz

Page 490: 1 GHz = 1000 MHz.

Topic: Hardware

Q36. Answer: B — Megahertz and Gigahertz

Page 490: The ticks per second of the internal clock are measured in megahertz and gigahertz.

Topic: Hardware

Q37. Answer: B — Faster the computer

Page 490: Higher the clock-speed, faster the computer.

Topic: Hardware

Q38. Answer: B — Electrical device

Page 490: Flip-flop is a simple electrical device and can either be a '0' or a '1', meaning current flow is either inward or outward.

Topic: Hardware

Q39. Answer: B — Certain voltages

Page 490: The most important characteristic of registers is that binary digits can be stored in them using certain voltages.

Topic: Hardware

Q40. Answer: B — The number of digits used

Page 490: The binary number system is similar to decimal. Only difference is in the number of digits used (2 vs 10).

Topic: Binary System

Q41. Answer: B — 0 and 1

Page 490: Binary uses only 0 and 1, while decimal uses 0-9.

Topic: Binary System

Q42. Answer: B — 256

Page 490: 8 bits = 256 combinations, enough for all alphabets, numbers and special characters.

Topic: Binary System

Q43. Answer: B — Kilobyte, Megabyte and Gigabyte

Page 490: For complex information we need larger number of bytes - kilobyte, megabyte and gigabyte.

Topic: Data Units

Q44. Answer: B — The input

Page 490: Logic gates are components that generate a 1 or a 0 depending on the input.

Topic: Logic Gates

Q45. Answer: B — 0

Page 490: A high is symbolically represented by a '1' and a low is represented by a '0'.

Topic: Binary System

Q46. Answer: B — Binary convention (1s and 0s)

Page 490: Computer scientists use binary convention: high=1, low=0.

Topic: Binary System

Q47. Answer: B — Two parts

Page 490: The term 'binary' refers to something that has two parts.

Topic: Binary System

Q48. Answer: B — General Public Licence (open and free to all)

Page 490: Linus released Linux under General Public Licence, making it open and free to all.

Topic: Software

Q49. Answer: B — Two main parts: Hardware and Software

Page 490: A computer may be used to control purely mechanical action. It has two main parts: Hardware and Software.

Topic: Hardware

Q50. Answer: B — Output devices

Page 490: The computer supplies processed information back to the users using special output devices.

Topic: Hardware

CPU & Memory Unit – Page 491

Source: Lucent's General Knowledge – Computer

Computer: CPU Components & Memory Types

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. The CPU is the most important part of the computer. It is called:

- A. Heart of computer
- B. Brain of the computer
- C. Eyes of computer
- D. Hands of computer

Ans: B

Q2. The CPU can be divided into three main components:

- A. RAM, ROM, Cache
- B. ALU, CU and Registers
- C. Input, Output, Storage
- D. Monitor, Keyboard, Mouse

Ans: B

Q3. The Arithmetic and Logic Unit (ALU) performs:

- A. Only storage
- B. All the mathematical and logical operations on the information supplied to the CPU
- C. Only printing
- D. Only networking

Ans: B

Q4. Control Unit (CU) directs the working of the CPU by fetching instructions from:

- A. Output devices
- B. Memory
- C. Printer
- D. Monitor

Ans: B

Q5. Registers are storage locations that hold instructions or data while the CPU is using them. They consist of:

- A. Transistors
- B. Flip-flops and registers used by the CPU
- C. Capacitors
- D. Inductors

Ans: B

Q6. CPU registers are the fastest memory elements in:

- A. Hard disk
- B. The computer
- C. Internet
- D. Printer

Ans: B

Q7. Data Width refers to:

- A. Physical size of CPU
- B. Number of bits of data that can be manipulated within the CPU at one given time
- C. Weight of computer
- D. Length of wires

Ans: B

Q8. The data width of a computer is also called its: A. Byte size B. Word size C. File size D. Disk size	Ans: B
Q9. Computers have data widths ranging from: A. 1 to 4 bits B. 8 to 64 bits C. 100 to 200 bits D. 1000 bits	Ans: B
Q10. A higher data width means the CPU is capable of processing data: A. Slower B. Faster C. Same speed D. Cannot process	Ans: B
Q11. Address Range refers to the amount of memory that can be: A. Printed B. Directly read or written by the CPU C. Deleted D. Copied only	Ans: B
Q12. The speed of CPU is known as: A. Baud Rate B. Clock Speed C. Bandwidth D. Frequency	Ans: B
Q13. The computer is essentially composed of tiny devices that can be put on or off to indicate: A. Colors B. 1 or 0 C. Sound D. Light	Ans: B
Q14. To synchronize the change of all components, the CPU uses: A. External clock B. An internal clock C. Alarm D. Timer	Ans: B

Q15. The memory unit stores all instructions and data for: A. The printer B. The CPU C. The monitor D. The keyboard	Ans: B
Q16. Memory is of two kinds: A. Fast and Slow B. Primary and Secondary C. Big and Small D. Old and New	Ans: B
Q17. Primary Memory is often referred to as the working memory or: A. Storage memory B. Main memory of a computer system C. External memory D. Backup memory	Ans: B
Q18. Primary memory is temporary in nature, i.e. Data stored in primary memory is lost when: A. New data comes B. The computer is switched off C. The disk is full D. Internet disconnects	Ans: B
Q19. Primary memory is also called volatile memory. Example of primary memory is: A. Hard disk B. RAM C. CD D. USB	Ans: B
Q20. The two basic kinds of primary memory are: A. HDD and SSD B. Random Access Memory (RAM) and Read Only Memory (ROM) C. CD and DVD D. USB and SD Card	Ans: B
Q21. RAM is a: A. Read only memory B. Read/write memory C. Non-volatile memory D. Permanent memory	Ans: B

Q22. RAM is volatile, which means: A. Data is permanent B. Data is lost when power is switched off C. Data cannot be written D. Data is always backed up	Ans: B
Q23. RAM capacity greatly influences the: A. Weight of computer B. Computing ability of the computer C. Color of monitor D. Size of keyboard	Ans: B
Q24. The ROM can NOT be: A. Read B. Altered C. Powered D. Installed	Ans: B
Q25. Information is stored on the ROM at the time of its: A. First use B. Manufacture C. Installation D. Update	Ans: B
Q26. The ROM is non-volatile and retains its information even after: A. Data deletion B. The power is turned off C. Formatting D. Virus attack	Ans: B
Q27. PROM stands for: A. Permanent Read Only Memory B. Programmable Read Only Memory C. Primary Read Only Memory D. Protected Read Only Memory	Ans: B
Q28. Cache is the primary memory that stores: A. Permanent data B. Temporary data for quick access C. Only images D. Only videos	Ans: B
Q29. Secondary Memory is used to store data for: A. Very short time B. A long term C. Only during shutdown D. Only during startup	Ans: B

Q30. Secondary memory is permanent in nature, so it is also called: A. Volatile memory B. Non-volatile memory C. Temporary memory D. Flash memory	Ans: B
Q31. Common examples of secondary memory include: A. RAM and ROM B. Floppy diskettes, hard discs, magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks C. Only RAM D. Only Cache	Ans: B
Q32. Secondary memory is NOT directly accessible to the CPU. Information is moved from secondary to: A. Output device B. Primary memory first and then to the CPU C. Printer D. Internet	Ans: B
Q33. A floppy diskette is a plastic disk coated with: A. Glass B. Magnetic material C. Gold D. Silicon	Ans: B
Q34. Special devices known as disk drives are capable of reading from and writing to floppies using: A. Laser B. Special magnetic 'head' C. Electrical current D. Light beam	Ans: B
Q35. Magnetic tapes are long plastic tapes coated with: A. Silicon B. Magnetic material C. Glass D. Copper	Ans: B
Q36. A problem with magnetic tapes is that information can NOT be: A. Stored B. Accessed directly (like in floppy diskettes) C. Written D. Erased	Ans: B

Q37. The third type of medium, called fixed or hard disks, contains: A. Single disk B. Several discs of a hard material C. Paper D. Only one platter	Ans: B
Q38. Primary memory is fast but expensive. To reduce storage costs, computers also use: A. More primary memory B. Secondary memory C. No memory D. Only cache	Ans: B
Q39. Any piece of information stored on a floppy diskette can be: A. Never accessed B. Directly accessed C. Only read once D. Only written once	Ans: B
Q40. The memory unit holds instructions and data: A. During printing B. Before or after the CPU processes these C. During shutdown D. Only at startup	Ans: B
Q41. The Control Unit controls the flow of data between the ALU and: A. Printer B. Other parts of the computer C. Monitor only D. Speaker only	Ans: B
Q42. Primary memory must be able to provide data: A. Slowly B. Very quickly C. Only when requested D. Only once	Ans: B
Q43. Primary memory is directly accessible to: A. The printer B. The CPU C. The monitor D. The keyboard	Ans: B

Q44. The information on ROM might be in the form of crucial instructions that govern the: A. Printer B. Working of the computer C. Internet connection D. Speaker volume	Ans: B
Q45. In PROM, the manufacturer of computer may choose to load a program designed by his company. The computer would use this PROM like: A. RAM B. Any other ROM C. Hard disk D. Cache	Ans: B
Q46. Magnetic tapes can store far larger amounts of data than: A. Hard disks B. Floppy diskettes C. SSDs D. DVDs	Ans: B
Q47. It makes all the required calculations and processes data. This refers to: A. Memory Unit B. CPU (it makes all the required calculations and processes data) C. Output Unit D. Input Unit	Ans: B
Q48. The Storage device of a computer system is known as: A. CPU B. Memory C. ALU D. CU	Ans: B
Q49. Memory Unit can receive data, hold it and deliver according to the instructions from: A. Output unit B. Control unit C. Printer D. Monitor	Ans: B
Q50. With every tick of the internal clock, all switches change their position in: A. Random order B. Perfect harmony C. No particular order D. Reverse order	Ans: B

Answer Key & Explanations

Q1. Answer: B — Brain of the computer

Page 491: The Central Processing Unit or CPU is the most important part of the computer. It is called the brain of the computer.

Topic: CPU

Q2. Answer: B — ALU, CU and Registers

Page 491: The CPU can be divided into three main components: (a) ALU, (b) CU and (c) Registers.

Topic: CPU

Q3. Answer: B — All the mathematical and logical operations on the information supplied to the CPU

Page 491: ALU performs all the mathematical and logical operations on the information supplied to the CPU.

Topic: CPU

Q4. Answer: B — Memory

Page 491: Control Unit directs the working of the CPU. It fetches instructions (Programs) from the memory.

Topic: CPU

Q5. Answer: B — Flip-flops and registers used by the CPU

Page 491: Registers consist of flip-flops and registers used by the CPU. They are the fastest memory elements in the computer.

Topic: CPU

Q6. Answer: B — The computer

Page 491: The registers used by the CPU are the fastest memory elements in the computer.

Topic: CPU

Q7. Answer: B — Number of bits of data that can be manipulated within the CPU at one given time

Page 491: Data Width: It refers to the number of bits of data that can be manipulated within the CPU at one given time.

Topic: CPU

Q8. Answer: B — Word size

Page 491: The data width of a computer is also called its word size.

Topic: CPU

Q9. Answer: B — 8 to 64 bits

Page 491: Computers have data widths ranging from 8 to 64 bits.

Topic: CPU

Q10. Answer: B — Faster

Page 491: A higher data width means the CPU is capable of processing data faster. A CPU with higher data width is more powerful.

Topic: CPU

Q11. Answer: B — Directly read or written by the CPU

Page 491: Address range refers to the amount of memory that can be directly read or written by the CPU.

Topic: CPU

Q12. Answer: B — Clock Speed

Page 491: Clock Speed: The speed of CPU is known as Clock Speed.

Topic: CPU

Q13. Answer: B — 1 or 0

Page 491: The computer is essentially composed of tiny devices that can be put on or off to indicate 1 or 0.

Topic: CPU

Q14. Answer: B — An internal clock

Page 491: To synchronize the change of all these components the CPU uses an internal clock.

Topic: CPU

Q15. Answer: B — The CPU

Page 491: The memory unit stores all instructions and data for the CPU.

Topic: Memory

Q16. Answer: B — Primary and Secondary

Page 491: Memory is of two kinds: (a) Primary and (b) Secondary.

Topic: Memory

Q17. Answer: B — Main memory of a computer system

Page 491: Primary Memory is often referred to as the working memory or the main memory of a computer system.

Topic: Memory

Q18. Answer: B — The computer is switched off

Page 491: Primary memory is temporary. Data is lost when computer is switched off. So it is also called volatile memory.

Topic: Memory

Q19. Answer: B — RAM

Page 491: Primary memory is volatile. Example of primary memory is RAM.

Topic: Memory

Q20. Answer: B — Random Access Memory (RAM) and Read Only Memory (ROM)

Page 491: Two basic kinds of primary memory are Random Access Memory (RAM) and Read Only Memory (ROM).

Topic: Memory

Q21. Answer: B — Read/write memory

Page 491: The RAM is a read/write memory. The CPU can change the contents of the RAM at any time.

Topic: Memory

Q22. Answer: B — Data is lost when power is switched off

Page 491: RAM is volatile. In addition, RAM is volatile - data lost when computer switched off.

Topic: Memory

Q23. Answer: B — Computing ability of the computer

Page 491: The RAM capacity greatly influences the computing ability of the computer. Capacity usually measured in kilobytes and megabytes.

Topic: Memory

Q24. Answer: B — Altered

Page 491: The ROM can not be altered. Information stored on ROM at time of manufacture.

Topic: Memory

Q25. Answer: B — Manufacture

Page 491: Information is stored on the ROM at the time of its manufacture.

Topic: Memory

Q26. Answer: B — The power is turned off

Page 491: The ROM is non-volatile and retains its information even after the power is turned off.

Topic: Memory

Q27. Answer: B — Programmable Read Only Memory

Page 491: PROM (Programmable Read Only Memory) has the option of being programmed.

Topic: Memory

Q28. Answer: B — Temporary data for quick access

Page 491: Cache is the primary memory that stores temporary data for quick access.

Topic: Memory

Q29. Answer: B — A long term

Page 491: Secondary Memory: It is used to store data for a long term. It operates at a much slower rate than primary memory.

Topic: Memory

Q30. Answer: B — Non-volatile memory

Page 491: Secondary memory is permanent in nature, so it is also called non-volatile. It is also cheaper than primary memory.

Topic: Memory

Q31. Answer: B — Floppy diskettes, hard discs, magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks

Page 491: Common examples of secondary memory are floppy diskettes, hard (fixed) discs and magnetic tapes, CDs, DVDs, Memory Cards, USB Sticks.

Topic: Memory

Q32. Answer: B — Primary memory first and then to the CPU

Page 491: It is not directly accessible to CPU. Information is moved from secondary memory to primary memory first and then to CPU.

Topic: Memory

Q33. Answer: B — Magnetic material

Page 491: A floppy diskette is a plastic disk coated with magnetic material.

Topic: Memory

Q34. Answer: B — Special magnetic 'head'

Page 491: Special devices known as disk drives are capable of reading from and writing to floppies using special magnetic 'head'.

Topic: Memory

Q35. Answer: B — Magnetic material

Page 491: Magnetic tapes are long plastic tapes coated with magnetic material.

Topic: Memory

Q36. Answer: B — Accessed directly (like in floppy diskettes)

Page 491: A problem with magnetic tapes is that information can not be accessed directly as in floppy diskettes.

Topic: Memory

Q37. Answer: B — Several discs of a hard material

Page 491: The third type of medium, called fixed or hard disks, are more or less similar to floppy. One hard disk drive contains several discs of hard material.

Topic: Memory

Q38. Answer: B — Secondary memory

Page 491: Primary memory is fast but expensive. To reduce storage costs, computers also use secondary memory.

Topic: Memory

Q39. Answer: B — Directly accessed

Page 491: Any piece of information stored on a floppy diskette can be directly accessed.

Topic: Memory

Q40. Answer: B — Before or after the CPU processes these

Page 491: The memory unit holds instructions and data before or after the CPU processes these.

Topic: Memory

Q41. Answer: B — Other parts of the computer

Page 491: Control Unit controls flow of data between the ALU and other parts of the computer.

Topic: CPU

Q42. Answer: B — Very quickly

Page 491: Primary memory is directly accessible to the CPU. It must be able to provide data very quickly.

Topic: Memory

Q43. Answer: B — The CPU

Page 491: Primary memory is directly accessible to the CPU.

Topic: Memory

Q44. Answer: B — Working of the computer

Page 491: Information on ROM might be crucial instructions that govern the working of the computer.

Topic: Memory

Q45. Answer: B — Any other ROM

Page 491: In PROM, manufacturer may load a program. The computer would use this PROM like any other ROM.

Topic: Memory

Q46. Answer: B — Floppy diskettes

Page 491: Magnetic tapes can store far larger amounts of data than the floppy diskette.

Topic: Memory

Q47. Answer: B — CPU (it makes all the required calculations and processes data)

Page 491: CPU makes all the required calculations and processes data.

Topic: CPU

Q48. Answer: B — Memory

Page 491: The storage device of a computer system is known as memory.

Topic: Memory

Q49. Answer: B — Control unit

Page 491: Memory Unit can receive data, hold it and deliver according to the instructions from the control unit.

Topic: Memory

Q50. Answer: B — Perfect harmony

Page 491: With every tick of this clock all switches that need to change their position do so in perfect harmony.

Topic: CPU

Software, Shortcuts & Glossary – Page 493

Source: Lucent's General Knowledge – Computer

Computer: Software, Keyboard Shortcuts & Glossary

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Computer programs are written using special languages called:

- A. Human languages
- B. Programming languages
- C. Sign languages
- D. Musical languages

Ans: B

Q2. There are several programming languages. Each has its own:

- A. Color
- B. Grammar called its syntax
- C. Sound
- D. Shape

Ans: B

Q3. Machine level and Assembly language are examples of:

- A. High-level languages
- B. Low-level languages
- C. Web languages
- D. Database languages

Ans: B

Q4. A special program called Assembler converts all instructions into:

- A. High-level language
- B. The binary format
- C. Natural language
- D. HTML

Ans: B

Q5. Because all such instructions must finally be converted to binary form, all high-level languages have their own translator programs called:

- A. Assemblers
- B. Compilers or Interpreters
- C. Browsers
- D. Editors

Ans: B

Q6. Software relates to set of programs. The software that makes the computer hardware parts operational is called:

- A. Application software
- B. Operational (system) software
- C. Utility software
- D. Game software

Ans: B

Q7. Examples of popular high-level languages are:

- A. Assembly only
- B. C, C++, Java, FORTRAN, Pascal
- C. Machine code only
- D. Binary only

Ans: B

Q8. Operating systems like Unix, MS-DOS, Linux, Windows, Mac OS are some of: A. Application software B. The most popular operating systems used by computers C. Games D. Viruses	Ans: B
Q9. The most important system's program is: A. Word processor B. Operating system C. Calculator D. Media player	Ans: B
Q10. Another popular type of application programs is the Database Management Systems (DBMS). A common term used for application programs that permit text editing is: A. Spreadsheet B. Word processors C. Calculator D. Media player	Ans: B
Q11. Software is generally divided into two kinds: A. Big and Small B. System Software and Application Software C. Free and Paid D. Old and New	Ans: B
Q12. The keyboard shortcut Ctrl + A is used for: A. Bold B. Select All C. Copy D. Paste	Ans: B
Q13. The keyboard shortcut Ctrl + B is used for: A. Select All B. Bold C. Copy D. Italic	Ans: B
Q14. The keyboard shortcut Ctrl + C is used for: A. Bold B. Copy C. Cut D. Paste	Ans: B

Q15. The keyboard shortcut Ctrl + D is used for: A. Copy B. Font Dialogue Box C. Delete D. Duplicate	Ans: B
Q16. The keyboard shortcut Ctrl + E is used for: A. Bold B. Centre Alignment C. Exit D. Edit	Ans: B
Q17. The keyboard shortcut Ctrl + I is used for: A. Insert B. Italics C. Image D. Indent	Ans: B
Q18. The keyboard shortcut Ctrl + J is used for: A. Jump B. Justified Alignment C. Join D. Junk	Ans: B
Q19. The keyboard shortcut Ctrl + N is used for: A. Next page B. Opens New blank document C. New folder D. Number	Ans: B
Q20. The keyboard shortcut Ctrl + O is used for: A. Output B. Opens existing document C. Options D. Outline	Ans: B
Q21. The keyboard shortcut Ctrl + R is used for: A. Redo B. Right Alignment C. Replace D. Refresh	Ans: B
Q22. The keyboard shortcut Ctrl + Esc is used for: A. Exit program B. Displays the Start Menu in the Windows OS C. Escape D. Close	Ans: B

Q23. The keyboard shortcut PrtScn is used for: A. Print document B. Print screen / Take a screen shot C. Preview D. Properties	Ans: B
Q24. The keyboard shortcut Alt + Tab is used for: A. Open new tab B. Switch to next opened program C. Close tab D. New tab	Ans: B
Q25. The keyboard shortcut Ctrl + End is used to: A. Delete B. Go to end of a File/Worksheet C. End program D. Shutdown	Ans: B
Q26. The keyboard shortcut Ctrl + Home is used to: A. Go to desktop B. Go to beginning of a File/Worksheet C. Home page D. Start menu	Ans: B
Q27. The keyboard shortcut Alt + Shift + Tab is used to: A. Open new window B. Switch to previous opened Program C. Close window D. Minimize	Ans: B
Q28. The keyboard shortcut Alt + F4 is used to: A. Open file B. Close windows/Shut down C. Find D. Format	Ans: B
Q29. The keyboard shortcut Ctrl + Z is used for: A. Zoom B. Undo C. Zero D. Zip	Ans: B
Q30. The keyboard shortcut Ctrl + Y is used for: A. Yield B. Redo C. Yes D. Year	Ans: B

Q31. The keyboard shortcut Ctrl + W is used to: A. Open window B. Close File C. Write D. Wide	Ans: B
Q32. The keyboard shortcut Ctrl + P is used for: A. Paste B. Print C. Preview D. Page	Ans: B
Q33. The keyboard shortcut Ctrl + U is used for: A. Undo B. Underline C. Upload D. Update	Ans: B
Q34. The keyboard shortcut Ctrl + K is used for: A. Kill B. Hyperlink C. Keyboard D. Keep	Ans: B
Q35. The keyboard shortcut Ctrl + S is used for: A. Select B. Save C. Search D. Sort	Ans: B
Q36. The keyboard shortcut F7 is used for: A. Find B. Spell Check C. Format D. File	Ans: B
Q37. The keyboard shortcut F2 is used for: A. Find B. Edit/Rename C. Format D. File	Ans: B
Q38. The keyboard shortcut F5 is used for: A. Find B. Go to C. Format D. File	Ans: B

Q39. ALU stands for: A. Advanced Logic Unit B. Arithmetic Logic Unit C. Automated Logic Unit D. Applied Logic Unit	Ans: B
Q40. BIOS stands for: A. Binary Input Output Signal B. Basic Input Output System C. Basic Internal Operating System D. Binary Integrated Operating Software	Ans: B
Q41. Boot Loader reads the main portion of the operating system from: A. RAM B. Secondary memory C. Cache D. ROM	Ans: B
Q42. Bit is the short form of: A. Binary Term B. Binary Digit C. Binary Data D. Byte Information	Ans: B
Q43. A program stored in ROM is called: A. Software B. BIOS (firmware) C. Application D. Virus	Ans: B
Q44. Batch Processing means: A. Single processing B. Data are processed in a batch C. No processing D. Random processing	Ans: B
Q45. CU stands for: A. Central Unit B. Control Unit C. Computing Unit D. Core Unit	Ans: B
Q46. Browsing means: A. Printing documents B. Viewing webpages on the website C. Deleting files D. Writing programs	Ans: B

Q47. Byte is defined as:

- A. 4 bits
- B. One byte is a collection of 8 bits
- C. 16 bits
- D. 32 bits

Ans: B

Q48. Cell in a spreadsheet refers to:

- A. A single row
- B. Boxes created by the intersection of rows and columns
- C. A single column
- D. A header

Ans: B

Q49. A Cracker is a person who breaks into a:

- A. Bank vault
- B. Computer system to steal information as programs for unauthorized use
- C. House
- D. Car

Ans: B

Q50. Copyright means the material and information are the property of:

- A. Everyone
- B. The owner or producer
- C. The government only
- D. No one

Ans: B

Answer Key & Explanations

Q1. Answer: B — Programming languages

Page 493: Computer programs are written using special languages called programming languages.

Topic: Software

Q2. Answer: B — Grammar called its syntax

Page 493: Each programming language has its own grammar called its syntax.

Topic: Software

Q3. Answer: B — Low-level languages

Page 493: Machine level and assembly language are examples of low-level languages.

Topic: Software

Q4. Answer: B — The binary format

Page 493: A special program called Assembler converts all instructions into the binary format.

Topic: Software

Q5. Answer: B — Compilers or Interpreters

Page 493: All high-level languages have their own form of translator—programs called compilers or interpreters.

Topic: Software

Q6. Answer: B — Operational (system) software

Page 493: Software relates to set of programs. Software that makes hardware parts operational is called operational/system software. In other words, it governs the operations of a computer system.

Topic: Software

Q7. Answer: B — C, C++, Java, FORTRAN, Pascal

Page 493: Examples of popular high-level languages are C, C++, Java, FORTRAN, Pascal.

Topic: Software

Q8. Answer: B — The most popular operating systems used by computers

Page 493: Unix, MS-DOS, Linux, Windows, Mac OS are some of the most popular operating systems.

Topic: Software

Q9. Answer: B — Operating system

Page 493: The most important system's program is the operating system.

Topic: Software

Q10. Answer: B — Word processors

Page 493: A common term for application programs for text editing is word processors.

Topic: Software

Q11. Answer: B — System Software and Application Software

Page 493: Software generally divided into two kinds of programs: system software (collectively called the hardware) and application programs.

Topic: Software

Q12. Answer: B — Select All

Page 493: Ctrl + A = Select All.

Topic: Shortcuts

Q13. Answer: B — Bold

Page 493: Ctrl + B = Bold.

Topic: Shortcuts

Q14. Answer: B — Copy

Page 493: Ctrl + C = Copy.

Topic: Shortcuts

Q15. Answer: B — Font Dialogue Box

Page 493: Ctrl + D = Font Dialogue Box.

Topic: Shortcuts

Q16. Answer: B — Centre Alignment

Page 493: Ctrl + E = Centre Alignment.

Topic: Shortcuts

Q17. Answer: B — Italics

Page 493: Ctrl + I = Italics.

Topic: Shortcuts

Q18. Answer: B — Justified Alignment

Page 493: Ctrl + J = Justified Alignment.

Topic: Shortcuts

Q19. Answer: B — Opens New blank document

Page 493: Ctrl + N = Opens New blank document.

Topic: Shortcuts

Q20. Answer: B — Opens existing document

Page 493: Ctrl + O = Opens existing document.

Topic: Shortcuts

Q21. Answer: B — Right Alignment

Page 493: Ctrl + R = Right Alignment.

Topic: Shortcuts

Q22. Answer: B — Displays the Start Menu in the Windows OS

Page 493: Ctrl + Esc = Displays the Start Menu in the Windows OS.

Topic: Shortcuts

Q23. Answer: B — Print screen / Take a screen shot

Page 493: PrtScn = Print screen / Take a screen shot.

Topic: Shortcuts

Q24. Answer: B — Switch to next opened program

Page 493: Alt + Tab = Switch to next opened program.

Topic: Shortcuts

Q25. Answer: B — Go to end of a File/Worksheet

Page 493: Ctrl + End = Go to end of a File/Worksheet.

Topic: Shortcuts

Q26. Answer: B — Go to beginning of a File/Worksheet

Page 493: Ctrl + Home = Go to beginning of a File/Worksheet.

Topic: Shortcuts

Q27. Answer: B — Switch to previous opened Program

Page 493: Alt + Shift + Tab = Switch to previous opened Program.

Topic: Shortcuts

Q28. Answer: B — Close windows/Shut down

Page 493: Alt + F4 = Close windows/Shut down.

Topic: Shortcuts

Q29. Answer: B — Undo

Page 493: Ctrl + Z = Undo.

Topic: Shortcuts

Q30. Answer: B — Redo

Page 493: Ctrl + Y = Redo.

Topic: Shortcuts

Q31. Answer: B — Close File

Page 493: Ctrl + W = Close File.

Topic: Shortcuts

Q32. Answer: B — Print

Page 493: Ctrl + P = Print.

Topic: Shortcuts

Q33. Answer: B — Underline

Page 493: Ctrl + U = Underline.

Topic: Shortcuts

Q34. Answer: B — Hyperlink

Page 493: Ctrl + K = Hyperlink.

Topic: Shortcuts

Q35. Answer: B — Save

Page 493: Ctrl + S = Save.

Topic: Shortcuts

Q36. Answer: B — Spell Check

Page 493: F7 = Spell Check.

Topic: Shortcuts

Q37. Answer: B — Edit/Rename

Page 493: F2 = Edit/Rename.

Topic: Shortcuts

Q38. Answer: B — Go to

Page 493: F5 = Go to.

Topic: Shortcuts

Q39. Answer: B — Arithmetic Logic Unit

Page 493: ALU: It stands for Arithmetic Logic Unit. All calculations in computer are done here.

Topic: Glossary

Q40. Answer: B — Basic Input Output System

Page 493: BIOS: It stands for Basic Input Output System.

Topic: Glossary

Q41. Answer: B — Secondary memory

Page 493: Boot Loader: It reads the main portion of the operating system from secondary memory.

Topic: Glossary

Q42. Answer: B — Binary Digit

Page 493: Bit: It is the short form of Binary Digit.

Topic: Glossary

Q43. Answer: B — BIOS (firmware)

Page 493: BIOS/program is stored in ROM.

Topic: Glossary

Q44. Answer: B — Data are processed in a batch

Page 493: Batch Processing: Data are processed in a batch.

Topic: Glossary

Q45. Answer: B — Control Unit

Page 493: CU: It stands for Control Unit. It controls the computer system.

Topic: Glossary

Q46. Answer: B — Viewing webpages on the website

Page 493: Browsing: Viewing webpages on the website.

Topic: Glossary

Q47. Answer: B — One byte is a collection of 8 bits

Page 493: Byte: One byte is a collection of 8 bits.

Topic: Glossary

Q48. Answer: B — Boxes created by the intersection of rows and columns

Page 493: Cell: Cells are boxes created by the intersection of rows and columns.

Topic: Glossary

Q49. Answer: B — Computer system to steal information as programs for unauthorized use

Page 493: Cracker: A cracker is a person who breaks into a computer system to steal information as programs for unauthorized use.

Topic: Glossary

Q50. Answer: B — The owner or producer

Page 493: Copyright: It means the material and information are the personal property of the owner or producer.

Topic: Glossary

Glossary & Abbreviations – Page 494

Source: Lucent's General Knowledge – Computer

Computer: Glossary, Definitions & Abbreviations

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. DBMS stands for:

- A. Data Binary Management System
- B. Database Management System
- C. Digital Base Management System
- D. Data Block Management System

Ans: B

Q2. DHTML stands for:

- A. Data Hyper Text Markup Language
- B. Dynamic Hyper Text Markup Language
- C. Direct Hyper Text Markup Language
- D. Digital Hyper Text Markup Language

Ans: B

Q3. DOS stands for:

- A. Data Operating System
- B. Disk Operating System
- C. Digital Operating System
- D. Direct Operating System

Ans: B

Q4. HTML stands for:

- A. High Text Markup Language
- B. Hyper Text Markup Language
- C. Home Text Markup Language
- D. Host Text Markup Language

Ans: B

Q5. IAB stands for Internet Activities Board, formed in 1983, now called:

- A. Internet Advisory Board
- B. Internet Architectural Board
- C. Internet Access Board
- D. Internet Application Board

Ans: B

Q6. IAS stands for Internet Application Software also known as:

- A. Search engines
- B. Browsers
- C. Databases
- D. Compilers

Ans: B

Q7. ISP stands for:

- A. Internet Software Provider
- B. Internet Service Provider
- C. Internet System Protocol
- D. Internet Security Provider

Ans: B

Q8. Internet Society was formed in 1992 to promote: A. Television B. The use of Internet C. Radio D. Newspapers	Ans: B
Q9. A Hacker is a person who breaks into a computer system to: A. Repair it B. Get access to the information stored there C. Clean it D. Update it	Ans: B
Q10. GML stands for: A. General Machine Language B. General Mark-up Language C. Global Machine Language D. Generic Markup Logic	Ans: B
Q11. MAC OS is an Operating System used in Macintosh Computer, developed by: A. Microsoft B. Apple C. Google D. IBM	Ans: B
Q12. Linux is a Unix like Operating System with: A. Command line only B. Graphical user interface C. No interface D. Voice interface	Ans: B
Q13. Modem is a device to connect different computer systems to the internet using: A. Radio waves B. Telephone or cable lines C. Satellite only D. Bluetooth	Ans: B
Q14. Piracy is the illegal reproduction and distribution of: A. Books B. Software application C. Food D. Clothes	Ans: B
Q15. Primary Key is a field with: A. Any value B. Unique value for each record C. Same value for all records D. No value	Ans: B

Q16. RDBMS stands for: A. Random Database Management System B. Relational Database Management System C. Remote Database Management System D. Real Database Management System	Ans: B
Q17. SOLARIS is a Unix like Operating System developed by: A. Microsoft B. Sun Microsystems C. Apple D. Google	Ans: B
Q18. Tag is part of HTML. It determines: A. Color only B. The way the browser displays text in a Web Page C. Sound D. Speed	Ans: B
Q19. Unix is a powerful multiuser operating system that uses: A. Graphical interface only B. Command line user interface C. Voice interface D. Touch interface	Ans: B
Q20. CDAC stands for: A. Centre for Data and Computing B. Centre for Development of Advanced Parallel Computing C. Computer Development and Control D. Central Data Analysis Centre	Ans: B
Q21. C-DOT stands for: A. Centre for Development of Technology B. Centre for Development Of Telematics C. Computer Department of Technology D. Central Data Office Technology	Ans: B
Q22. HTTP stands for: A. High Text Transfer Protocol B. Hyper Text Transfer Protocol C. Home Text Transfer Protocol D. Host Text Transfer Protocol	Ans: B
Q23. ROM stands for: A. Random Output Memory B. Read Only Memory C. Run Only Memory D. Real Output Memory	Ans: B

Q24. RAM stands for: A. Read All Memory B. Random Access Memory C. Run Access Memory D. Real Access Memory	Ans: B
Q25. BIOS stands for: A. Binary Input Output Signal B. Basic Input-Output System C. Base Internal Operating System D. Binary Integrated Output System	Ans: B
Q26. MODEM stands for: A. Modular Device Manager B. Modulation-Demodulation C. Modern Device Machine D. Module-Demonstration	Ans: B
Q27. CAD stands for: A. Computer Automated Data B. Computer Aided Design C. Central Access Device D. Control and Design	Ans: B
Q28. LAN stands for: A. Large Area Network B. Local Area Network C. Long Access Network D. Linear Area Network	Ans: B
Q29. WAN stands for: A. Wireless Area Network B. Wide Area Network C. World Access Network D. Web Area Network	Ans: B
Q30. MAN stands for: A. Main Area Network B. Metropolitan Area Network C. Module Access Network D. Multi Area Network	Ans: B
Q31. ENIAC stands for: A. Electronic Network Integrated And Calculator B. Electronic Numerical Integrator And Calculator C. Electronic Number Input And Computer D. Enhanced Numerical Integrator And Calculator	Ans: B

Q32. FORTRAN stands for: A. Form Translation B. Formula Translation C. Forward Translation D. Format Translation	Ans: B
Q33. COBOL stands for: A. Computer Oriented Binary Language B. Common Business Oriented Language C. Central Office Business Language D. Computer Object Based Language	Ans: B
Q34. FAX stands for: A. Fast Access Xerox B. Fascimile Automated Xerox C. File Access Exchange D. Fax Automated Transfer	Ans: B
Q35. FLOPS stands for: A. Floating Line Operations Per Second B. Floating Operations Per Second C. Fast Logic Operations Per Second D. File Logic Operations Per Second	Ans: B
Q36. HLL stands for: A. High Logic Language B. High Level Language C. Home Level Language D. Hybrid Logic Language	Ans: B
Q37. IBM stands for: A. Indian Business Machine B. International Business Machine C. Integrated Binary Machine D. Internal Business Module	Ans: B
Q38. IC stands for: A. Internal Computer B. Integrated Circuit C. Input Controller D. Internet Connection	Ans: B
Q39. IP stands for: A. Internal Processing B. Internet Protocol C. Input Port D. Integrated Program	Ans: B

Q40. PSTN stands for: A. Private Switched Telephone Network B. Public Switched Telephone Network C. Protected System Telephone Network D. Primary Switched Telephone Network	Ans: B
Q41. CDMA stands for: A. Computer Division Multiple Access B. Code Division Multiple Access C. Central Data Multiple Access D. Common Device Multiple Access	Ans: B
Q42. DTP stands for: A. Data Transfer Protocol B. Desk Top Publishing C. Digital Transfer Processing D. Direct Text Processing	Ans: B
Q43. DTS stands for: A. Digital Transfer System B. Desk Top System C. Data Translation Service D. Direct Technology System	Ans: B
Q44. CAM stands for: A. Computer Access Module B. Computer Aided Manufacturing C. Central Automated Machine D. Control and Management	Ans: B
Q45. CATScan stands for: A. Computer Automated Testing Scan B. Computerised Axial Tomography Scan C. Central Access Technology Scan D. Computer Analysis Testing Scan	Ans: B
Q46. COMAL stands for: A. Computer Oriented Machine Language B. Common Algorithmic Language C. Central Object Management Language D. Computer Analysis Language	Ans: B
Q47. DOS (as abbreviation) stands for: A. Data Operating System B. Disc Operating System C. Digital Output System D. Direct Operating Software	Ans: B

Q48. LDU stands for:

- A. Large Data Unit
- B. Liquid Display Unit
- C. Light Data Unit
- D. Linear Display Unit

Ans: B

Q49. CPU stands for:

- A. Computer Processing Unit
- B. Central Processing Unit
- C. Core Processing Unit
- D. Control Processing Unit

Ans: B

Q50. GAIS stands for:

- A. Global Access Internet System
- B. Gateway Internet Access Service
- C. General Automated Internet Service
- D. Government Access Internet Service

Ans: B

Answer Key & Explanations

Q1. Answer: B — Database Management System

Page 494: DBMS: It stands for Database Management System. It is a software package to manage database.

Topic: Glossary

Q2. Answer: B — Dynamic Hyper Text Markup Language

Page 494: DHTML: It stands for Dynamic Hyper Text Markup Language. Used to create dynamic content on web pages.

Topic: Glossary

Q3. Answer: B — Disk Operating System

Page 494: DOS: It stands for Disk Operating System. It is an operating system.

Topic: Glossary

Q4. Answer: B — Hyper Text Markup Language

Page 494: HTML: It stands for Hyper Text Markup Language. It is used to create web-pages.

Topic: Glossary

Q5. Answer: B — Internet Architectural Board

Page 494: IAB: Internet Activities Board formed in 1983, now called Internet Architectural Board.

Topic: Glossary

Q6. Answer: B — Browsers

Page 494: IAS: Internet Application Software also known as Browsers.

Topic: Glossary

Q7. Answer: B — Internet Service Provider

Page 494: ISP: It stands for Internet Service Provider.

Topic: Glossary

Q8. Answer: B — The use of Internet

Page 494: Internet Society: It was formed in 1992 to promote the use of Internet.

Topic: Glossary

Q9. Answer: B — Get access to the information stored there

Page 494: Hacker: A person who breaks into a computer system to get access to the information stored there.

Topic: Glossary

Q10. Answer: B — General Mark-up Language

Page 494: GML: It stands for General Mark-up Language.

Topic: Glossary

Q11. Answer: B — Apple

Page 494: MAC OS: Operating System used in Macintosh Computer, developed by Apple.

Topic: Glossary

Q12. Answer: B — Graphical user interface

Page 494: Linux: It is a Unix like Operating System with graphical user interface.

Topic: Glossary

Q13. Answer: B — Telephone or cable lines

Page 494: Modem: device to connect different computer systems to the internet using telephone or cable lines.

Topic: Glossary

Q14. Answer: B — Software application

Page 494: Piracy: illegal reproduction and distribution of software application.

Topic: Glossary

Q15. Answer: B — Unique value for each record

Page 494: Primary Key: It is a field with unique value for each record.

Topic: Glossary

Q16. Answer: B — Relational Database Management System

Page 494: RDBMS: Relational Database Management System.

Topic: Abbreviations

Q17. Answer: B — Sun Microsystems

Page 494: SOLARIS: It is a Unix like Operating System developed by Sun Microsystems.

Topic: Glossary

Q18. Answer: B — The way the browser displays text in a Web Page

Page 494: Tag: It is part of HTML. It determines the way the browser displays text in a Web Page.

Topic: Glossary

Q19. Answer: B — Command line user interface

Page 494: Unix: powerful multiuser operating system. It uses command line user interface.

Topic: Glossary

Q20. Answer: B — Centre for Development of Advanced Parallel Computing

Page 494: CDAC: Centre for Development of Advanced Parallel Computing.

Topic: Abbreviations

Q21. Answer: B — Centre for Development Of Telematics

Page 494: C-DOT: Centre for Development Of Telematics.

Topic: Abbreviations

Q22. Answer: B — Hyper Text Transfer Protocol

Page 494: HTTP: Hyper Text Transfer Protocol.

Topic: Abbreviations

Q23. Answer: B — Read Only Memory

Page 494: ROM: Read Only Memory.

Topic: Abbreviations

Q24. Answer: B — Random Access Memory

Page 494: RAM: Random Access Memory.

Topic: Abbreviations

Q25. Answer: B — Basic Input-Output System

Page 494: BIOS: Basic Input-Output System.

Topic: Abbreviations

Q26. Answer: B — Modulation-Demodulation

Page 494: MODEM: Modulation-Demodulation.

Topic: Abbreviations

Q27. Answer: B — Computer Aided Design

Page 494: CAD: Computer Aided Design.

Topic: Abbreviations

Q28. Answer: B — Local Area Network

Page 494: LAN: Local Area Network.

Topic: Abbreviations

Q29. Answer: B — Wide Area Network

Page 494: WAN: Wide Area Network.

Topic: Abbreviations

Q30. Answer: B — Metropolitan Area Network

Page 494: MAN: Metropolitan Area Network.

Topic: Abbreviations

Q31. Answer: B — Electronic Numerical Integrator And Calculator

Page 494: ENIAC: Electronic Numerical Integrator And Calculator.

Topic: Abbreviations

Q32. Answer: B — Formula Translation

Page 494: FORTRAN: Formula Translation.

Topic: Abbreviations

Q33. Answer: B — Common Business Oriented Language

Page 494: COBOL: Common Business Oriented Language.

Topic: Abbreviations

Q34. Answer: B — Fascimile Automated Xerox

Page 494: FAX: Fascimile Automated Xerox.

Topic: Abbreviations

Q35. Answer: B — Floating Operations Per Second

Page 494: FLOPS: Floating Operations Per Second.

Topic: Abbreviations

Q36. Answer: B — High Level Language

Page 494: HLL: High Level Language.

Topic: Abbreviations

Q37. Answer: B — International Business Machine

Page 494: IBM: International Business Machine.

Topic: Abbreviations

Q38. Answer: B — Integrated Circuit

Page 494: IC: Integrated Circuit.

Topic: Abbreviations

Q39. Answer: B — Internet Protocol

Page 494: IP: Internet Protocol.

Topic: Abbreviations

Q40. Answer: B — Public Switched Telephone Network

Page 494: PSTN: Public Switched Telephone Network.

Topic: Abbreviations

Q41. Answer: B — Code Division Multiple Access

Page 494: CDMA: Code Division Multiple Access.

Topic: Abbreviations

Q42. Answer: B — Desk Top Publishing

Page 494: DTP: Desk Top Publishing.

Topic: Abbreviations

Q43. Answer: B — Desk Top System

Page 494: DTS: Desk Top System.

Topic: Abbreviations

Q44. Answer: B — Computer Aided Manufacturing

Page 494: CAM: Computer Aided Manufacturing.

Topic: Abbreviations

Q45. Answer: B — Computerised Axial Tomography Scan

Page 494: CATScan: Computerised Axial Tomography Scan.

Topic: Abbreviations

Q46. Answer: B — Common Algorithmic Language

Page 494: COMAL: Common Algorithmic Language.

Topic: Abbreviations

Q47. Answer: B — Disc Operating System

Page 494: DOS: Disc Operating System.

Topic: Abbreviations

Q48. Answer: B — Liquid Display Unit

Page 494: LDU: Liquid Display Unit.

Topic: Abbreviations

Q49. Answer: B — Central Processing Unit

Page 494: CPU: Central Processing Unit.

Topic: Abbreviations

Q50. Answer: B — Gateway Internet Access Service

Page 494: GAIS: Gateway Internet Access Service.

Topic: Abbreviations

Internet & Memorable Facts – Page 495

Source: Lucent's General Knowledge – Computer

Computer: Internet, Web, Operating Systems & Memorable Facts

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. LISP stands for:

- A. Linear System Processing
- B. List Processing
- C. Logic Internal System Processing
- D. Light Speed Processing

Ans: B

Q2. MICR stands for:

- A. Machine Internal Character Recognition
- B. Magnetic Ink Character Recognition/Reader
- C. Micro Input Character Reader
- D. Module Ink Character Recognition

Ans: B

Q3. MIPS stands for:

- A. Mega Instructions Per Second
- B. Million Instructions Per Second
- C. Micro Input Per Second
- D. Module Instructions Per Second

Ans: B

Q4. UPS stands for:

- A. Universal Processing System
- B. Uninterruptable Power Supply
- C. United Power System
- D. Ultra Power Source

Ans: B

Q5. VDU stands for:

- A. Video Digital Unit
- B. Visual Display Unit
- C. Virtual Data Unit
- D. Voice Display Unit

Ans: B

Q6. VLSI stands for:

- A. Very Light Scale Integration
- B. Very Large Scale Integration
- C. Virtual Large System Integration
- D. Variable Logic System Integration

Ans: B

Q7. WWW stands for:

- A. Wide Web World
- B. World Wide Web
- C. Web World Wide
- D. Wireless World Web

Ans: B

Q8. MS Access is a powerful program to create and manage: A. Spreadsheets B. Databases C. Presentations D. Graphics	Ans: B
Q9. Internet is the network of computer networks with millions of computers attached to it. It was originally developed by: A. UK Ministry of Defence B. US Department of Defence C. Indian Government D. NASA	Ans: B
Q10. 'The World Wide Web' was invented by: A. Bill Gates B. Tim Berners-Lee C. Steve Jobs D. Mark Zuckerberg	Ans: B
Q11. FAT 32 and NTFS are the two major file systems in: A. Linux B. Microsoft Windows C. Mac OS D. Android	Ans: B
Q12. Websites are files in servers, which are: A. Weak computers B. Powerful computers C. Old computers D. Broken computers	Ans: B
Q13. The collection of all websites is called: A. Internet B. World Wide Web or WWW C. LAN D. WAN	Ans: B
Q14. Hyper text was first coined by Ted Nelson in: A. 1980s B. 1960s C. 1990s D. 1970s	Ans: B

Q15. HTML is a markup language used to create Web Pages. It uses commands called: A. Functions B. Tags C. Classes D. Modules	Ans: B
Q16. HTML documents are viewed in: A. Text editors B. Web browsers C. Compilers D. Databases	Ans: B
Q17. Most Popular Mobile Operating System is: A. iOS B. Android C. Windows D. Linux	Ans: B
Q18. The correct order of Android versions (old to new) starting from Cupcake is: Cupcake → Donut → Eclairs → Froyo → Ginger Bread → Honeycomb → Ice Cream Sandwich → Jelly Beans → Kit-kat → Lollipop → Marshmallow → Nougat → Oreo → Pie → Android 10 →: A. Android 9 B. Android 11 C. Android 8 D. Android 7	Ans: B
Q19. Some popular operating systems are: A. Only Windows B. DOS, UNIX, Windows, LINUX, Mac OS etc. C. Only Linux D. Only Mac OS	Ans: B
Q20. Most Popular Social Networking Sites include: A. Only WhatsApp B. Facebook, Twitter, LinkedIn and Instagram C. Only Telegram D. Only YouTube	Ans: B
Q21. Speed of Computer (or Clock speed) is measured in: A. Watts B. Hertz (Hz) C. Meters D. Kilograms	Ans: B

Q22. Speed of Modern Processors are often measured in: A. MHz B. Gigahertz (GHz) C. KHz D. Hz	Ans: B
Q23. Speed of communication over a data channel is measured in: A. Hertz B. Baud C. Watts D. Volts	Ans: B
Q24. DHCP (Dynamic Host Configuration Protocol) is a network management protocol used on TCP/IP networks to provide: A. Speed increase B. Quick, automatic and central management for distribution of IP addresses C. Virus protection D. Data backup	Ans: B
Q25. A Social Network can be represented as: A. Table B. Graph C. List D. Array	Ans: B
Q26. IPv6 protocol defines an IP address of: A. 32 bit B. 128 bit C. 64 bit D. 256 bit	Ans: B
Q27. Junk email is called: A. Virus B. Spam C. Bug D. Worm	Ans: B
Q28. Coding is the process of writing out: A. Letters B. Computer instructions C. Books D. Music	Ans: B

Q29. Four major digital initiatives to push e-education by the govt. are Swayam, Swayam Prabha, National Academic Depository and: A. PARAM B. National Digital Library C. ENIAC D. UNIVAC	Ans: B
Q30. Oracle is a: A. Programming language B. Database software C. Operating system D. Hardware	Ans: B
Q31. Most popular online Shopping Websites include: A. Only Google B. Amazon.com, Flipkart.com and ebay.com C. Only Wikipedia D. Only YouTube	Ans: B
Q32. A database is a collection of: A. Random files B. Interrelated data C. Images only D. Videos only	Ans: B
Q33. A relational database is a collection of formally described tables from which data can be accessed in many different ways without having to: A. Open it B. Reorganize the database tables C. Delete anything D. Connect to internet	Ans: B
Q34. The Windows Explorer program is more efficient for viewing folders in: A. Mac B. Windows C. Linux D. DOS	Ans: B
Q35. Windows Explorer is divided vertically into: A. Three panes B. Two panes C. Four panes D. One pane	Ans: B

Q36. The left side pane of Windows Explorer displays: A. File contents B. Disk drives and folders in a hierarchical order C. Programs only D. Pictures only	Ans: B
Q37. The process of linking text values in a series within a formula is called: A. Merging B. Concatenation C. Joining D. Combining	Ans: B
Q38. Data processing involves some activities like data computing, data manipulation and: A. Data deletion B. Information management C. Data corruption D. Data hiding	Ans: B
Q39. An operating system is an interface between: A. Two computers B. The human user and the computer hardware C. Two programs D. Two networks	Ans: B
Q40. An operating system manages computer resources and performs: A. Only one function B. Different functions C. No function D. Only display	Ans: B
Q41. An operating system is responsible for: A. Only games B. Process management, file management, etc. C. Only printing D. Only internet	Ans: B
Q42. Data are raw facts and: A. Programs B. Figures C. Software D. Hardware	Ans: B

Q43. Hardware comprises of the physical units of a: A. Network B. Computer system C. Internet D. Database	Ans: B
Q44. Software is a set of: A. Hardware B. Programs C. Cables D. Chips	Ans: B
Q45. Both hardware and software together make a computer system: A. Broken B. Functional C. Slow D. Heavy	Ans: B
Q46. SNOBOL stands for: A. System Notation for Binary Logic B. StriNg Oriented symBOLic Language C. Standard Network Binary Logic D. Simple Notation for Object Logic	Ans: B
Q47. NICNET stands for: A. Network Internet Computer Network B. National Informatics Centre Network C. National Internet Connection Network D. New Information Computer Network	Ans: B
Q48. OMR stands for: A. Optical Machine Reader B. Optical Mark Reader/Recognition C. Online Mark Reader D. Object Mark Recognition	Ans: B
Q49. MOPS stands for: A. Million Output Per Second B. Million Operations Per Second C. Micro Output Per Second D. Module Operations Per Second	Ans: B
Q50. Websites contain pages called: A. Documents B. Web Pages C. Files D. Folders	Ans: B

Answer Key & Explanations

Q1. Answer: B — List Processing

Page 495: LISP: List Processing.

Topic: Abbreviations

Q2. Answer: B — Magnetic Ink Character Recognition/Reader

Page 495: MICR: Magnetic Ink Character Recognition/Reader.

Topic: Abbreviations

Q3. Answer: B — Million Instructions Per Second

Page 495: MIPS: Million Instructions Per Second.

Topic: Abbreviations

Q4. Answer: B — Uninterruptable Power Supply

Page 495: UPS: Uninterruptable Power Supply.

Topic: Abbreviations

Q5. Answer: B — Visual Display Unit

Page 495: VDU: Visual Display Unit.

Topic: Abbreviations

Q6. Answer: B — Very Large Scale Integration

Page 495: VLSI: Very Large Scale Integration.

Topic: Abbreviations

Q7. Answer: B — World Wide Web

Page 495: WWW: World Wide Web.

Topic: Abbreviations

Q8. Answer: B — Databases

Page 495: MS Access is a powerful program to create and manage our databases.

Topic: Facts

Q9. Answer: B — US Department of Defence

Page 495: Internet was originally developed by the US Department of Defence.

Topic: Internet

Q10. Answer: B — Tim Berners-Lee

Page 495: 'The World Wide Web' was invented by Tim Berners-Lee.

Topic: Internet

Q11. Answer: B — Microsoft Windows

Page 495: FAT 32 and NTFS are the two major file systems in Microsoft Windows.

Topic: Facts

Q12. Answer: B — Powerful computers

Page 495: Websites are files in servers, which are powerful computers.

Topic: Internet

Q13. Answer: B — World Wide Web or WWW

Page 495: The collection of all websites is called World Wide Web or WWW.

Topic: Internet

Q14. Answer: B — 1960s

Page 495: Hyper text was first coined by Ted Nelson in 1960s.

Topic: Internet

Q15. Answer: B — Tags

Page 495: HTML is a markup language used to create Web Pages. It uses commands called Tags.

Topic: Internet

Q16. Answer: B — Web browsers

Page 495: Text editors are used to compose HTML documents. HTML documents are viewed in Web browsers.

Topic: Internet

Q17. Answer: B — Android

Page 495: Most Popular Mobile Operating System is Android, followed by iOS & Windows.

Topic: Facts

Q18. Answer: B — Android 11

Page 495: Android versions end with: ...Oreo → Pie → Android 10 → Android 11.

Topic: Facts

Q19. Answer: B — DOS, UNIX, Windows, LINUX, Mac OS etc.

Page 495: Some popular operating systems are DOS, UNIX, Windows, LINUX, Mac OS etc.

Topic: Facts

Q20. Answer: B — Facebook, Twitter, Linkedin and Instagram

Page 495: Most Popular Social Networking Sites: Facebook, Twitter, Linkedin and Instagram.

Topic: Facts

Q21. Answer: B — Hertz (Hz)

Page 495: Speed of Computer (or Clock speed) is measured in Hertz (Hz).

Topic: Facts

Q22. Answer: B — Gigahertz (GHz)

Page 495: Speed of Modern Processors are often measured in Gigahertz (GHz).

Topic: Facts

Q23. Answer: B — Baud

Page 495: Speed of communication over a data channel is measured in 'Baud'.

Topic: Facts

Q24. Answer: B — Quick, automatic and central management for distribution of IP addresses

Page 495: DHCP provides quick, automatic and central management for distribution of IP addresses within the network.

Topic: Internet

Q25. Answer: B — Graph

Page 495: A Social Network can be represented as graph.

Topic: Internet

Q26. Answer: B — 128 bit

Page 495: IPv6 protocol defines on IP address of 128 bit.

Topic: Internet

Q27. Answer: B — Spam

Page 495: Junk email is called 'spam'.

Topic: Internet

Q28. Answer: B — Computer instructions

Page 495: Coding is the process of writing out computer instructions.

Topic: Facts

Q29. Answer: B — National Digital Library

Page 495: Four major digital initiatives: Swayam, Swayam Prabha, National Academic Depository and National Digital Library.

Topic: Facts

Q30. Answer: B — Database software

Page 495: Oracle is a database software.

Topic: Facts

Q31. Answer: B — Amazon.com, Flipkart.com and ebay.com

Page 495: Most popular online Shopping Websites: Amazon.com, Flipkart.com and ebay.com.

Topic: Facts

Q32. Answer: B — Interrelated data

Page 495: A database is a collection of interrelated data.

Topic: Facts

Q33. Answer: B — Reorganize the database tables

Page 495: A relational database is organized as set of formally described tables from which data can be accessed without reorganizing.

Topic: Facts

Q34. Answer: B — Windows

Page 495: The Windows Explorer program is more efficient for viewing folders in Windows.

Topic: Facts

Q35. Answer: B — Two panes

Page 495: Windows Explorer is divided vertically into two parts or two panes.

Topic: Facts

Q36. Answer: B — Disk drives and folders in a hierarchical order

Page 495: The left side pane displays disk drives and folders in a hierarchical order.

Topic: Facts

Q37. Answer: B — Concatenation

Page 495: The process of linking text values in a series within a formula is called 'concatenation'.

Topic: Facts

Q38. Answer: B — Information management

Page 495: Data processing involves some activities like data computing, data manipulation and information management.

Topic: Facts

Q39. Answer: B — The human user and the computer hardware

Page 495: An operating system is an interface between the human user and the computer hardware.

Topic: Facts

Q40. Answer: B — Different functions

Page 495: An operating system manages computer resources and performs different functions.

Topic: Facts

Q41. Answer: B — Process management, file management, etc.

Page 495: An operating system is responsible for process management, file management, etc.

Topic: Facts

Q42. Answer: B — Figures

Page 495: Data are raw facts and figures.

Topic: Facts

Q43. Answer: B — Computer system

Page 495: Hardware comprises of the physical units of a computer system.

Topic: Facts

Q44. Answer: B — Programs

Page 495: Software is a set of programs.

Topic: Facts

Q45. Answer: B — Functional

Page 495: Both hardware and software together make a computer system functional.

Topic: Facts

Q46. Answer: B — StriNg Oriented symBOLic Language

Page 495: SNOBOL: StriNg Oriented symBOLic Language.

Topic: Abbreviations

Q47. Answer: B — National Informatics Centre Network

Page 495: NICNET: National Informatics Centre Network.

Topic: Abbreviations

Q48. Answer: B — Optical Mark Reader/Recognition

Page 495: OMR: Personal Computer Disk Operation System... wait. OMR: Optical Mark Reader/Recognition (from abbreviation table).

Topic: Abbreviations

Q49. Answer: B — Million Operations Per Second

Page 495: MOPS: Million Operations Per Second.

Topic: Abbreviations

Q50. Answer: B — Web Pages

Page 495: Websites contain pages called Web Pages.

Topic: Internet

Cloud Computing & Govt Initiatives – Page 496

Source: Lucent's General Knowledge – Computer

Computer: Cloud Computing, Encryption, Government Digital Initiatives

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Cloud Computing is on demand delivery of computer power, database storage, applications and other IT resources through:

- A. CD/DVD
- B. A cloud services platform mainly via the internet
- C. Floppy disk
- D. USB drive

Ans: B

Q2. Cloud Computing is a distributed computer architecture that provides software, infrastructure and:

- A. Hardware only
- B. Platforms just as required by applications/users
- C. Nothing
- D. Only storage

Ans: B

Q3. Spreadsheet/MS Excel is appropriate for performing:

- A. Only typing
- B. Numerical and statistic calculations
- C. Only drawing
- D. Only printing

Ans: B

Q4. CAD/CAM systems are based on:

- A. Simple text
- B. Interactive Computer Graphics
- C. Audio signals
- D. Video only

Ans: B

Q5. Data Encryption Standard (DES) is an implementation of a Feistel Cipher. It uses how many round Feistel structure?

- A. 8
- B. 16
- C. 32
- D. 64

Ans: B

Q6. The block size of DES (Data Encryption Standard) is:

- A. 32 bits
- B. 64 bits
- C. 128 bits
- D. 256 bits

Ans: B

Q7. The Banker algorithm is a resource allocation and deadlock avoidance algorithm developed by: A. Tim Berners-Lee B. Edsger W. Dijkstra C. Charles Babbage D. Alan Turing	Ans: B
Q8. Gateway is used to connect two different networks using: A. Same protocol B. Different protocols C. No protocol D. Single cable	Ans: B
Q9. A half byte is known as 'Nibble' and consists of: A. 8 bits B. 4 bits C. 2 bits D. 16 bits	Ans: B
Q10. MS Excel from Version 8.0 to 11.0 could handle 65,536 rows and: A. 128 columns B. 256 columns C. 512 columns D. 1024 columns	Ans: B
Q11. MS Excel Version 12.0 can handle 10,48,576 rows and: A. 256 columns B. 16,384 columns C. 1024 columns D. 65,536 columns	Ans: B
Q12. The robot named 'Pepe' was recently used by pupils at a primary school in: A. Tamil Nadu B. Kerala C. Karnataka D. Delhi	Ans: B
Q13. India's first indigenously developed payment ecosystem for transport is based on: A. UPI B. National Common Mobility Card (NCMC) standards C. Credit cards only D. Cash only	Ans: B

Q14. Mobile App 'cVIGIL' has been launched by: A. RBI B. Election Commission of India C. NITI Aayog D. Supreme Court	Ans: B
Q15. VISHWAS stands for Video Integration and State Wide Advanced: A. System B. Security C. Software D. Service	Ans: B
Q16. VISHWAS project aims at improving: A. Education B. Law and Order Management, Proactive traffic management and control, Crime detection C. Agriculture D. Healthcare	Ans: B
Q17. A gutter margin adds extra space to the side margin, top margin, or inside margins of a document that you plan to: A. Delete B. Bind C. Email D. Copy	Ans: B
Q18. An integrated circuit (IC) is small chip that can function as an amplifier, oscillator, timer, microprocessor, or even computer memory. It is usually made of: A. Copper B. Silicon C. Gold D. Iron	Ans: B
Q19. WorldWideWeb: First Web browser was developed in the mid and late 1980s. Tim Berners-Lee is credited with developing it in: A. 1985 B. 1990 C. 1995 D. 2000	Ans: B
Q20. The first web browser was called WorldWideWeb and was later renamed: A. Chrome B. Nexus C. Firefox D. Safari	Ans: B

Q21. Windows 10 was released as a preview on: A. October 5, 2021 B. 1st October, 2014 C. March 4, 2019 D. June 1, 2015	Ans: B
Q22. Windows 11 was released on: A. 1st October, 2014 B. 5th October, 2021 C. March 4, 2019 D. January 1, 2022	Ans: B
Q23. e-Pramaan is a framework for: A. e-Commerce B. e-Authentication for public services C. e-Learning D. e-Banking	Ans: B
Q24. Online e-Sign (e-Hastakshar) facility was formally launched in: A. 2010 B. 2015 C. 2020 D. 2005	Ans: B
Q25. e-Sign service facilitates instant signing of documents online by citizens in: A. Illegal form B. A legally acceptable form C. Temporary form D. Unverified form	Ans: B
Q26. National Knowledge Network (NKN) was established in: A. 2005 B. 2010 C. 2015 D. 2020	Ans: B
Q27. The objective of NKN is to interconnect all institutions of higher learning and research with: A. Low-speed network B. A high-speed data communication network C. Radio signals D. Television	Ans: B

Q28. NKN is the only network globally that carries research and evaluation, internet and e-Governance traffic as: A. One vertical B. Three independent verticals under one umbrella C. Separate networks D. Two verticals	Ans: B
Q29. UMANG stands for: A. Universal Mobile Access for New Generation B. Unified Mobile Application for New-Age Governance C. United Mobile App for National Government D. Universal Management Application for Governance	Ans: B
Q30. UMANG was developed and launched in: A. 2015 B. 2017 C. 2019 D. 2020	Ans: B
Q31. UMANG's core platform is integrated with: A. Only banking B. Aadhaar, DigiLocker, PayGov, Rapid Assessment System (RAS) etc. C. Only education D. Only health	Ans: B
Q32. UMANG supports how many Indian languages? A. 5 B. 12 C. 22 D. 8	Ans: B
Q33. Over how many services from central/states departments are on boarded into UMANG? A. 500 B. 2,033 C. 100 D. 5000	Ans: B
Q34. Digital Locker (Digilocker) is a key initiative under: A. Make in India B. Digital India C. Swachh Bharat D. Skill India	Ans: B

Q35. Digital Locker is a platform for insurance and verification of documents and certificates in: A. Paper form B. A digital way (eliminating use of physical documents) C. Audio form D. Video form	Ans: B
Q36. Indian residents who signup for a DigiLocker account get a dedicated: A. Physical locker B. Cloud storage space C. Bank account D. Email address	Ans: B
Q37. Over how many crore users are registered with DigiLocker platform? A. 10 B. 29.61 crore C. 50 crore D. 5 crore	Ans: B
Q38. Over how many crore documents are accessible via Digi Locker? A. 100 B. 675 crore C. 50 D. 1000	Ans: B
Q39. PRAGATI stands for: A. Progressive Governance And Technology Initiative B. Pro-Active Governance And Timely Implementation C. Public Resource And Government Technology Initiative D. Program for Active Government And Technology	Ans: B
Q40. PRAGATI is a video conferencing facility part of Digital India programme for e-Governance, launched in: A. 2010 B. 2015 C. 2017 D. 2020	Ans: B
Q41. PRAGATI enables faster implementation of central level schemes, state level projects and resolutions of grievances between: A. Two states only B. State and central level departments C. Two countries D. Private companies	Ans: B

Q42. MeitY has made e-Praamaan available for public usage with the help of: A. IIT Delhi B. C-DAC, Mumbai C. IISc Bangalore D. NIC Delhi	Ans: B
Q43. Orthographic 3D Projection (a form of parallel projection) is a means of representing 3D objects in: A. 3D B. 2D C. 4D D. 1D	Ans: B
Q44. TAG Heuer has associated with which company to develop a smart luxury watch? A. Apple B. Google C. Samsung D. Microsoft	Ans: B
Q45. NKN connects SDCs (State Data Centre), NDCs (National Data Centre), Ministries and departments across: A. Only one state B. The government (all educational institutes) C. Only private sector D. Only banks	Ans: B
Q46. India's first payment ecosystem for transport based on NCMC standards was launched on: A. January 26, 2019 B. 4th March, 2019 C. August 15, 2019 D. December 25, 2018	Ans: B
Q47. cVIGIL is an innovative mobile application for citizens to report: A. Traffic violations B. Model Code of Conduct and Expenditure violations during elections C. Tax fraud D. Pollution	Ans: B
Q48. An IC can hold anywhere from hundreds to: A. Thousands B. Millions of transistors, resistors and capacitors C. Tens D. Only one	Ans: B

Q49. e-Sign can be integrated with service delivery applications via:

- A. Email
- B. An Application Programming Interface (API)
- C. Phone call
- D. Letter

Ans: B

Q50. Digital India's flagship program is aimed at transforming India into a digitally empowered society and:

- A. Agricultural economy
- B. Knowledge economy
- C. Industrial economy
- D. Trading economy

Ans: B

Answer Key & Explanations

Q1. Answer: B — A cloud services platform mainly via the internet

Page 496: Cloud Computing – On demand delivery of resources through a cloud services platform mainly via the internet.

Topic: Cloud Computing

Q2. Answer: B — Platforms just as required by applications/users

Page 496: It is a distributed computer architecture that provides software, infrastructure and platforms just as required by applications/users.

Topic: Cloud Computing

Q3. Answer: B — Numerical and statistic calculations

Page 496: Spreadsheet/MS Excel is appropriate for performing numerical and statistic calculations.

Topic: Computer Facts

Q4. Answer: B — Interactive Computer Graphics

Page 496: CAD/CAM systems are based on Interactive Computer Graphics.

Topic: Computer Facts

Q5. Answer: B — 16

Page 496: DES uses 16 round Feistel structure and the block size is 64 bits.

Topic: Computer Facts

Q6. Answer: B — 64 bits

Page 496: DES uses 16 round Feistel structure and the block size is 64 bits.

Topic: Computer Facts

Q7. Answer: B — Edsger W. Dijkstra

Page 496: The Banker algorithm is a resource allocation and deadlock avoidance algorithm developed by Edsger W. Dijkstra.

Topic: Computer Facts

Q8. Answer: B — Different protocols

Page 496: Gateway is used to connect two different networks using different protocols.

Topic: Computer Facts

Q9. Answer: B — 4 bits

Page 496: A half byte is known as 'Nibble' (consists 4 bits).

Topic: Computer Facts

Q10. Answer: B — 256 columns

Page 496: MS Excel Version 8.0 to 11.0 could handle 65,536 rows and 256 columns.

Topic: Computer Facts

Q11. Answer: B — 16,384 columns

Page 496: Later Version 12.0 can handle 10,48,576 rows and 16,384 columns.

Topic: Computer Facts

Q12. Answer: B — Kerala

Page 496: The robot named Pepe was recently used by pupils at a primary school in Kerala.

Topic: Computer Facts

Q13. Answer: B — National Common Mobility Card (NCMC) standards

Page 496: India's first indigenously developed payment ecosystem for transport, based on National Common Mobility Card (NCMC) standards, launched on 4th March, 2019.

Topic: Govt Initiatives

Q14. Answer: B — Election Commission of India

Page 496: Mobile App 'cVIGIL' has been launched by Election Commission of India for citizens to report Model Code of Conduct and Expenditure violations during elections.

Topic: Govt Initiatives

Q15. Answer: B — Security

Page 496: VISHWAS is acronym for Video Integration and State Wide Advanced Security.

Topic: Govt Initiatives

Q16. Answer: B — Law and Order Management, Proactive traffic management and control, Crime detection

Page 496: VISHWAS aims at improving Law and Order Management, Proactive traffic management and control, Crime detection through video analytics.

Topic: Govt Initiatives

Q17. Answer: B — Bind

Page 496: A gutter margin adds extra space to margins of a document you plan to bind.

Topic: Computer Facts

Q18. Answer: B — Silicon

Page 496: An IC is usually made of silicon, can hold hundreds to millions of transistors, resistors and capacitors.

Topic: Computer Facts

Q19. Answer: B — 1990

Page 496: Tim Berners-Lee is credited with developing in 1990, both the first web server and the first web browser called WorldWideWeb (later renamed Nexus).

Topic: Internet

Q20. Answer: B — Nexus

Page 496: The first web browser was called WorldWideWeb (without spaces) and later renamed Nexus.

Topic: Internet

Q21. Answer: B — 1st October, 2014

Page 496: Windows 10 was released as a preview on 1st October, 2014.

Topic: Computer Facts

Q22. Answer: B — 5th October, 2021

Page 496: Windows 11 was released on 5th October, 2021.

Topic: Computer Facts

Q23. Answer: B — e-Authentication for public services

Page 496: e-Pramaan: Framework for e-Authentication for public services. Objective is to deliver government services securely.

Topic: Govt Initiatives

Q24. Answer: B — 2015

Page 496: Online e-Sign (e-Hastakshar) facility was formally launched in 2015.

Topic: Govt Initiatives

Q25. Answer: B — A legally acceptable form

Page 496: e-Sign facilitates instant signing of documents online by citizens in a legally acceptable form.

Topic: Govt Initiatives

Q26. Answer: B — 2010

Page 496: National Knowledge Network (NKN) was established in 2010.

Topic: Govt Initiatives

Q27. Answer: B — A high-speed data communication network

Page 496: NKN objective: interconnect all institutions of higher learning and research with a high-speed data communication network.

Topic: Govt Initiatives

Q28. Answer: B — Three independent verticals under one umbrella

Page 496: NKN is the only network globally that carries research/evaluation, internet and e-Governance traffic as three independent verticals under one umbrella.

Topic: Govt Initiatives

Q29. Answer: B — Unified Mobile Application for New-Age Governance

Page 496: UMANG: Unified Mobile Application for New-Age Governance.

Topic: Govt Initiatives

Q30. Answer: B — 2017

Page 496: UMANG has been developed and launched in 2017 as a single mobile platform to deliver major government services.

Topic: Govt Initiatives

Q31. Answer: B — Adhaar, DigiLocker, PayGov, Rapid Assessment System (RAS) etc.

Page 496: UMANG's core platform is integrated with Adhaar, DigiLocker, PayGov, Rapid Assessment System (RAS) etc.

Topic: Govt Initiatives

Q32. Answer: B — 12

Page 496: UMANG supports 12 Indian languages, in addition to English.

Topic: Govt Initiatives

Q33. Answer: B — 2,033

Page 496: Over 2,033 services from 207 central/states departments are on boarded into UMANG.

Topic: Govt Initiatives

Q34. Answer: B — Digital India

Page 496: Digital Locker (Digilocker) is a key initiative under Digital India, the Government of India's flagship program.

Topic: Govt Initiatives

Q35. Answer: B — A digital way (eliminating use of physical documents)

Page 496: Digital Locker is a platform for insurance and verification of documents and certificates in a digital way, eliminating use of physical documents.

Topic: Govt Initiatives

Q36. Answer: B — Cloud storage space

Page 496: Indian residents who signup get a dedicated cloud storage space.

Topic: Govt Initiatives

Q37. Answer: B — 29.61 crore

Page 496: Over 29.61 crore users are registered with this platform.

Topic: Govt Initiatives

Q38. Answer: B — 675 crore

Page 496: Over 675 crore documents are accessible via Digi Locker.

Topic: Govt Initiatives

Q39. Answer: B — Pro-Active Governance And Timely Implementation

Page 496: PRAGATI: Pro-Active Governance And Timely Implementation.

Topic: Govt Initiatives

Q40. Answer: B — 2015

Page 496: PRAGATI video conferencing facility, part of Digital India programme, e-Governance Reforming Government through Technology, was launched in 2015.

Topic: Govt Initiatives

Q41. Answer: B — State and central level departments

Page 496: PRAGATI enables faster implementation of central level schemes/projects and resolutions of grievances between state and central level departments.

Topic: Govt Initiatives

Q42. Answer: B — C-DAC, Mumbai

Page 496: MeitY has made e-Pramaan available for public usage with the help of C-DAC, Mumbai.

Topic: Govt Initiatives

Q43. Answer: B — 2D

Page 496: Orthographic 3D Projection (parallel projection) is a means of representing 3D objects in 2D.

Topic: Computer Facts

Q44. Answer: B — Google

Page 496: TAG Heuer, a premium watchmaker, has associated with Google to develop a smart luxury watch.

Topic: Computer Facts

Q45. Answer: B — The government (all educational institutes)

Page 496: NKN connects all SDCs, NDCs, Ministries and departments across the government and educational institutions.

Topic: Govt Initiatives

Q46. Answer: B — 4th March, 2019

Page 496: India's first payment ecosystem for transport based on NCMC was launched on 4th March, 2019.

Topic: Govt Initiatives

Q47. Answer: B — Model Code of Conduct and Expenditure violations during elections

Page 496: cVIGIL is for citizens to report Model Code of Conduct and Expenditure violations during elections.

Topic: Govt Initiatives

Q48. Answer: B — Millions of transistors, resistors and capacitors

Page 496: An IC can hold anywhere from hundreds to millions of transistors, resistors and capacitors.

Topic: Computer Facts

Q49. Answer: B — An Application Programming Interface (API)

Page 496: e-Sign is an online electronic signature service which can be integrated with service delivery applications via an API.

Topic: Govt Initiatives

Q50. Answer: B — Knowledge economy

Page 496: Digital India's flagship program aimed at transforming India into a digitally empowered society and knowledge economy.

Topic: Govt Initiatives